

AP EAMCET-2019-Engineering-20 Apr-Shift-1 Mathematics

1. $f(x) = \frac{x}{e^x - 1} + \frac{x}{2} + 2\cos^3 \frac{x}{2}$ on $\mathbb{R} - \{0\}$ is

- (1) One one function
- (2) Bijection
- (3) Algebraic function
- (d) Even function

2. Consider the following lists

List I

A) $f(x) = \frac{|x+2|}{x+2}, x \neq -2$

B) $g(x) = [x], x \in \mathbb{R}$

C) $h(x) = |x - [x]|, x \in \mathbb{R}$

D) $f(x) = \frac{1}{2 - \sin 3x}, x \in \mathbb{R}$

List II

I) $\left[\frac{1}{3}, 1\right]$

(II) $\mathbb{Z}$

(III) $\mathbb{W}$

(IV) $[0, 1)$

(V) $\{-1, 1\}$

	A	B	C	D
(1)	V	III	II	I

	A	B	C	D
(2)	III	II	IV	I

	A	B	C	D
(3)	V	III	IV	I

	A	B	C	D
(4)	I	II	III	IV

3. Assertion (A):

$$(1) + (1 + 2 + 4) + (4 + 6 + 9) + (9 + 12 + 16) + \dots + (81 + 90 + 100) = 1000$$

Reason (R): $\sum_{r=1}^n (r^3 - (r-1)^3) = n^3$ for any natural number n .

(1) Both (A) and (R) are true and (R) is the correct explanation of (A).

(2) Both (A) and (R) are true and (R) is not the correct explanation of (A).

(3) (A) is true but (R) is false.

(4) (A) is false but (R) is true.

4. If $A = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$, $P = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$ and $X = APA^T$ then $A^T X^{50} A =$

(1) $\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$

(2) $\begin{bmatrix} 2 & 1 \\ 0 & -1 \end{bmatrix}$

(3) $\begin{bmatrix} 25 & 1 \\ 1 & -25 \end{bmatrix}$

(4) $\begin{bmatrix} 1 & 50 \\ 0 & 1 \end{bmatrix}$

5. If $[x]$ is the greatest integer less than or equal to x and $|x|$ is the modulus of x , then the system of three equations $2x + 3|y| + 5|z| = 0$, $x + |y| - 2|z| = 4$, $x + |y| + [z] = 1$ has

(1) a unique solution

(2) finitely many solutions

(3) infinitely many solutions

(4) no solutions

6. Investigate the λ and μ for the system $x + 2y + 3z = 6$,
 $x + 3y + 5z = 9$, $2x + 5y + \lambda z = \mu$ and match the following values
 in List I with the items in List II

List I

A) $\lambda = 8, \mu \neq 15$

B) $\lambda \neq 8, \mu \in \mathbb{R}$

C) $\lambda = 8, \mu = 15$

List II

I) Infinitely many solutions

II) No solutions

III) Unique solutions

The correct match is

(1)

A	B	C
I	II	III

(2)

A	B	C
II	III	I

(3)

A	B	C
III	I	II

(4)

A	B	C
III	II	I

7. If $z_n = x + iy$, $x, y \in \mathbb{R}$, $(x, y) \neq (0, -4)$ and $\text{Arg}\left(\frac{2z-3}{z+4i}\right) = \frac{\pi}{4}$

then the locus of z is

(1) $2x^2 + 2y^2 + 5x + 5y - 12 = 0$

$$(2) \ 2x^2 - 3xy + y^2 + 5x + y - 12 = 0$$

$$(3) \ 2x^2 + 3xy + y^2 + 5x + y + 12 = 0$$

$$(4) \ 2x^2 + 2y^2 - 11x + 7y - 12 = 0$$

8. If $z = x + iy$, $x, y \in R$ and the imaginary part of $\frac{\bar{z} - 1}{z - i}$ is 1 then

locus of z is

$$(1) \ x + y + 1 = 0$$

$$(2) \ x + y + 1 = 0, (x, y) \neq (0, -1)$$

$$(3) \ x^2 + y^2 - x + 3y + 2 = 0$$

$$(4) \ x^2 + y^2 - x + 3y + 2 = 0, (x, y) \neq (0, -1)$$

9. If ω represents a complex cube root of unity, then

$$\left(1 + \frac{1}{\omega}\right)\left(1 + \frac{1}{\omega^2}\right) + \left(2 + \frac{1}{\omega}\right)\left(2 + \frac{1}{\omega^2}\right) + \dots + \left(n + \frac{1}{\omega}\right)\left(n + \frac{1}{\omega^2}\right) =$$

$$(1) \ \frac{n(n^2 + 1)}{3}$$

$$(2) \ \frac{n(n^2 + 2)}{3}$$

$$(3) \frac{n(n^2 - 2)}{3}$$

$$(4) \frac{n^2(n-1)}{6}$$

10. If ω represents a complex cube root of unity, then

$$\sum_{r=1}^9 r(r+1-\omega)(r+1-\omega^2) =$$

(1) 5025

(2) 4020

(3) 2016

(4) 3015

11. If α and β are the roots of $x^2 + 7x + 3 = 0$ and $\frac{2\alpha}{3-4\alpha}, \frac{2\beta}{3-4\beta}$

are the roots of $ax^2 + bx + c = 0$ and GCD of a, b, c is 1 then

$$a + b + c =$$

(1) 11

(2) 0

(3) 243

(4) 81

12. If α, β are the roots of $x^2 + bx + c = 0$, γ, δ are the roots of

$x^2 + b_1x + c_1 = 0$ and $\gamma < \alpha < \delta < \beta$ then $(c - c_1)^2 <$

(1) $(b_1 - b)(bc_1 - b_1c)$

(2) 1

(3) $(b - b_1)^2$

(4) $(c - c_1)(b_1c - b_1c_1)$

13. Let a, b, c be the sides of scalane triangle. If λ is a real number

such that root of the equation

$x^2 + 2(a + b + c)x + 3\lambda(ab + bc + ca) = 0$ are real then the

interval in which λ lies is

(1) $\left(-\infty, \frac{4}{3}\right)$

(2) $\left(\frac{5}{3}, \infty\right)$

(3) $\left(\frac{1}{3}, \frac{5}{3}\right)$

(4) $\left(\frac{4}{3}, \infty\right)$

14. The polynomial equation of degree 4 having real coefficients with three of its roots as $2 \pm \sqrt{3}$ and $1 + 2i$, is

(1) $x^4 - 6x^3 - 14x^2 + 22x + 5 = 0$

(2) $x^4 - 6x^3 - 19x + 22x - 5 = 0$

(3) $x^4 - 6x^3 + 19x - 22x + 5 = 0$

(4) $x^4 - 6x^3 + 14x^3 - 22x + 5 = 0$

15. All letters of the word ANIMAL are permitted in all possible ways and the permutations thus formed are arranged in dictionary order. If the rank of word ANIMAL is x . Then the permutation with rank x , among the permutation obtained by permuting the letter of the word PERSON and arranging the permutations thus formed in the dictionary order is

(1) ENOPRS

(2) NOSPRE

(3) NOEPRS

(4) ESORNP

16. A student is allowed to choose atmost n books from a collection of $2n + 1$ books. If the total number of ways in which he can select atleast one book is 255, then the value of n is

- (1) 4
- (2) 5
- (3) 6
- (4) 7

17. The sum of all the coefficients in the binomial expansion of

$(1 + 2x)^n$ is 6561. Let $R = (1 + 2x)^n = I + F$ where $I \in \mathbb{N}$ and

$0 < F < 1$. If $x = \frac{1}{\sqrt{2}}$, then $1 - \frac{F}{1 + (\sqrt{2} - 1)^4} =$

- (1) $(3\sqrt{2} - 4)$
- (2) $4(3\sqrt{2} + 4)$
- (3) $(\sqrt{2} - 1)^4$
- (4) 1

18. If $\frac{(1 - px)^{-1}}{1 - qx} = a_0 + a_1x + a_2x^2 + a_3x^3 + \dots$, then $a_n =$

$$(1) \frac{p^{n+1} - q^{n+1}}{q - p}$$

$$(2) \frac{p^{n+1} - q^{n+1}}{p - q}$$

$$(3) \frac{p^n - q^n}{q - p}$$

$$(4) \frac{p^n - q^n}{p - q}$$

19. If $\frac{3}{(x-1)(x^2+x+1)} = \frac{1}{x-1} - \frac{x+2}{(x^2+x+1)} = f_1(x) - f_2(x)$ and

$$\frac{x+1}{(x-1)^2(x^2+x+1)} = Af_1(x) + \left(B + \frac{D}{x-1}\right)f_2(x) + \frac{C}{(x-1)^2} \quad \text{then}$$

$$A + B + C + D =$$

(1) 1

(2) $-\frac{1}{3}$

(3) 0

(4) $\frac{1}{3}$

20. Let M and m respectively denote the maximum and minimum values of $[f(\theta)]^2$ where

$$f(\theta) = \sqrt{a^2 \cos^2 \theta + b^2 \sin^2 \theta} + \sqrt{a^2 \sin^2 \theta + b^2 \cos^2 \theta}. \quad \text{Then}$$

$$M - m =$$

(1) $a^2 + b^2$

(2) $(a - b)^2$

(3) $a^2 b^2$

(4) $(a + b)^2$

21. If $\cos A = -\frac{60}{61}$ and $\tan B = -\frac{7}{24}$ and neither A nor B is in the

second quadrant, then the angle $A + \frac{B}{2}$ lies in the quadrant

(1) 1

(2) 2

(3) 3

(4) 4

22. $\cos^2 5^\circ - \cos^2 15^\circ - \sin^2 15^\circ + \sin^2 35^\circ$
 $+ \cos 15^\circ \sin 15^\circ - \cos 5^\circ \sin 35^\circ =$

(1) 0

(2) 1

(3) $\frac{3}{2}$

(4) 2

23. If $\cos \theta \neq 0$ and $\sec \theta - 1 = (\sqrt{2} - 1) \tan \theta$ then $\theta =$

(1) $n\pi + \frac{\pi}{8}, n \in Z$

(2) $2n\pi + \frac{\pi}{4}$ (or) $2n\pi, n \in Z$

(3) $2n\pi + \frac{\pi}{8}, n \in Z$

(4) $2n\pi - \frac{\pi}{4}$ (or) $2n\pi, n \in Z$

24. $\cot \left[\sum_{n=3}^{32} \cot^{-1} \left(1 + \sum_{k=1}^n 2k \right) \right] =$

(1) $\frac{10}{3}$

(2) $\frac{8}{3}$

(3) $\frac{14}{3}$

(4) $\frac{16}{3}$

25. If $\sin x \cos y = \cos \theta$, $\cos x \sin y = \sin \theta$ and $4 \tan x = 3$. Then

$\sin^2 y =$

(1) $\frac{4}{5}$

(2) $\frac{9}{16}$

(3) $\frac{9}{25}$

(4) $\frac{16}{25}$

26. In triangle ABC, if $\frac{b+c}{9} = \frac{c+a}{10} = \frac{a+b}{11} = k$ then

$\frac{\cos A + \cos B}{\cos C} =$

(1) $\frac{9}{10}$

(2) $\frac{10}{11}$

(3) $\frac{11}{12}$

(4) $\frac{12}{13}$

27. In triangle ABC, with usual notation, match the items in list I with the items in list II and choose the correct option

List I

List II

A)

$$r_1 r_2 \sqrt{\left(\frac{4R - r_1 - r_2}{r_1 + r_2} \right)}$$

I)

$$b$$

B)

$$\frac{r_2(r_3 + r_2)}{\sqrt{r_1 r_2 + r_2 r_3 + r_3 r_1}}$$

II)

 a^2, b^2, c^2 are in A.P

C)

$$\frac{a}{c} = \frac{\sin(A - B)}{\sin(B - C)}$$

III) Δ

D)

$$bc \cos^2 \frac{A}{2}$$

IV) $Rr_1 r_2 r_3$ V) $s(s - a)$

The correct match is

	A	B	C	D
(1)	IV	III	I	V

	A	B	C	D
(2)	V	IV	III	II

	A	B	C	D
(3)	III	I	II	V

	A	B	C	D
(4)	IV	V	II	I

28. If a, b, c are the sides of $\triangle ABC$ for which $r_1 = 8$, $r_2 = 12$ and $r_3 = 24$ then the ordered triad $(a, b, c) =$

(1) $(12, 20, 16)$

(2) $(12, 16, 20)$

(3) $(16, 12, 20)$

(4) $(20, 16, 12)$

29. If $4\bar{i} + 7\bar{j} + 8\bar{k}$, $2\bar{i} + 3\bar{j} + 4\bar{k}$, $2\bar{i} + 5\bar{j} + 7\bar{k}$ are respectively the position vectors of vertices A, B, C of triangle ABC, then the position vectors of the point where the bisector of angle meets $\overline{BC}$ is

$$(1) \ 2\bar{i} + \frac{13}{3}\bar{j} + 2\bar{k}$$

$$(2) \ 2\bar{i} - \frac{13}{3}\bar{j} + 6\bar{k}$$

$$(3) \ 2\bar{i} + 13\bar{j} + 6\bar{k}$$

$$(4) \ 2\bar{i} + \frac{13}{3}\bar{j} + 6\bar{k}$$

30. The equation of the plane passing through the points $\bar{i} + 2\bar{j} - \bar{k}$ and perpendicular to the line of intersection of the plane $\bar{r} \cdot (3\bar{i} - \bar{j} + \bar{k}) = 1$ and $\bar{r} \cdot (\bar{i} + 4\bar{j} - 2\bar{k}) = 2$ is

$$(1) \ \bar{r} \cdot (-2\bar{i} - 5\bar{j} + \bar{k}) = 0$$

$$(2) \ \bar{r} \cdot (\bar{i} + 7\bar{j} + 4\bar{k}) = 0$$

$$(3) \ \bar{r} \cdot (2\bar{i} - 7\bar{j} - 13\bar{k}) = 1$$

$$(4) \ \bar{r} \cdot (-2\bar{i} + 7\bar{j} + 13\bar{k}) = 0$$

31. If the position vector of the vertices, A, B and C of ΔABC are $\bar{i} + 2\bar{j} - 5\bar{k}$, $-2\bar{i} + 2\bar{j} + \bar{k}$ and $2\bar{i} + \bar{j} - \bar{k}$ respectively then $\angle B =$

$$(1) \ \cos^{-1}\left(\frac{7}{3\sqrt{10}}\right)$$

$$(2) \cos^{-1}\left(\frac{8}{\sqrt{105}}\right)$$

$$(3) \cos^{-1}\left(\frac{1}{\sqrt{42}}\right)$$

$$(4) \cos^{-1}\left(-\frac{7}{3\sqrt{10}}\right)$$

32. If the position vector of the vertices of ΔABC are $\overrightarrow{OA} = 3\vec{i} + \vec{j} + 2\vec{k}$, $\overrightarrow{OB} = \vec{i} + 2\vec{j} + 3\vec{k}$ and $\overrightarrow{OC} = 2\vec{i} + 3\vec{j} + \vec{k}$ then the length of altitude of triangle ABC drawn from A is

$$(1) \sqrt{\frac{3}{2}}$$

$$(2) \frac{3}{\sqrt{2}}$$

$$(3) \frac{\sqrt{3}}{2}$$

$$(4) \frac{3}{2}$$

33. A new tetrahedron is formed by joining the faces of a given tetrahedron OABC. Then the ratio of volume of new tetrahedron to that of given tetrahedron is

(1) $\frac{3}{25}$

(2) $\frac{1}{27}$

(3) $\frac{5}{62}$

(4) $\frac{1}{162}$

34. Let $\vec{A} = 2\vec{i} + \vec{j} - 2\vec{k}$ and $\vec{B} = \vec{i} + \vec{j}$ if $\vec{C}$ is such that $\vec{A} \cdot \vec{C} = |\vec{C}|$, $|\vec{C} - \vec{A}| = 2\sqrt{2}$ and the angle between $\vec{A} \times \vec{B}$ and $\vec{C}$ is 30° then the value of $|(\vec{A} \times \vec{C}) \times \vec{C}|$ is

(1) $\frac{2}{3}$

(2) $\frac{3}{2}$

(3) 3

(4) 2

35. If $a_0, a_1, \dots, a_{11}$ are in an arithmetic progression with common difference d , then their mean deviation from arithmetic mean is

(1) $\frac{30}{11}|d|$

(2) $2|d|$

(3) $3|d|$

(4) $12|d|$

36. The variance of the following continuous frequency distribution is

Class Interval	0-10	10-20	20-30	30-40
Frequency	2	3	4	1

(1) 201

(2) 62

(3) 19

(4) 84

37. If two sections of strength 30 and 45 are formed from 75 students who are admitted in a school, then the probability that two particular students are always together in the same section is

(1) $\frac{66}{185}$

(2) $\frac{19}{37}$

(3) $\frac{29}{185}$

(4) $\frac{18}{37}$

38. A bag contains $2n$ coins out of which $n-1$ are unfair with head on both sides and remaining are fair. One coin is picked from a bag at random and tossed. If the probability that head falls in the toss is $\frac{41}{56}$ then the number of unfair coins in the bag is

(1) 18

(2) 15

(3) 13

(4) 14

39. Bag A contains 6 Green and 8 Red balls and bag B contains 9 Green and 5 Red balls. A card is drawn at random from a well shuffled pack of 52 playing cards. If it is a spade, two balls are drawn at random from bag A, otherwise two balls are drawn at random from bag B. If the two balls drawn are found to be of the

same colour, then the probability that they are drawn from bag A is

(1) $\frac{43}{181}$

(2) $\frac{1}{4}$

(3) $\frac{48}{131}$

(4) $\frac{43}{138}$

40. A random variable X has the probability distribution,

$X = x_i$	1	2	3	4	5	6
$P(X = x_i)$	0.2	0.3	0.12	0.1	0.2	0.08

If $A = \{x_i / x_i \text{ is a prime number}\}$, $B = \{x_i / x_i < 4\}$ are two events then $P(A \cup B) =$

(1) 0.31

(2) 0.62

(3) 0.82

(4) 0.41

41. In a Poisson distribution with mean $\sum_{x=0}^{\infty} |x - \bar{x}| P(X = x) =$

(1) e

(2) $\frac{1}{e}$

(3) $\frac{2}{e}$

(4) $\frac{2}{3e}$

42. Two straight rods of length $2a$ and $2b$ move along the coordinate axis in such a way that their extremities are always concyclic.

Then the locus of center of such circles is

(1) $2(x^2 + y^2) = a^2 + b^2$

(2) $2(x^2 - y^2) = a^2 + b^2$

(3) $x^2 + y^2 = a^2 + b^2$

(4) $x^2 - y^2 = a^2 - b^2$

43. When the coordinate axes are rotated around the origin in the positive direction through an angle $\frac{\pi}{4}$, if the equation

$25x^2 + 9y^2 = 225$ is transformed to $\alpha x^2 + \beta xy + \gamma y^2 = \delta$ the
 $\alpha + \beta + \gamma - (\sqrt{\delta})^2 =$

- (1) 3
- (2) 9
- (3) 4
- (4) 16

44. The equation of the line through the point of intersection of the lines $3x - 4y + 1 = 0$ and $5x + y - 1 = 0$ and making equal non-zero intercepts on the coordinate axes is

- (1) $2x + 2y = 3$
- (2) $23x + 23y = 6$
- (3) $23x + 23y = 11$
- (4) $2x + 2y = 7$

45. The line through $P(a, 2)$ where $a \neq 0$, making an angle 45° with positive X-axis meets the curve $\frac{x^2}{9} + \frac{y^2}{4} = 1$ at A and D and the coordinate axis at B and C. If PA, PB, PC and PD are in geometric progression then $2a =$

- (1) 13
- (2) 7
- (3) 1
- (4) -13

46. The equation of the perpendicular bisector of the sides AB and AC of a $\triangle ABC$ are $x - y = 5$ and $x + 2y = 0$ respectively. If A is $(1, -2)$ then the equation of straight line BC is

- (1) $14x + 23y - 40 = 0$
- (2) $12x + 17y - 28 = 0$
- (3) $14x - 29y - 30 = 0$
- (4) $7x - 12y + 15 = 0$

47. If each line of a pair of lines original line passing through origin is at a perpendicular distance of 4 units from the point $(3, 4)$, then the equation of the pair of lines is

- (1) $7x^2 + 24xy = 0$
- (2) $7y^2 + 24xy = 0$
- (3) $7y^2 - 24xy = 0$
- (4) $7x^2 - 24xy = 0$

48. Variable straight lines $y = mx + c$ make intercepts on the curve $y^2 - 4ax = 0$ which subtend a right angle at the origin. Then the point of concurrence of these lines $y = mx + c$ is

- (1) $(4a, 0)$
- (2) $(2a, 0)$
- (3) $(-4a, 0)$
- (4) $(-2a, 0)$

49. The abscissae of two points P, Q are the roots of the equation $2x^2 + 4x - 7 = 0$ and their ordinates are the roots of the equation $3x^2 - 12x - 1 = 0$. Then the centre of the circle with PQ as a diameter is

- (1) $(-1, 2)$
- (2) $(-2, 6)$
- (3) $(1, -2)$
- (4) $(2, -6)$

50. If the angle between a pair of tangents drawn from a point P to the circle $x^2 + y^2 + 4x - 6y + 9\sin^2 \alpha + 13\cos^2 \alpha = 0$ is 2α , then the equation of the locus of P is

(1) $x^2 + y^2 + 4x - 6y + 4 = 0$

(2) $x^2 + y^2 + 4x - 6y - 9 = 0$

(3) $x^2 + y^2 - 4x + 6y - 4 = 0$

(4) $x^2 + y^2 + 4x - 6y + 9 = 0$

51. The equation of the circle whose radius is 3 and which touches internally the circle $x^2 + y^2 - 4x - 6y - 12 = 0$ at the point $(-1, -1)$ is

(1) $5x^2 + 5y^2 + 9x - 6y - 7 = 0$

(2) $5x^2 + 5y^2 - 8x - 14y - 32 = 0$

(3) $5x^2 + 5y^2 - 6x + 8y - 8 = 0$

(4) $5x^2 + 5y^2 + 6x - 8y - 12 = 0$

52. Suppose that the circle $x^2 + y^2 + 2gx + 2fy + c = 0$ has its centre on $2x + 3y - 7 = 0$ and cut the circles $x^2 + y^2 - 4x - 6y + 11 = 0$

and $x^2 + y^2 - 10x - 4y + 21 = 0$ orthogonally. Then

$$5g - 10f + 3c =$$

- (1) 0
- (2) 1
- (3) 3
- (4) 9

53. If the radical axis of the circle $x^2 + y^2 + 2gx + 2fy + c = 0$ and $2x^2 + 2y^2 + 3x + 8y + 2c = 0$ touches the circle $x^2 + y^2 + 2x + 2y + 1 = 0$ then $(4g - 3)(f - 2) =$

- (1) 0
- (2) -1
- (3) 1
- (4) 2

54. The parabola $x^2 = 4ay$ makes an intercept of length $\sqrt{40}$ units on the line $y = 1 + 2x$ then a value of $4a$ is

- (1) 2
- (2) -2
- (3) -1

(4) 2

55. The locus of the point of intersection of perpendicular normals to the parabola $y^2 = ax$ is

(1) $y^2 - 2ax + a^2 = 0$

(2) $y^2 + ax + 2a^2 = 0$

(3) $y^2 - ax + 2a^2 = 0$

(4) $y^2 - ax + 3a^2 = 0$

56. P is a variable point on the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ with foci F_1 and F_2 . If A is the area of the triangle PF_1F_2 , then the maximum value of A is

(1) $\frac{e}{ab}$

(2) $\frac{ae}{b}$

(3) aeb

(4) $\frac{ab}{e}$

57. If the line joining the points $A(\alpha)$ and $B(\beta)$ on the ellipse

$\frac{x^2}{25} + \frac{y^2}{9} = 1$ is a focal chord, then one possible value of

$\cot \frac{\alpha}{2} \cot \frac{\beta}{2}$ is

(1) -3

(2) 3

(3) -9

(4) 9

58. The equation of a tangent to the hyperbola

$16x^2 - 25y^2 - 96x + 100y - 356 = 0$ which makes an angle 45°

with its transverse axis is

(1) $x - y + 2 = 0$

(2) $x - y + 4 = 0$

(3) $x + y + 2 = 0$

(4) $x + y + 4 = 0$

59. If $P(0, 7, 10)$, $Q(-1, 6, 6)$ and $R(-4, 9, 6)$ are three points in the space, then PQR is

- (1) Right angled isosceles triangle
- (2) Equilateral triangle
- (3) Isosceles but not right angled triangle
- (4) Scalene triangle

60. $A(2,3,5)$, $B(\alpha,3,3)$ and $C(7,5,\beta)$ are the vertices of a triangle.

If the median through A is equally inclined with the coordinate

axes then $\cos^{-1}\left(\frac{\alpha}{\beta}\right) =$

(1) $\cos^{-1}\left(\frac{-1}{9}\right)$

(2) $\frac{\pi}{2}$

(3) $\frac{\pi}{3}$

(4) $\cos^{-1}\left(\frac{2}{5}\right)$

61. The plane $3x + 4y + 6z + 7 = 0$ is rotated about the line

$\vec{r} = (\vec{i} + 2\vec{j} - 3\vec{k}) + t(2\vec{i} - 3\vec{j} + \vec{k})$ until the plane passes through

the origin. The equation of the plane in new position is

(1) $x + y + z = 0$

(2) $6x + 3y - 4z = 0$

(3) $4x - 5y - 2z = 0$

(4) $x + 2y + 4z = 0$

62. If $\lim_{x \rightarrow \infty} \left\{ \frac{x^3 + 1}{x^2 + 1} - (\alpha x + \beta) \right\}$ exist and equal to 2 then the ordered pair (α, β) of real numbers is

(1) $(1, -1)$

(2) $(-2, 1)$

(3) $(-1, 1)$

(4) $(1, -2)$

63. For $k > 0$, $\sum_{x=0}^{\infty} \frac{k^x}{x!} \lim_{x \rightarrow \infty} \frac{n!}{(n-x)!} \left(1 - \frac{k}{n}\right)^{n-x} \left(\frac{1}{n}\right)^x =$

(1) 0

(2) k

(3) x

(4) 1

64. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be the function defined by

$$f(x) = \begin{cases} 5, & \text{if } x \leq 1 \\ a + bx, & \text{if } 1 < x < 3 \\ b + 5x, & \text{if } 3 \leq x < 5 \\ 30, & \text{if } x \geq 5 \end{cases} \text{ then } f \text{ is}$$

- (1) continuous if $a = 5$ and $b = 5$
- (2) continuous if $a = 0$, $b = 5$
- (3) continuous if $a = -5$, $b = 10$
- (4) not continuous for any value of a and b

65. Let $[x]$ denote the greatest integer less than or equal to x . Then the number of points where the function $y = [x] + |1 - x|$, $1 \leq x \leq 3$ is not differentiable, is

- (1) 1
- (2) 2
- (3) 3
- (4) 4

66. If $\sqrt{1 - x^6} + \sqrt{1 - y^6} = a(x^3 - y^3)$, then $y^2 \frac{d^2 y}{dx^2} =$

$$(1) \sqrt{\frac{1-y^6}{1-x^6}}$$

$$(2) x\sqrt{\frac{1-y^6}{1-x^6}}$$

$$(3) x^2\sqrt{\frac{1-y^6}{1-x^6}}$$

$$(4) \frac{1}{x^2}\sqrt{\frac{1-y^6}{1-x^6}}$$

67. If $y = f(x)$ is twice differentiable function such that a point P,

$$\frac{dy}{dx} = 4, \frac{d^2y}{dx^2} = -3, \text{ then } \left(\frac{d^2x}{dy^2} \right)_P =$$

$$(1) \frac{64}{3}$$

$$(2) \frac{16}{3}$$

$$(3) \frac{3}{16}$$

$$(4) \frac{3}{64}$$

68. The time T of oscillation of a simple pendulum of length L is governed by $T = 2\pi\sqrt{\frac{L}{g}}$, where g is constant. The percentage by which length be changed in order to correct an error of loss equal to 2 minutes of time per day is

(1) $-\frac{5}{18}$

(2) $-\frac{2}{9}$

(3) $\frac{1}{6}$

(4) $\frac{1}{9}$

69. Let A , G , H and S respectively denote the arithmetic mean, geometric mean, harmonic mean and the sum of the numbers $a_1, a_2, a_3, \dots, a_n$. Then the value of x at which the function

$f(x) = \sum_{k=1}^n (x - a_k)^2$ has minimum is

(1) S

(2) H

(3) G

(4) A

70. For $m > 1$, $n > 1$, the value of c for which the Rolle's theorem is applicable for the function $f(x) = x^{2m-1}(a-x)^{2n}$ in $(0, a)$ is

(1) $\frac{2am-1}{m+2n-1}$

(2) $\frac{a(m-n+1)}{2m+2n}$

(3) $\frac{a(2m-1)}{2m+2n-1}$

(4) $\frac{a(2m+1)}{m+n-1}$

71. If the function $f: [-1, 1] \rightarrow \mathbb{R}$ defined by

$$f(x) = \begin{cases} 2^x + 1, & x \in [-1, 0] \\ 1, & x = 0 \\ 2^x - 1, & x \in (0, 1] \end{cases} \quad \text{then in } [-1, 1], f(x) \text{ has}$$

(1) a maximum

(2) a minimum

(3) both maximum and minimum

(4) neither maximum nor minimum

$$72. \int \frac{x-1}{(x+1)\sqrt{x^3+x^2+x}} dx =$$

$$(1) 2 \tan^{-1} \left(\sqrt{\frac{1+x+x^2}{x}} \right) + c$$

$$(2) \tan^{-1} \left(\sqrt{\frac{1+x+x^2}{x}} \right) + c$$

$$(3) 2 \tan^{-1} \left(\sqrt{\frac{x}{1+x+x^2}} \right) + c$$

$$(4) \tan^{-1} \left(\sqrt{\frac{1+x^2}{x}} \right) + c$$

$$73. \text{ If } I(x) = \int x^2 (\log x)^2 dx \text{ and } I(1) = 0, \text{ then } I(x) =$$

$$(1) \frac{x^3}{18} \left[8(\log x)^2 - 3 \log x \right] + \frac{7}{18}$$

$$(2) \frac{x^3}{27} \left[9(\log x)^2 + 6 \log x \right] - \frac{2}{27}$$

$$(3) \frac{x^3}{27} \left[9(\log x)^2 - 6 \log x + 2 \right] - \frac{2}{27}$$

$$(4) \frac{x^3}{27} \left[9(\log x)^2 - 6\log x - 2 \right] + \frac{2}{27}$$

$$74. \int \frac{x^5 dx}{(x^2 + x + 1)(x^6 + 1)(x^4 - x^3 + x - 1)} =$$

$$(1) \log_e \left| \frac{x^6 - 1}{x^6 + 1} \right| + c$$

$$(2) \frac{1}{12} \log_e \left| \frac{x^6 - 1}{x^6 + 1} \right| + c$$

$$(3) \frac{1}{12} \log_e \left| \frac{x^4 + 1}{x^4 - 1} \right| + c$$

$$(4) \log_e \left| \frac{x^8 + 1}{x^6 - 1} \right| + c$$

$$75. \int \frac{dx}{x + \sqrt{x-1}} =$$

$$(1) \log_e |x + \sqrt{x-1}| - \frac{1}{\sqrt{3}} \tan^{-1} \left(\frac{2\sqrt{x-1} + 1}{\sqrt{3}} \right) + c$$

$$(2) \frac{1}{\sqrt{3}} \log_e |x + \sqrt{x-1}| - \tan^{-1} \left(\frac{2\sqrt{x-1} + 1}{\sqrt{3}} \right) + c$$

$$(3) \frac{2}{\sqrt{3}} \log_e |x + \sqrt{x-1}| - \tan^{-1} \left(\frac{2\sqrt{x-1}+1}{\sqrt{3}} \right) + c$$

$$(4) \log_e |x + \sqrt{x-1}| - \frac{2}{\sqrt{3}} \tan^{-1} \left(\frac{2\sqrt{x-1}+1}{\sqrt{3}} \right) + c$$

$$76. \int_{\log_e 2}^x \frac{dt}{\sqrt{e^t - 1}} = \frac{\pi}{6} \Rightarrow x =$$

$$(1) 2 \cdot \log_e 2$$

$$(2) 3 \cdot \log_e 2$$

$$(3) 4 \cdot \log_e 2$$

$$(4) 8 \cdot \log_e 2$$

$$77. \int_0^1 \frac{\log_e (1+x)}{1+x^2} dx =$$

$$(1) \frac{\pi}{4} \log_e 2$$

$$(2) \frac{\pi}{6} \log_e 6$$

$$(3) \frac{\pi}{2} \log_e 8$$

(4) $\frac{\pi}{8} \log_e 2$

78. If the area of the circle $x^2 + y^2 = 2$ is divided into parts by the parabola $y = x^2$, then the area (in sq. units) of the larger part is

(1) $\frac{3\pi}{2} - \frac{1}{3}$

(2) $6\pi - \frac{4}{3}$

(3) $\frac{4\pi}{3} - \frac{2}{3}$

(4) $4\pi - \frac{1}{4}$

79. If c is a parameter, then the differential equation of the family of curves $x^2 = c(y + c)^2$ is

(1) $x \left(\frac{dy}{dx} \right)^3 + y \left(\frac{dy}{dx} \right)^2 - 1 = 0$

(2) $x \left(\frac{dy}{dx} \right)^3 - y \left(\frac{dy}{dx} \right)^2 + 1 = 0$

$$(3) \ x \left(\frac{dy}{dx} \right)^3 + y \left(\frac{dy}{dx} \right)^2 + 1 = 0$$

$$(4) \ x \left(\frac{dy}{dx} \right)^3 - y \left(\frac{dy}{dx} \right)^2 - 1 = 0$$

80. If $f(x)$, $f'(x)$, $f''(x)$ are positive functions and $f(0)=1$, $f'(0)=2$, then the solution of the differential equation

$$\begin{vmatrix} f(x) & f'(x) \\ f'(x) & f''(x) \end{vmatrix} = 0 \text{ is}$$

(1) e^{2x}

(2) $2 \sin x + 1$

(3) $\sin^2 x + 2x + 1$

(4) e^{4x}

AP EAMCET-2019-Engineering-20 Apr-Shift-1 Physics

81. If the charge of electron 'e', mass of electron 'm', speed of light in vacuum is 'c' and Planck's constant 'h' are taken as fundamental quantities, then the permeability of vacuum ' μ_0 ' can be expressed as

(1) $\frac{h}{mc^2}$

(2) $\frac{hc}{me^2}$

(3) $\frac{h}{ce^2}$

(4) $\frac{mc^2}{he^2}$

82. The velocity of an object moving in a straight line path is given as a function of time by $v = 6t - 3t^2$, where v is in ms^{-1} , t is in s. The average velocity of the object between $t = 0$ and $t = 2$ seconds is

(1) 0 ms^{-1}

(2) 3 ms^{-1}

(3) 2 ms^{-1}

(4) 4 ms^{-1}

83. A gun and a target are at the same horizontal level separated by a distance of 600 m. The bullet is fired from the gun with a velocity of 500 ms^{-1} . In order to hit the target, the gun should be aimed to a height h above the target. The value of h is

(Acceleration due to gravity = 10 ms^{-2})

(1) 2.4 m

(2) 3.6 m

(3) 7.2 m

(4) 10.8 m

84. A projectile is thrown in the upward direction making an angle of 60° with the horizontal with a velocity of 140 ms^{-1} . Then, the time after which its velocity makes 45° with the horizontal is

(Acceleration due to gravity = 10 ms^{-2})

(1) 0.5124 s

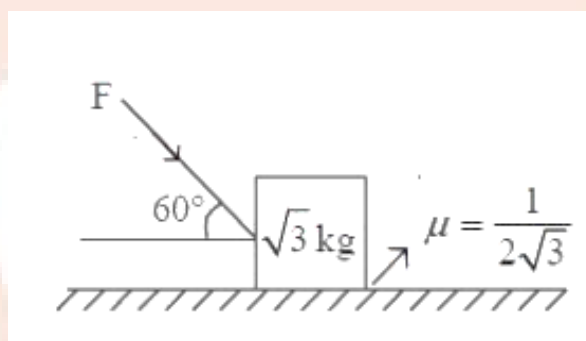
(2) 51.24 s

(3) 5.124 s

(4) 512.4 s

85. The maximum value of the applied force F such that the block as shown in the arrangement does not move is

(Acceleration due to gravity $= 10 \text{ ms}^{-2}$)



(1) 20 N

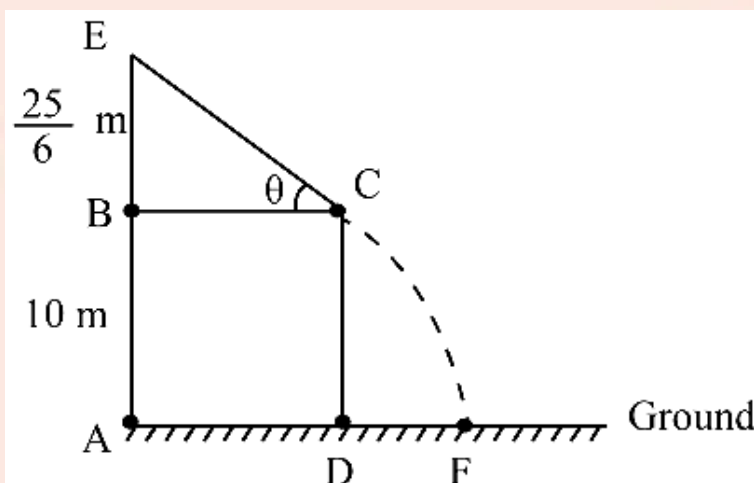
(2) 15 N

(3) 25 N

(4) 10 N

86. A rough inclined plane BCE of height $\left(\frac{25}{6}\right) \text{ m}$ is kept on a rectangular wooden block ABCD of height 10 m, as shown in the figure. A small block is allowed to slide down from the top E of the inclined plane. The coefficient of kinetic friction

between the block and the inclined plane is $\frac{1}{8}$ and the angle of inclination of the inclined plane is $\sin^{-1}(0.6)$. If the small block finally reaches the ground at point F. then DF =
(Acceleration due to gravity = 10 ms^{-2})



- (1) $\frac{5}{3}$ m
- (2) $\frac{10}{3}$ m
- (3) $\frac{13}{3}$ m
- (4) $\frac{20}{3}$ m

87. Two particles P and Q each of mass $3m$ lie at rest on the X-axis at points $(-a, 0)$ and $(+a, 0)$ respectively. A third particle R of mass $2m$ initially at the origin moves towards particle Q. If all the collisions of the system of 3 particles are elastic and head-on the total number of collisions in the system is

- (1) 2
- (2) 3
- (3) 4
- (4) 5

88. A motor engine pumps 1800 litre of water per minute from a well of depth 30 m and allows to pass through a pipe of cross sectional area 30 cm^2 . Then the power of the engine is

(Acceleration due to gravity $= 10 \text{ ms}^{-2}$)

- (1) 20.5 kW
- (2) 15.5 kW
- (3) 10.5 kW
- (4) 9.5 kW

89. A solid sphere of mass 100 kg and radius 10 m moving in a space becomes a circular disc of radius 20 m in one hour. Then the rate of change of moment of inertia in the process is

(1) $\frac{40}{9} \text{ kg m}^2 \text{ s}^{-1}$

(2) $\frac{10}{9} \text{ kg m}^2 \text{ s}^{-1}$

(3) $\frac{50}{9} \text{ kg m}^2 \text{ s}^{-1}$

(4) $\frac{25}{9} \text{ kg m}^2 \text{ s}^{-1}$

90. A semicircular plate of mass ' m ' has radius ' r ' and centre ' c '. The centre of mass of the plate is at a distance ' x ' from its centre ' c '. Its moment of inertia about an axis passing through its centre of mass and perpendicular to its plane is

(1) $\frac{mr^2}{2}$

(2) $\frac{mr^2}{4}$

(3) $\frac{mr^2}{2} + mx^2$

$$(4) \frac{mr^2}{2} - mx^2$$

91. Two bodies of masses m_1 and m_2 initially at rest at infinite distance apart move towards each other under gravitational force of attraction. Their relative velocity of approach when they are separated by a distance 'r' is

(G-Universal gravitational constant)

$$(1) \left[\frac{2G(m_1 - m_2)}{r} \right]^{1/2}$$

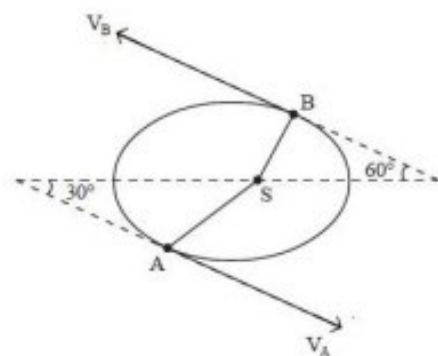
$$(2) \left[\frac{2G(m_1 + m_2)}{r} \right]^{1/2}$$

$$(3) \left[\frac{r}{2G(m_1 m_2)} \right]^{1/2}$$

$$(4) \left[\frac{2G}{r} m_1 m_2 \right]^{1/2}$$

92. A planet is revolving around the sun as shown in the figure. The radius vectors joining the sun and the planet at points A and B are 90×10^6 km and 60×10^6 km respectively. The ratio of

velocities of the planet at A and B when its velocities make 30° and 60° with major axis of the orbit is



(1) $\frac{3}{2\sqrt{3}}$

(2) $\frac{2}{\sqrt{3}}$

(3) $\frac{1}{\sqrt{3}}$

(4) $\frac{\sqrt{3}}{2}$

93. A solid copper cube of 7 cm edge is subjected to a hydraulic pressure of 8000 kPa. The volume contraction of the copper cube is

(Bulk modulus of copper = 140 GPa)

(1) $196 \times 10^{-3} \text{ cm}^3$

(2) $19.6 \times 10^{-6} \text{ cm}^3$

(3) $19.6 \times 10^{-3} \text{ cm}^3$

(4) $196 \times 10^3 \text{ cm}^3$

94. A long cylindrical glass vessel has a pin hole of diameter 0.2 mm at its bottom. The depth to which the vessel can be lowered vertically in a deep water bath without the water entering into the vessel is

(Surface tension of water = 0.07 Nm^{-1} , Acceleration due to gravity = 10 ms^{-2})

(1) 14 cm

(2) 7 cm

(3) 21 cm

(4) 28 cm

95. The focal length of a spherical mirror made of steel is 150 cm, If the temperature of the mirror increases by 200 K, its focal length becomes

(Coefficient of linear expansion of steel = $12 \times 10^{-6} \text{ }^\circ\text{C}^{-1}$)

(1) 186.3 cm

- (2) 153.6 cm
- (3) 150.036 cm
- (4) 150.36 cm

96. A metal rod of length 10 cm and area of cross section $2.8 \times 10^{-4} \text{ m}^2$ is covered with a non-conducting substance, One end of it is maintained at 80°C , while the other end is put in ice at 0°C . It is found that 20 g of ice melts in 5 min. The thermal conductivity of the metal in $\text{Js}^{-1} \text{ m}^{-1} \text{ K}^{-1}$ is

(Latent heat of ice is 80 cal g^{-1})

- (1) 70
- (2) 80
- (3) 90
- (4) 100

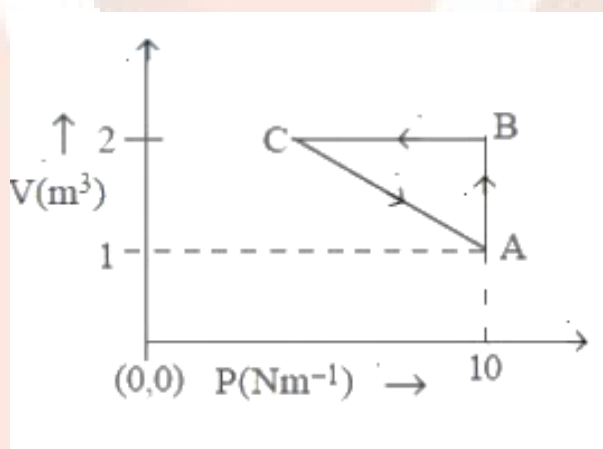
97. A gas expands with temperature according to the relation $V = kT^{2/3}$, where k is a constant. Work done when the temperature changes by 60 K is

(R-Universal gas constant)

- (1) 10 R

- (2) 20 R
- (3) 50 R
- (4) 40 R

98. An ideal gas is taken through the cycle $A \rightarrow B \rightarrow C \rightarrow A$ as shown in the figure. If the net heat supplied to the gas in the cycle is 5 J, the magnitude of work done during the process $C \rightarrow A$ is



- (1) 5 J
 - (2) 10 J
 - (3) 15 J
 - (4) 20 J
99. The average translational kinetic energy of a molecule in a gas becomes equal to 0.69 eV at a temperature about

[Boltzmann constant = $1.38 \times 10^{-23} \text{ JK}^{-1}$]

(1) 3370 °C

(2) 3388 °C

(3) 5333 °C

(4) 5060 °C

100. An earthquake generates both transverse (S) and longitudinal (P) waves in the earth with speeds 4.5 km s^{-1} and 8.0 km s^{-1} respectively. A seismograph records that the first P-wave arrives 3.5 minutes earlier than the first S-wave. From the seismograph, the epicenter of the earthquake is located at a distance

(1) 1080 km

(2) 2468 km

(3) 2160 km

(4) 4320 km

101. An observer moves towards a stationary source of sound with a speed $\frac{1}{5}$ of the speed of sound. The wavelength and frequency of the waves emitted by the source are ' λ ' and ' f ' respectively.

The apparent frequency and wavelength heard by the observer are respectively.

- (1) $1.2 f, \lambda$
- (2) $f, 1.2 \lambda$
- (3) $0.8 f, 0.8 \lambda$
- (4) $1.2 f, 1.2 \lambda$

102. An object is placed 0.1m in front of a convex lens of focal length 20 cm made of a material of refractive index 1.5. The surface of the lens away from the object is silvered. If the radius of curvature of the silvered surface is 22 cm, then the distance of the final image from the silvered surface is

- (1) 10 cm
- (2) 11 cm
- (3) 12 cm
- (4) 13 cm

103. In Young's double slit experiment, if the slit separation is twice the wavelength of light used, then the maximum number of interference maxima is

- (1) 0
- (2) 3
- (3) 5
- (4) 7

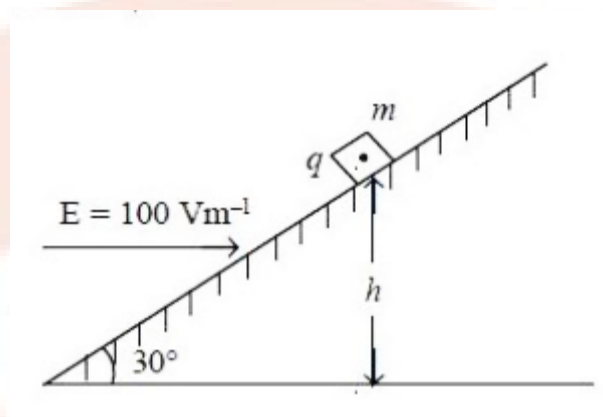
104. Three charges of each magnitude $100 \mu\text{C}$ are placed at the corners A, B and C of an equilateral triangle of side 4 m. If the charges at A and C are positive and the charge at point B is negative, then the magnitude of total force acting on charge at 'C' and angle made by it with $\overline{AC}$ are

- (1) 5.625 N , 60°
- (2) 0.5625 N , 60°
- (3) 5.625 N , 30°
- (4) 0.5625 N , 30°

105. All inclined plane making an angle 30° with the horizontal is placed in a uniform horizontal electric field of 100 Vm^{-1} as shown in figure, A small block of mass 1 kg and charge 0.01 C is allowed to slide down from rest from a height $h = 1 \text{ m}$. If the

coefficient of friction is 0.2, then the acceleration of the block is nearly

(Acceleration due to gravity = 10 ms^{-2})



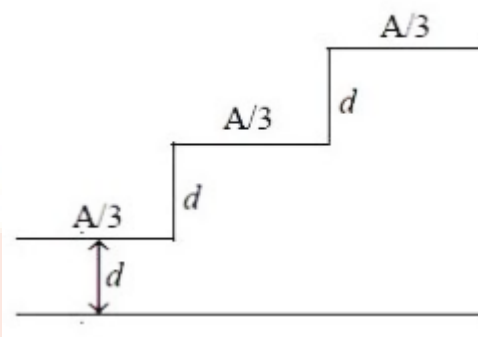
(1) 1.3 ms^{-2}

(2) 2.3 ms^{-2}

(3) 3.3 ms^{-2}

(4) 4.3 ms^{-2}

106. A capacitor is made of a flat plate of area A and a second plate of stair-like structure as shown in the figure. The area each stair is $\frac{A}{3}$ and the height is d . The capacitance of the arrangement is



(1) $\frac{\epsilon_0 A}{3d}$

(2) $\frac{6\epsilon_0 A}{11d}$

(3) $\frac{3\epsilon_0 A}{d}$

(4) $\frac{11\epsilon_0 A}{18d}$

107. The radius of a soap bubble is r and the surface tension of the soap solution is S , The electric potential to which the soap bubble be raised by charging it so that the pressure inside the bubble becomes equal to the pressure outside the bubble is (ϵ_0 - permittivity of free space)

(1) $\sqrt{\frac{Sr}{8\epsilon_0}}$

$$(2) \sqrt{\frac{Sr}{4\epsilon_0}}$$

$$(3) \sqrt{\frac{4Sr}{\epsilon_0}}$$

$$(4) \sqrt{\frac{8Sr}{\epsilon_0}}$$

108. The ratio of heats generated through shunt and galvanometer is 7:5 when they are connected to make an ammeter. If the resistance of the galvanometer is $112\ \Omega$ then the resistance of the shunt is

(1) $80\ \Omega$

(2) $8\ \Omega$

(3) $15.6\ \Omega$

(4) $1.56\ \Omega$

109. When an inductor of inductance $\frac{6}{\pi}$ H, a capacitor of capacitance $\frac{50}{\pi}\ \mu\text{F}$ and resistor of resistance R are connected in

series with an ac supply of rms voltage 220 V and frequency 50 Hz, the rms current through the circuit is 440 mA. Match the inductive reactance (X_L), the capacitive reactance (X_C), the resistance (R) and the impedance (Z) of the circuit given in list-I with the corresponding values given in list-II.

List-I		List-II	
A)	X_L	I)	$200\ \Omega$
B)	X_C	II)	$300\ \Omega$
C)	R	III)	$500\ \Omega$
D)	Z	IV)	$600\ \Omega$

	A	B	C	D
(1)	A IV	B II	C I	D III
(2)	A IV	B III	C I	D II
(3)	A IV	B I	C II	D III
(4)	A I	B IV	C III	D II

110. Assertion (A): When a proton and a neutron enter into a transverse magnetic field with equal speeds, then they trace circular paths of equal radii.

Reason (R): In a transverse magnetic field, the period of revolution of a charged particle in a circular path is directly proportional to the mass of the particle.

- (1) Both (A) and (R) are correct and (R) is the correct explanation of (A)
- (2) Both (A) and (R) are correct but (R) is not the correct explanation of (A)
- (3) (A) is correct but (R) is not correct
- (4) (A) is not correct but (R) is correct

111. If only $\frac{1}{51}^{\text{th}}$ of the main current is to be passed through a galvanometer then the shunt required is R_1 and if only $\frac{1}{11}^{\text{th}}$ of the main voltage is to be developed across the galvanometer, then the resistance required is R_2 . Then $\frac{R_2}{R_1} =$

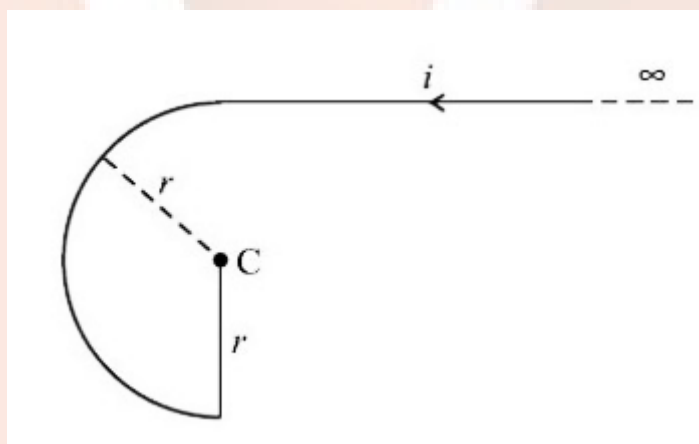
(1) $\frac{1}{500}$

(2) $\frac{50}{9}$

(3) $\frac{500}{3}$

(4) 500

112. The magnetic field at the centre C of the arrangement shown in the figure is



(1) $\frac{\mu_0 i}{2\pi r}(1 + \pi)$

(2) $\frac{\mu_0 i}{4\pi r}(1 + \pi)$

(3) $\frac{\mu_0 i}{\pi r}(1 + \pi)$

$$(4) \frac{\mu_0 i}{r} (1 + \pi)$$

113. To measure a magnetic field between the magnetic poles of a loud speaker, a small coil having 30 turns and 2.5 cm^2 area is placed perpendicular to the field and removed immediately. If the total charge flown through the coil is $7.5 \times 10^{-3} \text{ C}$, and the total resistance of wire and galvanometer is 0.3Ω , then the magnitude of the magnetic field is

- (1) 0.03 T
- (2) 0.3 T
- (3) 3 T
- (4) $3 \times 10^2 \text{ T}$

114. In an oscillating LC circuit, the maximum charge on the capacitor is Q . The charge on the capacitor when the energy is stored equally between the electric and magnetic fields is

- (1) $\frac{Q}{2}$
- (2) $\frac{Q}{\sqrt{3}}$

(3) Q

(4) $\frac{Q}{\sqrt{2}}$

115. An electromagnetic wave of frequency 1×10^{14} Hz is propagating along Z-axis. The amplitude of electric field is 4 Vm^{-1} , then energy density of electric field will be

(Permittivity of free space $= 8.8 \times 10^{-12} \text{ C}^2 \text{N}^{-1} \text{m}^{-2}$)

(1) $35.2 \times 10^{-13} \text{ Jm}^{-3}$

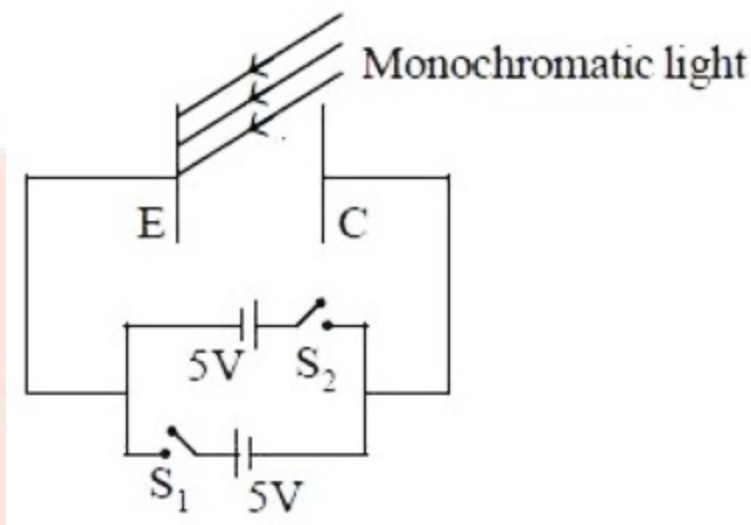
(2) $70.4 \times 10^{-13} \text{ Jm}^{-3}$

(3) $70.4 \times 10^{-12} \text{ Jm}^{-3}$

(4) $35.2 \times 10^{-12} \text{ Jm}^{-3}$

116. In a photoelectric experiment, a monochromatic light is incident on the emitter plate E, as shown in the figure. When switch S_1 is closed and switch S_2 is open, the photoelectrons strike the collector plate C with a maximum kinetic energy of 1 eV. If switch S_1 is open and switch S_2 is closed and the frequency of the incident light is doubled, the photoelectrons

strike the collector plate with a maximum kinetic energy of 20 eV. The threshold wavelength of the emitter plate is



- (1) 5233.3 Å
- (2) 4133.3 Å
- (3) 4166.7 Å
- (4) 5336.7 Å

117. In a system, a particle A of mass m and charge $-2q$ is moving in the nearest orbit around a very heavy particle B having charge $+q$. Assuming Bohr's model of the atom to be applicable to this system, the orbital angular velocity of the particle A is

(1)
$$\frac{2\pi m^2 q^2}{\epsilon_0 h^4}$$

$$(2) \frac{3\pi m^3 q^2}{\epsilon_0^3 h^2}$$

$$(3) \frac{2\pi m q^4}{\epsilon_0^2 h^3}$$

$$(4) \frac{5\pi m^2 q^3}{\epsilon_0^3 h^2}$$

118. In a nuclear reactor the activity of a radioactive substance is 2000/s. If the mean life of the products is 50 minutes, then in the steady power generation, the number of radio nuclides is

(1) 12×10^5

(2) 60×10^5

(3) 90×10^5

(4) 15×10^5

119. In a p-type semiconductor the donor level is at 50 meV above the valence band. To produce one electron, the maximum wave length of light photon required is

(Planck's constant = 6.6×10^{-34} J-s and speed of light in vacuum = 3×10^8 m/s)

- (1) $0.0248 \mu\text{m}$
- (2) $0.248 \mu\text{m}$
- (3) $2.48 \mu\text{m}$
- (4) $24.8 \mu\text{m}$

120. A signal of frequency 10 kHz and peak voltage 10 V is used to amplitude modulate a carrier of frequency 1 MHz and peak voltage 20 V . The side band frequencies in kHz are

- (1) $1010, 990$
- (2) $910, 1090$
- (3) $10, 11$
- (4) $1.01, 0.99$

AP EAMCET-2019-Engineering-20 Apr-Shift-1
Chemistry

121. The wavelength of a microscope particle of mass 9.1×10^{-31} kg is 182 nm, its kinetic energy in J is

$(h = 6.625 \times 10^{-34} \text{ J s})$

(1) 7.28×10^{-23}

(2) 7.28×10^{-24}

(3) 3.64×10^{-23}

(4) 3.64×10^{-24}

122. The energy of an electron in an orbit of hydrogen like ion with an orbit radius of 52.9 pm in J is (ground state energy of electron in hydrogen atom is -2.18×10^{-18} J)

(1) -4.36×10^{-18}

(2) -1.09×10^{-17}

(3) -8.72×10^{-18}

(4) -6.54×10^{-18}

123. In which of the following the electron gain enthalpy of elements is correctly arranged?

- (1) $S > Se > Te > O$
- (2) $F > Cl > Br > I$
- (3) $Na > Li > K > Pb$
- (4) $O > S > Se > Te$

124. Which of the following orders are correct against the property given?

- I) dipole moment $NF_3 > NH_3 > BF_3$
- II) Covalent bond length $C-O > N-O > O-H$
- III) Bond order $C_2 > B_2 > He_2$

- (1) I, II only
- (2) II, III only
- (3) I, III only
- (4) I, II, III

125. The molecule/ion having diamagnetic nature and a bond order of 1.0 is

- (1) He_2^+
- (2) Li_2^+

(3) B_2

(4) C_2

126. If the kinetic energy of O_2 gas is 4.0 kJmol^{-1} , its RMS speed in cm s^{-1} is

(1) 5.0×10^2

(2) 5.0×10^3

(3) 5.0×10^4

(4) 5.0×10^{-4}

127. The volume of 0.02 M acidified permanganate solution required for complete reaction of 60 mL of 0.01 M I^- ion solution to form I_2 in mL is

(1) 60

(2) 20

(3) 40

(4) 6

128. 6 g of graphite is burnt in a bomb calorimeter at 25°C and 1 atm pressure. The temperature of water increased from 25°C to

31 °C . If ΔH of this reaction is -248 kJ mol^{-1} . Find out

C_v (in kJ K^{-1}) of bomb calorimeter.

- (1) 20.667
- (2) 41.33
- (3) 1488
- (4) 0.145

129. In which of the following, the solubility of AgCl will be minimum?

- (1) 0.1 M KNO_3
- (2) 0.1 M KCl
- (3) 0.2 M KNO_3
- (4) Water

130. The number of species of the following that can act both as Bronsted acids and bases is

HCl , ClO_4^- , OH^- , H^+ , H_2O , HSO_4^- , SO_4^{2-} , H_2SO_4 , Cl^-

- (1) 4
- (2) 3
- (3) 1

(4) 2

131. Which one of the following properties has same value for H_2 and D_2 ?

- (1) density
- (2) enthalpy of bond dissociation
- (3) bond length
- (4) melting point

132. Identify the correct statements from the following.

- I) Tendency to form halide hydrates gradually increases from Be to Ba down the group.
- II) Tendency to form stable super oxides increases from Li to Cs down the group.
- III) Low solubility of LiF is due its high lattice energy.
- IV) Solubility of carbonates of group-2 elements increases down the group.

- (1) I, II
- (2) III, IV
- (3) II, III

(4) I, III

133. Which of the following metals exist in liquid state during summer season?

(1) Ga

(2) Al

(3) Pb

(4) Sn

134. Formic acid is heated with conc. H_2SO_4 at 100°C to form A and B. When Fe_2O_3 is heated strongly with B, the products formed are C and D. C can also be obtained by reacting CaCO_3 with dil HCl. What is D?

(1) Fe

(2) CO

(3) CO_2

(4) Fe_3O_4

135. The non-biodegradable waste formed in fertilizer industries is

(1) fly-ash

- (2) Carbon monoxide
- (3) gypsum
- (4) lead

136. Which one of the following has maximum number of hybrid orbitals?

- (1) C_6H_6
- (2) $(CH_3)_4C$
- (3) $(CH_3)_2C=O$
- (4) $CH_3-CH=CH-CN$

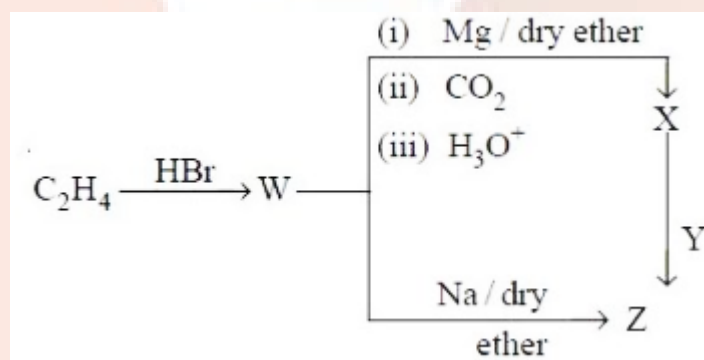
137. In Dumas method one gram of carbon compound gives 50 mL of N_2 at 300 K and 740 mm Hg pressure. If the aqueous tension at 300 K is 15 mm Hg, what is the percentage of nitrogen in it?

- (1) 5.42
- (2) 10.84
- (3) 21.68
- (4) 2.71

138. Sodium acetate was electrolysed by Kolbe's method to form two gases A and B at anode. C and D are formed when B is heated with regulated supply of O_2 or air in the presence of $(CH_3COO)_2Mn$. C reacts with NaOH to form a salt. A and D are respectively.

- (1) CO_2 , CH_3COOH
- (2) CO_2 , H_2O
- (3) C_2H_6 , H_2O
- (4) CO_2 , H_2O_2

139. What are Y and Z in the following reaction sequence?



Y

Z

- | | |
|-----------------------|--------------------|
| (1) NaOH/CaO | $CH_3(CH_2)_2CH_3$ |
| (2) NaOH/electrolysis | H_3C-CH_3 |
| (3) NaOH/CaO | H_3C-CH_3 |

(4) NaOH/electrolysis $\text{CH}_3(\text{CH}_2)_2\text{CH}_3$

140. At $T(\text{K})$ copper (atomic mass = 63.5 U) has fcc unit cell structure with edge length of $x \text{ \AA}$. What is the approximate density of Cu in g cm^{-3} at that temperature?

$(N_A = 6.0 \times 10^{23} \text{ mol}^{-1})$

(1) $\frac{42.3}{x^3}$

(2) $\frac{4.23}{x^3}$

(3) $\frac{423}{x^3}$

(4) $\frac{212}{x^3}$

141. The number of moles of solute present in the solution of I, II and III is respectively.

I) 500 mL of 0.2 M NaOH

II) 200 mL of 0.1 N H_2SO_4

III) 6 g of urea in 1 kg of water

(1) -0.1, 0.01, 0.1

(2) 0.1, 0.02, 0.1

(3) -0.2, 0.01, 0.1

(4) 0.1, 0.01, 0.2

142. 6 g of a mixture of naphthalene ($C_{10}H_8$) and anthracene ($C_{14}H_{10}$) is dissolved in 300 g of benzene. If the depression in freezing point is 0.70 K, the composition of naphthalene and anthracene in the mixture respectively in g are (molal depression constant of benzene is $5.1 \text{ K kg mol}^{-1}$)

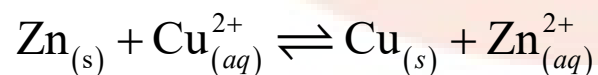
(1) 2.60, 3.40

(2) 3.40, 2.60

(3) 2.90, 3.10

(4) 3.10, 2.90

143. Under which of the following conditions E value of the cell for the cell reaction given is maximum?



$$\left(\frac{2.303 RT}{F} \text{ at } 298 \text{ K} = 0.059 \text{ V}, E_{\text{Zn}^{2+}|\text{Zn}}^0 = -0.76 \text{ V}, E_{\text{Cu}^{2+}|\text{Cu}}^0 = +0.34 \text{ V} \right)$$

(1) $C_1 = 0.1 \text{ M}, C_2 = 0.01 \text{ M}$

(2) $C_1 = 0.01 \text{ M}, C_2 = 0.1 \text{ M}$

(3) $C_1 = 0.1 \text{ M}, C_2 = 0.2 \text{ M}$

(4) $C_1 = 0.2 \text{ M}, C_2 = 0.1 \text{ M}$

144. In the first thermal decomposition of

$\text{C}_2\text{H}_5\text{I}_{(g)} \rightarrow \text{C}_2\text{H}_{4(g)} + \text{HI}_{(g)}$ the reactant in the beginning exerts a pressure of 2 bar in a closed vessel at 600 K. If the partial pressure of the reactant is 0.1 bar after 1000 minutes at the same temperature the rate constant in min^{-1} is

(log 2=0.30)

(1) 6.0×10^{-4}

(2) 6.0×10^{-3}

(3) 3.0×10^{-3}

(4) 3.0×10^{-4}

145. Identify the correct statements from the following.

I) Sulphur sol is an example of multimolecular colloid.

II) Tyndall effect is observed when the diameter of the dispersed particles is not much smaller than the wavelength of the light used.

III) The process of removing a dissolved substance from a colloidal solution by means of diffusion through a suitable membrane is called peptization.

IV) Eosin, gelatin are examples of negatively charged sols.

(1) I, II, III

(2) I, II, IV

(3) I, III, IV

(4) II, III, IV

146. Which of the following are carbonate ores?

I) Magnetite

II) Kaolinite

III) Siderite

IV) Calamine

(1) I, II, III

(2) II, III, IV

(3) I, II only

(4) III, IV only

147. Which of the following statements is not correct?

- (1) From SO_2 to TeO_2 reducing power decreases
- (2) The order of boiling points of hydrides of 16th group elements is $\text{H}_2\text{S} < \text{H}_2\text{Se} < \text{H}_2\text{Te} < \text{H}_2\text{O}$
- (3) Rhombic sulphur has S_8 molecules while monoclinic sulphur has S_6 molecules.
- (4) The bond angle in Ozone molecule is 117° .

148. Noble metals like gold and platinum are soluble in which of the following mixtures?

- (1) 1:1 mixture of conc. HNO_3 and conc. H_2SO_4
- (2) 1:3 mixture of conc. HCl and conc. HNO_3
- (3) 1:3 mixture of conc. HNO_3 and conc. HCl
- (4) 1:3 mixture of conc. H_2SO_4 and conc. HCl

149. Identify the set of acidic oxides.

- (1) Na_2O , CaO , BaO
- (2) ZnO , PbO , BeO
- (3) CO , NO , N_2O

(4) Mn_2O_7 , CrO_3 , V_2O_5

150. The wavelength of the light absorbed by the complexes

$[\text{Ni}(\text{H}_2\text{O})_6]^{2+}$, $[\text{Ni}(\text{en})_3]^{2+}$, $[\text{Ni}(\text{H}_2\text{O})_4\text{en}]^{2+}$ are λ_1 , λ_2 , λ_3

respectively. The correct order of wavelengths is

(1) $\lambda_1 > \lambda_2 > \lambda_3$

(2) $\lambda_3 > \lambda_2 > \lambda_1$

(3) $\lambda_1 > \lambda_3 > \lambda_2$

(4) $\lambda_2 > \lambda_3 > \lambda_1$

151. KMnO_4 oxidizes $\text{S}_2\text{O}_3^{2-}$ to SO_4^{2-} in medium x and NO_2^- to NO_3^- in medium y . x and y are respectively.

(1) acidic, basic

(2) acidic, acidic

(3) acidic, neutral

(4) neutral, acidic

152. Match the following

List-I

List-II

- | | |
|----------------------------|----------------------------|
| A) Teflon | I) SnCl_2 |
| B) Anionic polymerisation | II) C_2F_4 |
| C) Cationic polymerisation | III) Bakelite |
| D) Thermosetting polymer | IV) Polystyrene |
| | V) RLi |

	<u>A</u>	<u>B</u>	<u>C</u>	<u>D</u>
(1)	II	I	V	III

	<u>A</u>	<u>B</u>	<u>C</u>	<u>D</u>
(2)	II	V	I	IV

	<u>A</u>	<u>B</u>	<u>C</u>	<u>D</u>
(3)	II	V	I	III

	<u>A</u>	<u>B</u>	<u>C</u>	<u>D</u>
(4)	V	II	I	IV

153. Identify the correct set of monosaccharides present in sucrose (X), lactose (Y) and maltose (Z).

<i>X</i>	<i>Y</i>	<i>Z</i>
(1) glucose, fructose	galactose, glucose	glucose, fructose
(2) glucose, fructose	galactose, glucose	glucose, glucose
(3) glucose, glucose	galactose, glucose	glucose, glucose

(4) glucose, glucose glucose, fructose glucose, glucose

154. Which of the following are broad spectrum antibiotics?

Penicillin G	Chloramphenicol	Ofloxacin	Ampicillin
I	II	III	IV

- (1) I, II only
- (2) I, II, III
- (3) II, III, IV
- (4) I, III only

155. Arrange the following organic halides in correct order of reactivity towards S_N2 displacement.

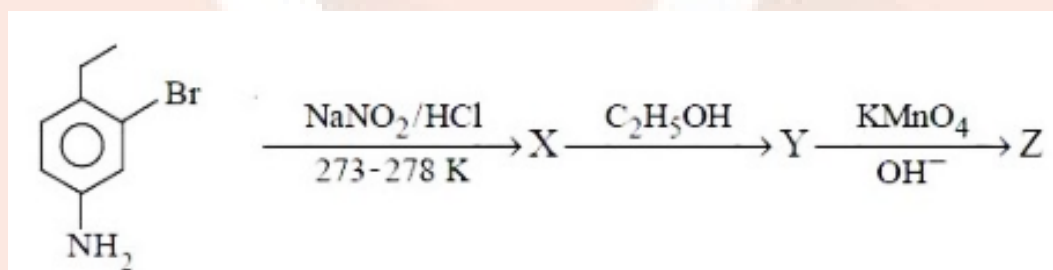
$(CH_3)_2C(Br)CH_2CH_3$	$BrCH_2(CH_2)_3CH_3$	$CH_3CH(Br)CH_2CH_3$
(P)	(Q)	(R)

- (1) $P > Q > R$
- (2) $R > P > Q$
- (3) $P > R > Q$
- (4) $Q > R > P$

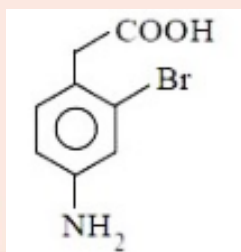
156. The bond angle between C-O and O-H bonds in alcohols is close to

- (1) 109°
- (2) 120°
- (3) 180°
- (4) 90°

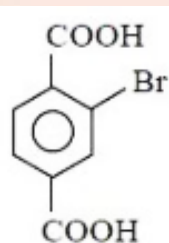
157. Identify Z in the following sequence of reactions.

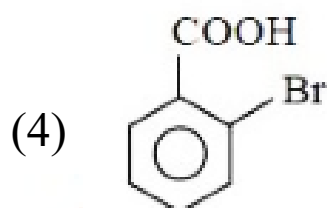
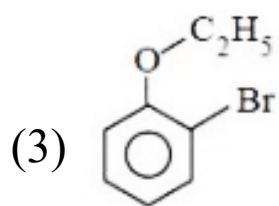


(1)

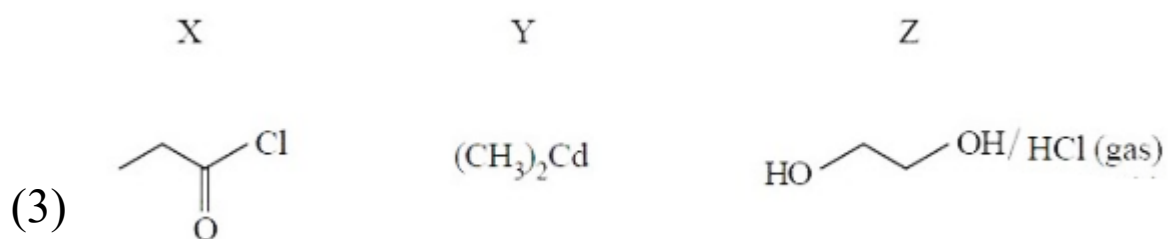
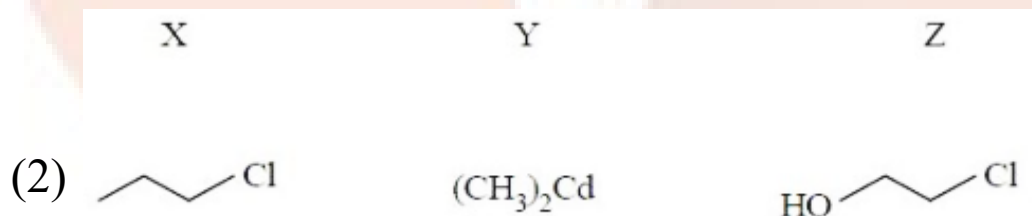
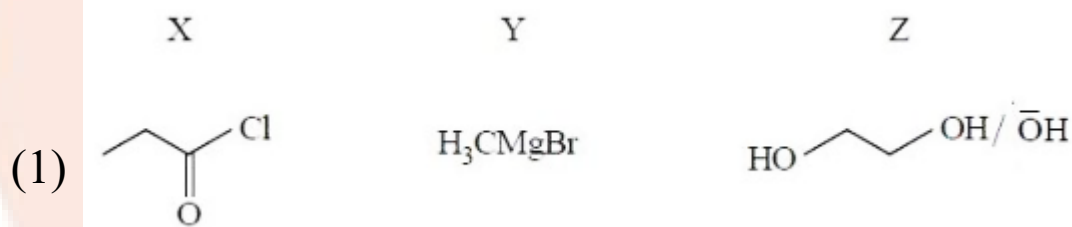
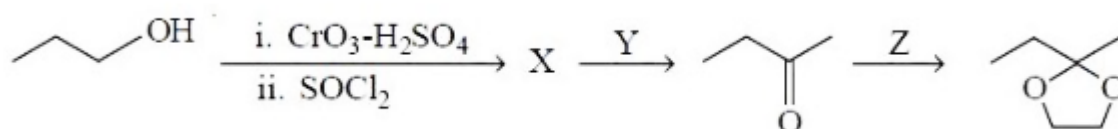


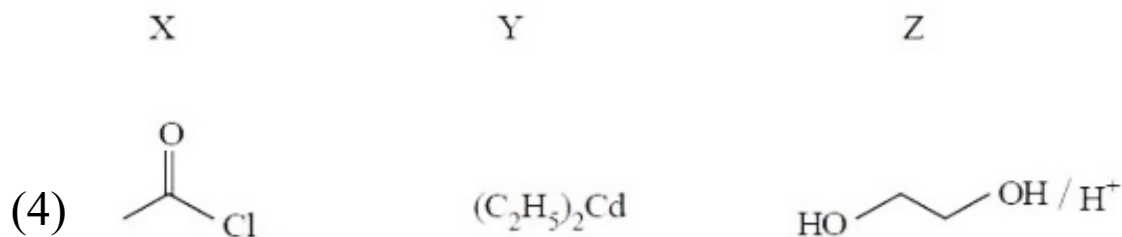
(2)



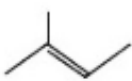
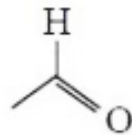
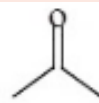


158. Identify X,Y,Z in the following reaction sequence.





159. 2-Methyl 2-butane on hydration gave an alcohol X. Isomer of X could be prepared from which of the following?

- (1)  + $\text{HBr}, \text{H}_2\text{O}$
- (2)  +  $\text{MgBr}, \text{H}_2\text{O} / \text{H}^+$
- (3)  +  $\text{MgBr}, \text{H}_2\text{O} / \text{H}^+$
- (4)  +  $\text{MgBr}, \text{H}_2\text{O} / \text{H}^+$

160. Acetic acid on heating with NH_3 forms A. When A reacts with LiAlH_4 followed by hydrolysis gives B. When B is heated with chloroform in KOH medium gives C. What are B and C respectively?

- (1) $\text{CH}_3\text{CONH}_2, \text{CH}_3\text{CH}_2\text{NC}$

(2) $\text{CH}_3\text{CH}_2\text{NH}_2$, $\text{CH}_3\text{CH}_2\text{NC}$

(3) $\text{CH}_3\text{CH}_2\text{NH}_2$, CH_3COOH

(4) $\text{CH}_3\text{CH}_2\text{CH}_2\text{NH}_2$, $\text{CH}_3\text{CH}_2\text{NC}$



AP EAMCET-2019-Engineering-20 Apr-Shift-1
Answer Key

1	4	33	2	65	4	97	4	129	2
2	3	34	2	66	3	98	1	130	4
3	1	35	3	67	4	99	4	131	3
4	4	36	4	68	1	100	3	132	3
5	3	37	2	69	4	101	1	133	1
6	2	38	3	70	3	102	2	134	1
7	1	39	1	71	4	103	3	135	3
8	4	40	3	72	1	104	1	136	2
9	2	41	3	73	3	105	2	137	1
10	4	42	4	74	2	106	4	138	3
11	4	43	2	75	4	107	4	139	4
12	1	44	3	76	1	108	1	140	3
13	1	45	1	77	4	109	3	141	1
14	4	46	1	78	1	110	4	142	2
15	4	47	2	79	4	111	4	143	1
16	1	48	1	80	1	112	2	144	3
17	3	49	1	81	3	113	2	145	2

18	2	50	4	82	3	114	4	146	4
19	3	51	2	83	3	115	3	147	3
20	2	52	4	84	3	116	2	148	3
21	1	53	1	85	1	117	3	149	4
22	1	54	2	86	4	118	2	150	3
23	2	55	4	87	1	119	1	151	4
24	1	56	3	88	3	120	1	152	3
25	4	57	3	89	1	121	2	153	2
26	3	58	1	90	4	122	3	154	3
27	3	59	1	91	2	123	1	155	4
28	2	60	1	92	2	124	2	156	1
29	4	61	1	93	3	125	*	157	4
30	3	62	4	94	1	126	3	158	3
31	2	63	4	95	4	127	4	159	3
32	2	64	4	96	4	128	1	160	2

AP EAMCET-2019-Engineering-20 Apr-Shift-1
Solutions – Mathematics

1. Consider the function,

$$f(x) = \frac{x}{e^x - 1} + \frac{x}{2} + 2\cos^3 \frac{x}{2}$$

Then,

$$\begin{aligned} f(-x) &= \frac{-x}{e^{-x} - 1} - \frac{x}{2} + 2\cos^3 \left(-\frac{x}{2} \right) \\ &= \frac{-xe^x}{1 - e^x} - \frac{x}{2} + 2\cos^3 \frac{x}{2} \\ &= \frac{xe^x}{e^x - 1} - \frac{x}{2} + 2\cos^3 \frac{x}{2} \\ &= \frac{x(e^x - 1 + 1)}{e^x - 1} - \frac{x}{2} + 2\cos^3 \frac{x}{2} \end{aligned}$$

Further simplify the above,

$$\begin{aligned} f(-x) &= x + \frac{x}{e^x - 1} - \frac{x}{2} + 2\cos^3 \frac{x}{2} \\ &= \frac{x}{e^x - 1} + \frac{x}{2} + 2\cos^3 \frac{x}{2} \\ &= f(x), \forall x \in R - \{0\} \end{aligned}$$

This implies $f(x)$ is an even function.

2. For A, Consider the function,

$$f(x) = \frac{|x+2|}{x+2}, x \neq -2$$

Then,

$$\begin{aligned} f(x) &= \begin{cases} \frac{x+2}{x+2}, & x > -2 \\ -\frac{x+2}{x+2}, & x < -2 \end{cases} \\ &= \begin{cases} 1, & x > -2 \\ -1, & x < -2 \end{cases} \end{aligned}$$

This implies, the range of $f(x)$ is $\{-1, 1\}$.

For B,

$$g(x) = \lfloor x \rfloor, x \in \mathbb{R}$$

Since $\lfloor x \rfloor \in I$ so $\lfloor x \rfloor \in W$

This implies the range of $g(x)$ is W .

For C,

$$h(x) = |x - \lfloor x \rfloor|, x \in \mathbb{R}$$

$$= \{x\} \in [0, 1)$$

This implies the range is $[0, 1)$.

For D,

$$f(x) = \frac{1}{2 - \sin 3x}, x \in \mathbb{R}$$

Then,

$$-1 \leq \sin 3x \leq 1$$

$$2 - 1 \leq 2 - \sin 3x \leq 2 + 1$$

$$\frac{1}{3} \leq \frac{1}{2 - \sin 2x} \leq \frac{1}{1}$$

This implies the range of $f(x)$ is $\left[\frac{1}{3}, 1\right]$.

3. Consider the expression,

$$(1) + (1 + 2 + 4) + (4 + 6 + 9) + (9 + 12 + 16) + \dots + (81 + 90 + 100)$$

Then,

$$\begin{aligned} & (1) + \left(\begin{matrix} 1^2 \\ + (2 \times 1) + 2^2 \end{matrix} \right) + \left(\begin{matrix} 2^2 \\ + (2 \times 3) + 3^2 \end{matrix} \right) + \left(\begin{matrix} 3^2 + \\ (3 \times 4) + 4^2 \end{matrix} \right) + \dots \\ & + \left(\begin{matrix} 9^2 + \\ (9 \times 10) + 10^2 \end{matrix} \right) \end{aligned}$$

This implies,

$$\begin{aligned}
 \sum_{r=1}^{10} \left[(r-1)^2 + r(r-1) + r^2 \right] &= \sum_{r=1}^{10} \left[r - (r-1)(r-1)^2 + r(r-1) + r^2 \right] \\
 &= \sum_{r=1}^{10} r^3 - (r-1)^3 \\
 &= (1^3 - 0^3) + (2^3 - 1^3) + (3^3 - 2^3) + \dots + \\
 &\quad (10^3 - 9^3) \\
 &= 10^3 - 0^3
 \end{aligned}$$

Further simplify the above,

$$\sum_{r=1}^{10} \left[(r-1)^2 + r(r-1) + r^2 \right] = 1000$$

This implies, both (A) and (R) are true and (R) is the correct explanation of (A).

4. Consider the matrix,

$$A = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$

Since above matrix satisfies $AA^T = I$ so, it is an orthogonal matrix then,

$$\begin{aligned}A^T X^{50} A &= A^T X^{49} (A P A^T) A \\&= A^T X^{49} A P (A^T A) \\&= A^T X^{49} A P \\&= A^T X^{48} A P^2 \dots\end{aligned}$$

Further simplify the above,

$$\begin{aligned}A^T X^{50} A &= A^T A P^{50} \\&= I P^{50} \\&= P^{50}\end{aligned}$$

This implies,

$$\begin{aligned}P &= \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \\P^2 &= \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix} \\P^3 &= \begin{bmatrix} 1 & 3 \\ 0 & 1 \end{bmatrix} \\\vdots & \\P^{50} &= \begin{bmatrix} 1 & 50 \\ 0 & 1 \end{bmatrix}\end{aligned}$$

Therefore, the correct option is (4).

5. Consider the system of equations,

$$2x + 3|y| + 5[z] = 0$$

$$x + |y| - 2[z] = 4$$

$$x + |y| + [z] = 1$$

Then by Cramer's rule,

$$x = \frac{\Delta_1}{\Delta}$$

$$|y| = \frac{\Delta_2}{\Delta}$$

$$[z] = \frac{\Delta_3}{\Delta}$$

Where,

$$\begin{aligned}\Delta &= \begin{vmatrix} 2 & 3 & 5 \\ 1 & 1 & -2 \\ 1 & 1 & 1 \end{vmatrix} \\ &= 2(1+2) - 3(1+2) + 5(1-1) \\ &= -3\end{aligned}$$

and,

$$\begin{aligned}\Delta_1 &= \begin{vmatrix} 0 & 3 & 5 \\ 4 & 1 & -2 \\ 1 & 1 & 1 \end{vmatrix} \\ &= 0(1+2) - 3(4+2) + 5(4-1) \\ &= -3\end{aligned}$$

and,

$$\begin{aligned}\Delta_2 &= \begin{vmatrix} 2 & 0 & 5 \\ 1 & 4 & -2 \\ 1 & 1 & 1 \end{vmatrix} \\ &= 2(4+2) - 0(1+2) + 5(1-4) \\ &= -3\end{aligned}$$

and,

$$\begin{aligned}\Delta_3 &= \begin{vmatrix} 2 & 3 & 0 \\ 1 & 1 & 4 \\ 1 & 1 & 1 \end{vmatrix} \\ &= 2(1-4) - 3(1-4) + 0(1-1) \\ &= 3\end{aligned}$$

This implies,

$$\begin{aligned}x &= \frac{-3}{-3} \\ &= 1\end{aligned}$$

And,

$$\begin{aligned}|y| &= \frac{-3}{-3} \\ &= 1\end{aligned}$$

And,

$$\begin{aligned}[z] &= \frac{-3}{3} \\ &= -1\end{aligned}$$

This implies,

$$z \in [-1, 0)$$

So, the system has infinitely many solutions.

6. Consider the system of equations

$$x + 2y + 3z = 6$$

$$x + 3y + 5z = 9$$

$$2x + 5y + \lambda z = \mu$$

Now from the Cramer's rule,

$$\begin{aligned}\Delta &= \begin{vmatrix} 1 & 2 & 3 \\ 1 & 3 & 5 \\ 2 & 5 & \lambda \end{vmatrix} \\ &= 1(3\lambda - 25) - 2(\lambda - 10) + 3(5 - 6) \\ &= \lambda - 8\end{aligned}$$

And,

$$\begin{aligned}\Delta_1 &= \begin{vmatrix} 6 & 2 & 3 \\ 9 & 3 & 5 \\ \mu & 5 & \lambda \end{vmatrix} \\ &= 6(3\lambda - 25) - 2(9\lambda - 5\mu) + 3(45 - 3\mu) \\ &= \mu - 15\end{aligned}$$

And,

$$\begin{aligned}\Delta_2 &= \begin{vmatrix} 1 & 6 & 3 \\ 1 & 9 & 5 \\ 2 & \mu & \lambda \end{vmatrix} \\ &= 1(9\lambda - 5\mu) - 6(\lambda - 10) + 3(\mu - 18) \\ &= 3\lambda - 2\mu + 6\end{aligned}$$

And,

$$\begin{aligned}\Delta_3 &= \begin{vmatrix} 1 & 2 & 6 \\ 1 & 3 & 9 \\ 2 & 5 & \mu \end{vmatrix} \\ &= 1(3\mu - 45) - 2(\mu - 18) + 6(5 - 6) \\ &= \mu - 15\end{aligned}$$

Now, if $\lambda = 8$ and $\mu \neq 15$, then system of linear equation has no solution.

If $\lambda \neq 8$ and $\mu \in \mathbb{R}$, then system of linear equation has a unique solution.

If $\lambda = 8$ and $\mu = 15$, then system of linear equation has infinite number of solutions. So, $\Delta_2 = 3\lambda - 2\mu + 6$ is also zero therefore, the correct option is (2).

7. Consider the expression,

$$z = x + iy, x, y \in R, (x, y) \neq (0, -4)$$

This implies,

$$\begin{aligned} \frac{2z-3}{z+4i} &= \frac{(2x-3)+2iy}{x+i(y+4)} \times \frac{x-i(y+4)}{x-i(y+4)} \\ &= \frac{(2x^2-3x+2y^2+8y)+i(2xy-2xy+3y-8x+12)}{x^2+(y+4)^2} \\ &= \frac{(2x^2-3x+2y^2+8y)+i(12+3y-8x)}{x^2+(y+4)^2} \end{aligned}$$

Then,

$$\begin{aligned} \text{Arg}\left(\frac{2z-3}{z+4i}\right) &= \text{Tan}^{-1}\left(\frac{12+3y-8x}{2x^2-3x+2y^2+8y}\right) \\ &= \frac{\pi}{4} \end{aligned}$$

This implies,

$$\frac{12 + 3y - 8x}{2x^2 - 3x + 2y^2 + 8y} = 1$$

$$2x^2 + 2y^2 + 5x + 5y - 12 = 0$$

8. The given expression is,

$$z = x + iy$$

Then,

$$\begin{aligned} \frac{\bar{z} - 1}{z - i} &= \frac{x - iy - 1}{x - iy - 1} \times \frac{x + i(y + 1)}{x + i(y + 1)} \\ &= \frac{[x(x - 1) + y(y + 1)] + i[(y + 1)(x - 1) - xy]}{x^2 + (y + 1)^2} \\ \operatorname{Im}\left(\frac{\bar{z} - 1}{z - i}\right) &= \frac{xy - y + x - 1 - xy}{x^2 + (y + 1)^2} \\ &= 1 \end{aligned}$$

This implies,

$$x^2 + y^2 - x + 3y + 2 = 0, (x, y) \neq (0, -1).$$

9. Consider the expression,

$$1 + \omega + \omega^2 = 0$$

And,

$$\omega^3 = 1$$

Then,

$$\begin{aligned}\left(r + \frac{1}{\omega}\right)\left(r + \frac{1}{\omega^2}\right) &= r^2 + r\left(\frac{1}{\omega} + \frac{1}{\omega^2}\right) + \frac{1}{\omega^3} \\ &= r^2 + \left(\frac{\omega^2 + \omega}{\omega^3}\right)r + \frac{1}{\omega^3} \\ &= r^2 - r + 1\end{aligned}$$

Then,

$$\begin{aligned}\sum_{r=1}^n \left(r + \frac{1}{\omega}\right)\left(r + \frac{1}{\omega^2}\right) &= \sum_{r=1}^n (r^2 - r + 1) \\ &= \frac{n(n+1)(2n+1)}{6} - \frac{n(n+1)}{2} + n \\ &= \frac{n(n+1)}{6} [2n+1-3] + n \\ &= \frac{n(n+1)}{6} (2n-2) + n\end{aligned}$$

Further simplify the above,

$$\begin{aligned}\sum_{r=1}^n \left(r + \frac{1}{\omega}\right)\left(r + \frac{1}{\omega^2}\right) &= \frac{n}{3} (n^2 - 1) + n \\ &= \frac{n}{3} [n^2 - 1 + 3] \\ &= \frac{n(n^2 + 2)}{3}\end{aligned}$$

10. Since ω is the complex cube root of unity then,

$$\begin{aligned} r(r+1-\omega)(r+1-\omega^2) &= r[(r+1)^2 - (\omega + \omega^2)(r+1) + \omega^3] \\ &= r[(r+1)^2 + (r+1) + 1] \\ &= r(r^2 + 3r + 3) \\ &= r^3 + 3r^2 + 3r \end{aligned}$$

This implies,

$$\begin{aligned} \sum_{r=1}^9 r(r+1-\omega)(r+1-\omega^2) &= \sum_{r=1}^9 (r^3 + 3r^2 + 3r) \\ &= \left(\frac{9(10)}{2}\right)^2 + 3\left(\frac{9(10)(19)}{6}\right) + 3\left(\frac{9 \times 10}{2}\right) \\ &= 45^2 + (45)(19) + (3)(45) \\ &= 3015 \end{aligned}$$

11. Let consider,

$$y = \frac{2\alpha}{3-4\alpha}$$

Then,

$$2\alpha = 3y - 4\alpha y$$

$$\alpha = \frac{3y}{2+4y}$$

Here, α is the root of equation $x^2 + 7x + 3 = 0$

So,

$$\left(\frac{3y}{2+4y}\right)^2 + 7\left(\frac{3y}{2+4y}\right) + 3 = 0$$

$$9y^2 + 84y^2 + 42y + 48y^2 + 48y + 12 = 0$$

$$141y^2 + 90y + 12 = 0$$

$$47y^2 + 30y + 4 = 0$$

So,

$$ax^2 + bx + c = 0$$

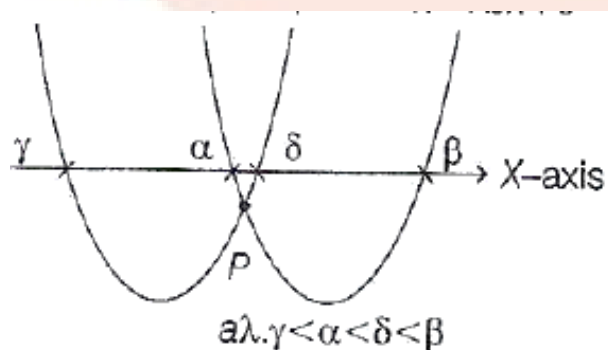
Comparison gives,

$$a = 47, b = 30 \text{ and } c = 4$$

The GCD of 47, 30, 4 is 1. So,

$$\begin{aligned} a + b + c &= 47 + 30 + 4 \\ &= 81 \end{aligned}$$

12. Consider the diagram,



For x coordinate of P on subtraction of given quadratic equations,

$$x^2 + b_1x + c_1 = 0$$

$$x^2 + bx + c = 0$$

Then,

$$(b_1 - b)x + (c_1 - c) = 0$$

$$x = \frac{c - c_1}{b_1 - b}$$

This implies,

$$f(x) = x^2 + bx + c$$

$$f\left(\frac{c - c_1}{b_1 - b}\right) < 0$$

$$\left(\frac{c - c_1}{b_1 - b}\right)^2 + b\left(\frac{c - c_1}{b_1 - b}\right) + c < 0$$

$$(c - c_1)^2 < b(c - c_1)(b_1 - b) - c(b_1 - b)^2$$

Then,

$$(c - c_1)^2 < (b_1 - b)[bc_1 - bc - cb_1 + cb]$$

$$(c - c_1)^2 < (b_1 - b)(bc_1 - cb_1)$$

13. The roots of the given equation are real so,

$$D \geq 0$$

This implies,

$$4(a+b+c)^2 - 4 \times 3\lambda(ab+bc+ca) \geq 0$$

$$(a+b+c)^2 - 3\lambda(ab+bc+ca) \geq 0$$

$$\lambda \leq \frac{(a+b+c)^2}{3(ab+bc+ca)}$$

For scalene triangle,

$$|b-c| < a, |c-a| < b, |a-b| < c$$

So,

$$(b-c)^2 + (c-a)^2 + (a-b)^2 < a^2 + b^2 + c^2$$

$$a^2 + b^2 + c^2 < 2(ab+bc+ca)$$

$$a^2 + b^2 + c^2 + 2(ab+bc+ca) < a^2 + b^2 + c^2 + 4(ab+bc+ca)$$

$$\frac{(a+b+c)^2}{3(ab+bc+ca)} < \frac{4}{3}$$

This implies,

$$\lambda < \frac{4}{3}$$

14. The quadratic equation that has root $2 \pm \sqrt{3}$ is,

$$x^2 + 4x - 1$$

And the equation with root $1 + 2i$ is,

$$x^2 - 2x + 5 = 0$$

So, the required polynomial equation is,

$$(x^2 - 4x + 1)(x^2 - 2x + 5) = 0$$

$$x^4 - 6x^3 + 14x^2 - 22x + 5 = 0$$

15. When the word ANIMAL is arranged in alphabetical order, we get AAILMN, then rank of word ANIMAL is,

$$AA.....4!$$

$$AI.....4!$$

$$AL.....4!$$

$$AM.....4!$$

$$ANA.....3!$$

$$ANIA.....2!$$

$$ANIL.....2!$$

$$ANIMAL.....1$$

Then rank is $x = 107$.

Similarly for word PERSON the alphabetical arrangement is ESORNP then rank of PERSON is,

$EN.....4!$

$EO.....4!$

$EP.....4!$

$ER.....4!$

$ESN.....3!$

$ESON.....2!$

$ESOP.....2!$

$ESORNP.....1$

Then rank is $x = 107$.

16. Consider the expression,

$${}^{2n+1}C_1 + {}^{2n+1}C_2 + {}^{2n+1}C_3 + \dots + {}^{2n+1}C_n = x$$

Then,

$${}^{2n+1}C_0 + {}^{2n+1}C_1 + {}^{2n+1}C_2 + \dots + {}^{2n+1}C_n + \dots + {}^{2n+1}C_{2n+1} = 2^{2n+1}$$

$$2x + {}^{2n+1}C_0 + {}^{2n+1}C_{2n+1} = 2^{2n+1}$$

$$2x + 1 + 1 = 2^{2n+1}$$

$$x = 2^{2n} - 1$$

Further simplify the above,

$$2^{2n} - 1 = 255$$

$$2^{2n} = 256$$

$$2^{2n} = 2^8$$

$$n = 4$$

17. It is given that, expansion of $(1 + 2x)^n$ for $x = 1$ is

$$(1 + 2)^n = 6561$$

$$3^n = 6561$$

$$n = 8$$

$$\text{For } x = \frac{1}{\sqrt{2}}$$

$$\begin{aligned} R &= (1 + 2x)^n \\ &= I + F \end{aligned}$$

$$(\sqrt{2} - 1)^8 = F'$$

$$(\sqrt{2} + 1)^8 + (\sqrt{2} - 1)^8 = I + (F + F')$$

This implies,

$$\begin{aligned} &2 \left[(\sqrt{2})^8 \right] + {}^8C_2 (\sqrt{2})^6 + {}^8C_4 (\sqrt{2})^4 + {}^8C_6 (\sqrt{2})^2 + {}^8C_8 \\ &= I + (F + F') \end{aligned}$$

So,

$$F + F' = 1$$

$$F = 1 - F'$$

$$= 1 - (\sqrt{2} - 1)^8$$

So,

$$\begin{aligned}
 1 - \frac{F}{1 + (\sqrt{2} - 1)^4} &= 1 - \frac{1 - (\sqrt{2} - 1)^8}{1 + (\sqrt{2} - 1)^4} \\
 &= 1 - \frac{\left[1 + (\sqrt{2} - 1)^4\right] \left[1 - (\sqrt{2} - 1)^4\right]}{1 + (\sqrt{2} - 1)^4} \\
 &= (\sqrt{2} - 1)^4
 \end{aligned}$$

18. Consider the expression,

$$\frac{(1 - px)^{-1}}{1 - qx} = a_0 + a_1x + a_2x^2 + a_3x^3 + \dots$$

$$(1 - px)^{-1} = 1 + px + p^2x^2 + p^3x^3 + \dots + p^nx^n + \dots$$

And,

$$(1 - qx)^{-1} = 1 + qx + q^2x^2 + q^3x^3 + \dots + q^nx^n + \dots$$

Then coefficient of x^n in the expansion of

$$(1 - px)^{-1} (1 - qx)^{-1} = p^n + p^{n-1}q + p^{n-2}q^2 + p^{n-3}q^3 + \dots + q^n$$

Then,

$$\begin{aligned}
 a_n &= \frac{p^n \left(1 - \left(\frac{q}{p} \right)^{n+1} \right)}{1 - \frac{q}{p}} \\
 &= \frac{p^n (p^{n+1} - q^{n+1}) p}{(p - q) p^{n+1}} \\
 &= \frac{p^{n+1} - q^{n+1}}{p - q}
 \end{aligned}$$

19. Consider the expression,

$$\begin{aligned}
 \frac{3}{(x-1)(x^2+x+1)} &= \frac{1}{x-1} - \frac{x+2}{(x^2+x+1)} \\
 &= f_1(x) - f_2(x)
 \end{aligned}$$

And

$$\frac{x+1}{(x-1)^2(x^2+x+1)} = Af_1(x) + \left(B + \frac{D}{x-1} \right) f_2(x) + \frac{C}{(x-1)^2}$$

From above,

$$f_1(x) = \frac{1}{x-1}$$

$$f_2(x) = \frac{x+2}{x^2+x+1}$$

So,

$$\frac{x+1}{(x-1)^2(x^2+x+1)} = \frac{A}{x-1} + \frac{(Bx-B+D)(x+2)}{(x^2+x+1)(x-1)} + \frac{C}{(x-1)^2}$$

$$x+1 = A(x-1)(x^2+x+1) + B(x-1)$$

$$(x+2)(x-1) + D(x+2)(x-1) + C(x^2+x+1)$$

Comparing the coefficients,

$$A + B = 0$$

And,

$$A - A + 2B - 2B + D + C = 0$$

$$D + C = 0$$

This implies $A + B + C + D = 0$.

20. Consider the expression.

$$f(\theta) = \sqrt{a^2 \cos^2 \theta + b^2 \sin^2 \theta} + \sqrt{a^2 \sin^2 \theta + b^2 \cos^2 \theta}$$

$$[f(\theta)]^2 = \left(a^2 \cos^2 \theta + b^2 \sin^2 \theta + a^2 \sin^2 \theta + b^2 \cos^2 \theta + 2\sqrt{(a^2 \cos^2 \theta + b^2 \sin^2 \theta)(a^2 \sin^2 \theta + b^2 \cos^2 \theta)} \right)$$

The above is maximum when $\sin^2 \theta = \cos^2 \theta = \frac{1}{2}$ and minimum

when these are zero.

So,

$$\begin{aligned} M &= (a^2 + b^2) + 2\sqrt{\left(\frac{a^2}{2} + \frac{b^2}{2}\right)^2} \\ &= 2(a^2 + b^2) \end{aligned}$$

And,

$$\begin{aligned} m &= (a^2 + b^2) + 2\sqrt{a^2b^2} \\ &= a^2 + b^2 + 2ab \end{aligned}$$

This implies,

$$\begin{aligned} M - m &= 2(a^2 + b^2) - [a^2 + b^2 + 2ab] \\ &= a^2 + b^2 - 2ab \\ &= (a - b)^2 \end{aligned}$$

21. Consider the expression,

$$\tan B = -\frac{7}{24}$$

Then,

$$\frac{2 \tan \frac{B}{2}}{1 - \tan^2 \frac{B}{2}} = -\frac{7}{24}$$

$$7 \tan^2 \frac{B}{2} - 48 \tan \frac{B}{2} - 7 = 0$$

$$\left(7 \tan \frac{B}{2} + 1\right) \left(\tan \frac{B}{2} - 7\right) = 0$$

$$\tan \frac{B}{2} = -\frac{1}{7}$$

Then,

$$\tan \left(A + \frac{B}{2}\right) = \frac{\tan A + \tan \frac{B}{2}}{1 - \tan A \tan \frac{B}{2}}$$

$$\begin{aligned} &= \frac{\frac{11}{60} - \frac{1}{7}}{1 + \frac{11}{420}} \\ &= \frac{17}{431} \end{aligned}$$

This implies,

$$A \in \text{III}^{\text{rd}} \text{ quadrant and } \frac{B}{2} \in \left(\frac{3\pi}{4}, \pi\right)$$

Now,

$\tan\left(A + \frac{B}{2}\right)$ is positive so,

$$\left(A + \frac{B}{2}\right) \in \text{Ist quadrant}$$

22. Consider the expression,

$$\begin{aligned} & \cos^2 5^\circ - \cos^2 15^\circ - \sin^2 15^\circ + \sin^2 35^\circ + \cos^2 15^\circ \sin 15^\circ - \cos 5^\circ \sin 35^\circ \\ &= \cos 5^\circ (\cos 5^\circ - \sin 35^\circ) - \cos 15^\circ (\cos 15^\circ - \sin 15^\circ) + \sin^2 35^\circ - \sin^2 15^\circ \\ &= \cos 5^\circ (\cos 5^\circ - \cos 55^\circ) - \cos 15^\circ (\cos 15^\circ - \cos 75^\circ) + \sin^2 50^\circ \sin^2 20^\circ \\ &= \cos 5^\circ (2 \sin 30^\circ \sin 25^\circ) - \cos 15^\circ (2 \sin 45^\circ \sin 30^\circ) + \sin 50^\circ \sin 20^\circ \\ &= \left(-\frac{\cos 15^\circ}{\sqrt{2}} + \cos 5^\circ \sin 25^\circ \right) \\ &\quad + \sin 50^\circ \sin 20^\circ \\ &= \left(-\frac{\cos 15^\circ}{\sqrt{2}} + \frac{1}{2} (2 \cos 5^\circ \cos 65^\circ) + \frac{1}{2} (2 \sin 50^\circ \sin 20^\circ) \right) \\ &= \left(-\frac{\cos 15^\circ}{\sqrt{2}} + \frac{1}{2} (\cos 70^\circ + \cos 60^\circ + \cos 30^\circ - \cos 70^\circ) \right) \\ &= -\frac{\cos 15^\circ}{\sqrt{2}} + \frac{1}{2} [2 \cos 45^\circ \cos 15^\circ] \\ &= -\frac{\cos 15^\circ}{\sqrt{2}} + \frac{\cos 15^\circ}{\sqrt{2}} \\ &= 0 \end{aligned}$$

23. If $\cos \theta \neq 0$

And,

$$\sec \theta - 1 = (\sqrt{2} - 1) \tan \theta$$

$$\frac{1 - \cos \theta}{\cos \theta} = (\sqrt{2} - 1) \frac{\sin \theta}{\cos \theta}$$

$$2 \sin^2 \frac{\theta}{2} = (\sqrt{2} - 1) 2 \sin \frac{\theta}{2} \cos \frac{\theta}{2}$$

$$\sin \frac{\theta}{2} = 0 \text{ or } \tan \frac{\theta}{2} = \sqrt{2} - 1$$

This implies,

$$\frac{\theta}{2} = n\pi, n \in \mathbb{Z}$$

$$\frac{\theta}{2} = n\pi + \frac{\pi}{8}, n \in \mathbb{Z}$$

$$\theta = 2n\pi + \frac{\pi}{4}$$

$$2n\pi, n \in \mathbb{Z}$$

24. Consider the expression,

$$\sum_{k=1}^n 2k = n(n+1)$$

Then,

$$\cot\left(\sum_{n=3}^{32} \cot^{-1}(1+n(n+1))\right) = \sum_{n=3}^{32} \tan^{-1}\left(\frac{(n+1)-n}{1+(n+1)n}\right)$$

This implies,

$$\begin{aligned} \sum_{n=3}^{32} \tan^{-1}\left(\frac{(n+1)-n}{1+(n+1)n}\right) &= \sum_{n=3}^{32} \tan^{-1}(n+1) - \tan^{-1} n \\ &= (\tan^{-1} 4 - \tan^{-1} 3 + (\tan^{-1} 5 - \tan^{-1} 4)) + \dots \\ &\quad + (\tan^{-1} 33 - \tan^{-1} 32) \\ &= \tan^{-1} 33 - \tan^{-1} 3 \\ &= \tan^{-1} \left[\frac{33-3}{1+99} \right] \end{aligned}$$

Further simplify the above,

$$\begin{aligned} \sum_{n=3}^{32} \tan^{-1}\left(\frac{(n+1)-n}{1+(n+1)n}\right) &= \tan^{-1}\left(\frac{3}{10}\right) \\ \cot\left(\sum_{n=3}^{32} \tan^{-1}\left(\frac{(n+1)-n}{1+(n+1)n}\right)\right) &= \cot\left(\tan^{-1}\left(\frac{3}{10}\right)\right) \\ \cot\left[\cot^{-1}\left(\frac{10}{3}\right)\right] &= \frac{10}{3} \end{aligned}$$

25. Consider the expression,

$$\tan x = \frac{3}{4}$$

Then,

$$\sin^2 x = \frac{9}{25}$$

And

$$\cos^2 x = \frac{16}{25}$$

Then,

$$\cos^2 \theta + \sin^2 \theta = 1$$

$$(\sin \cosh y)^2 + (\cos x \sinh y)^2 = 1$$

$$9 + 9 \sinh^2 y + 16 \sinh^2 y = 25$$

$$\sinh^2 y = \frac{16}{25}$$

26. Let,

$$\frac{b+c}{9} = \frac{c+a}{10} = \frac{a+b}{11} = k$$

This implies,

$$a = 6k, b = 5k, c = 4k$$

Then,

$$\begin{aligned}
 \frac{\cos A + \cos B}{\cos C} &= \frac{\frac{b^2 + c^2 - a^2}{2bc} + \frac{a^2 + c^2 - b^2}{2ac}}{\frac{a^2 + b^2 - c^2}{2ab}} \\
 &= \frac{\frac{25 + 16 - 36}{40} + \frac{36 + 16 - 25}{48}}{\frac{36 + 25 - 16}{60}} \\
 &= \frac{\frac{5}{40} + \frac{27}{48}}{\frac{45}{60}} \\
 &= \frac{11}{12}
 \end{aligned}$$

27. For A,

Consider $r_1 r_2 \sqrt{\frac{4R - r_1 - r_2}{r_1 + r_2}}$

$$= \left[\frac{\Delta^2}{(s-a)(s-b)} \times \left(\sqrt{\frac{4R - 4R \cos \frac{C}{2} \left(\sin \frac{A}{2} \cos \frac{B}{2} + \sin \frac{B}{2} \cos \frac{A}{2} \right)}{4R \cos \frac{C}{2} \left(\sin \frac{A}{2} \cos \frac{B}{2} + \sin \frac{B}{2} \cos \frac{A}{2} \right)}} \right) \right]$$

$$= \frac{\Delta^2}{(s-a)(s-b)} \sqrt{\frac{4R \left(1 - \cos^2 \frac{C}{2} \right)}{4R \cos^2 \frac{C}{2}}}$$

$$= \frac{\Delta^2}{(s-a)(s-b)} \tan \frac{C}{2}$$

$$= \frac{\Delta^2}{(s-a)(s-b)} \sqrt{\frac{(s-a)(s-b)}{s(s-c)}}$$

Then,

$$\begin{aligned} r_1 r_2 \sqrt{\frac{4R - r_1 - r_2}{r_1 + r_2}} &= \frac{\Delta^2}{\Delta} \\ &= \Delta \end{aligned}$$

For B,

$$\begin{aligned}
 \frac{r_2(r_3 + r_2)}{\sqrt{r_1 r_2 + r_2 r_3 + r_3 r_1}} &= \frac{\frac{\Delta}{s-b} \left(\frac{\Delta}{s-c} + \frac{\Delta}{s-a} \right)}{\sqrt{\frac{(s-a) + (s-b) + (s-c)}{(s-a)(s-b)(s-c)}}} \\
 &= \frac{\frac{\Delta}{s-b} \left(\frac{2S - a - c}{(s-a)(s-c)} \right)}{\sqrt{\frac{3S - a - b - c}{(s-a)(s-b)(s-c)}}} \\
 &= \frac{\Delta(b)}{\sqrt{s(s-a)(s-b)(s-c)}} \\
 &= b
 \end{aligned}$$

For C,

$$\begin{aligned}
 \frac{a}{c} &= \frac{\sin(A-B)}{\sin(B-C)} \\
 \frac{\sin A}{\sin C} &= \frac{\sin(A-B)}{\sin(B-C)}
 \end{aligned}$$

$$\sin A \sin(B-C) = \sin(A-B) \sin C$$

$$2 \sin A \cos B \sin C = \sin A \sin B \cos C + \sin B \cos A \sin C$$

This implies,

$$2 \frac{a}{2R} \left(\frac{a^2 + c^2 - b^2}{2ac} \right) \left(\frac{c}{2R} \right) = \left(\frac{a}{2R} \left(\frac{b}{2R} \right) \left(\frac{a^2 + b^2 - c^2}{2abc} \right) \right) +$$

$$\left(\frac{b}{2R} \left(\frac{b^2 + c^2 - a^2}{2bc} \right) \left(\frac{c}{2R} \right) \right)$$

$$2(a^2 + c^2 - b^2) = a^2 + b^2 - c^2 + b^2 + c^2 - a^2$$

$$2a^2 + 2c^2 - 2b^2 = 2b^2$$

$$2b^2 = a^2 + c^2$$

Therefore the correct option is (3).

28. It is given that,

$$r_1 = 8 = \frac{\Delta}{s - a}$$

$$r_2 = 12 = \frac{\Delta}{s - b}$$

$$r_3 = 24 = \frac{\Delta}{s - c}$$

From first two equations,

$$\frac{s - b}{s - a} = \frac{2}{3}$$

$$3s - 3b = 2s - 2a$$

$$5a + c = 5b \quad (I)$$

From r_2 and r_3 ,

$$\frac{s-c}{s-b} = \frac{1}{2}$$

$$2s - 2c = s - b$$

$$a + 3b = 3c \quad (II)$$

From r_1 and r_3 ,

$$\frac{s-c}{s-a} = \frac{1}{3}$$

$$3s - 3c = s - a$$

$$2a + b = 2c \quad (III)$$

From (I), (II), and (III),

$$(a, b, c) = (12, 16, 20)$$

29. The bisector of angle A divides the BC in ratio $c : b$, where

$$c = AB$$

$$= \sqrt{4 + 16 + 16}$$

$$= 6$$

And,

$$b = AC$$

$$= \sqrt{4 + 4 + 1}$$

$$= 3$$

The position vector of the point where bisector of A meet BC is,

$$\frac{6(2\bar{i} + 5\bar{j} + 7\bar{k}) + 3(2\bar{i} + 3\bar{j} + 4\bar{k})}{6 + 3} = \frac{18\bar{i} + 39\bar{j} + 54\bar{k}}{9}$$

$$= 2\bar{i} + \frac{13}{3}\bar{j} + 6\bar{k}$$

30. The equation of the plane passes through the point $(\bar{i} + 2\bar{j} - \bar{k}) = (1, 2, -1)$ having direction ratios normal to a, b, c is,

$$a(x - 1) + b(y - 2) + c(z + 1) = 0 \quad \text{(I)}$$

The plane in equation (I) is perpendicular to line of intersection of planes $\bar{r} \cdot (3\bar{i} - \bar{j} + \bar{k}) = 1$ and $\bar{r} \cdot (\bar{i} + 4\bar{j} - 2\bar{k}) = 2$ so,

$$3a - b + c = 0$$

$$a + 4b - 2c = 0$$

This implies,

$$\frac{a}{2 - 4} = \frac{-b}{-6 - 1} = \frac{c}{12 + 1}$$

$$\frac{a}{2} = \frac{b}{-7} = \frac{c}{-13}$$

So, equation of required plane is,

$$2(x-1) - 7(y-2) - 13(z+1) = 0$$

$$2x - 7y - 13z - 2 + 14 - 13 = 0$$

$$2x - 7y - 13z = 1$$

$$\vec{r} \cdot (2\vec{i} - 7\vec{j} - 13\vec{k}) = 1$$

31. Consider,

$$\overrightarrow{BA} = 3\vec{i} - 6\vec{k}$$

And,

$$\overrightarrow{BC} = 4\vec{i} - \vec{j} - 12\vec{k}$$

Then,

$$\begin{aligned}\cos(\angle B) &= \frac{|\overrightarrow{BA} \cdot \overrightarrow{BC}|}{|\overrightarrow{BA}| |\overrightarrow{BC}|} \\ &= \frac{12 + 12}{\sqrt{9 + 36} \sqrt{16 + 1 + 4}} \\ &= \frac{8}{\sqrt{105}}\end{aligned}$$

$$\angle B = \cos^{-1}\left(\frac{8}{\sqrt{105}}\right)$$

32. The length of altitude drawn from A is given by,

$$h = \frac{\frac{1}{2}|\overline{AB} \times \overline{AC}|}{\frac{1}{2}|\overline{BC}|}$$

$$\overline{AB} = -2\bar{i} + \bar{j} + \bar{k}$$

$$\overline{AC} = -\bar{i} + 2\bar{j} - \bar{k}$$

$$\overline{BC} = \bar{i} + \bar{j} - 2\bar{k}$$

So,

$$\begin{aligned}\overline{AB} \times \overline{AC} &= \begin{vmatrix} \bar{i} & \bar{j} & \bar{k} \\ -2 & 1 & 1 \\ -1 & 2 & -1 \end{vmatrix} \\ &= \bar{i}(-1-2) - \bar{j}(2+1) + \bar{k}(-4+1) \\ &= -3\bar{i} - 3\bar{j} - 3\bar{k}\end{aligned}$$

This implies,

$$|\overline{AB} \times \overline{AC}| = 3\sqrt{3}$$

And,

$$|\overline{BC}| = \sqrt{6}$$

Then,

$$h = \frac{3\sqrt{3}}{\sqrt{2}\sqrt{3}}$$

$$= \frac{3}{\sqrt{2}}$$

33. The volume of tetrahedron OABC is given by,

$$OABC = \frac{1}{6} [\bar{a} \bar{b} \bar{c}]$$

Now the position vectors are,

$$\frac{\bar{a} + \bar{b}}{3}, \frac{\bar{b} + \bar{c}}{3}, \frac{\bar{c} + \bar{a}}{3}, \frac{\bar{a} + \bar{b} + \bar{c}}{3}$$

So, conterminous edge vectors of new tetrahedron are $\frac{\bar{a}}{3}, \frac{\bar{b}}{3}, \frac{\bar{c}}{3}$

So, the volume of new tetrahedron is,

$$\frac{1}{6} \left[\frac{\bar{a}}{3} \frac{\bar{b}}{3} \frac{\bar{c}}{3} \right] = \frac{1}{6 \times 27} [\bar{a} \bar{b} \bar{c}]$$

So, the required ratio is $\frac{1}{27}$.

34. From the given vectors,

$$\begin{aligned}
 \overline{A} \times \overline{B} &= \begin{vmatrix} \bar{i} & \bar{j} & \bar{k} \\ 2 & 1 & -2 \\ 1 & 1 & 0 \end{vmatrix} \\
 &= \bar{i}(2) - \bar{j}(2) + \bar{k}(2-1) \\
 &= 2\bar{i} - 2\bar{j} + \bar{k}
 \end{aligned}$$

This implies,

$$\begin{aligned}
 |(\overline{A} \times \overline{B}) \times \overline{C}| &= |(\overline{A} \times \overline{B})| |\overline{C}| \sin 30^\circ \\
 &= \sqrt{4+4+1} |\overline{C}| \left(\frac{1}{2}\right) \\
 &= \frac{3}{2} |\overline{C}| \\
 |\overline{C} - \overline{A}| &= 2\sqrt{2}
 \end{aligned}$$

This implies,

$$\begin{aligned}
 |\overline{C}|^2 + |\overline{A}|^2 - 2\overline{C} \cdot \overline{A} &= 8 \\
 |\overline{C}|^2 + 9 - 2|\overline{C}| &= 8 \\
 (|\overline{C}| - 1)^2 &= 0 \\
 |\overline{C}| &= 1
 \end{aligned}$$

So,

$$|(\overline{A} \times \overline{B}) \times \overline{C}| = \frac{3}{2}$$

35. The mean of given data is,

$$\begin{aligned}\bar{x} &= \frac{a_0 + a_1 + a_2 + \dots + a_{11}}{12} \\ &= \frac{a_0 + a_{11}}{2} \\ &= \frac{a_1 + a_{10}}{2} \\ &= \dots\end{aligned}$$

Then the deviation from mean is given by,

$$\begin{aligned}|\bar{x} - a_0| &= \frac{|a_{11} - a_0|}{2} \\ &= \frac{11|d|}{2}\end{aligned}$$

And,

$$\begin{aligned}|\bar{x} - a_1| &= \frac{|a_{10} - a_1|}{2} \\ &= \frac{9|d|}{2}, \dots\end{aligned}$$

So, sum of deviations is,

$$2 \left[\frac{11|d|}{2} \cdot 6 + \frac{9|d|}{2} + \frac{7|d|}{2} + \frac{5|d|}{2} + \frac{3|d|}{2} + \frac{|d|}{2} \right] = 36|d|$$

The mean deviation from arithmetic mean is given by,

$$\frac{36|d|}{12} = 3|d|$$

36. Consider given data,

Class Interval	Frequency (f_i)	x_i	$x_i f_i$	$(\bar{x} - x_i)^2$	$f_i (\bar{x} - x_i)^2$
0-10	2	5	10	196	392
10-20	3	5	45	16	48
20-30	4	25	100	36	144
30-40	1	35	35	256	256
$\sum f_i = 10$		$\sum x_i f_i = 190$			
		$\sum f_i (\bar{x} - x_i)^2 = 840$			

In the above table,

$$\begin{aligned}\bar{x} &= \frac{\sum x_i f_i}{N} \\ &= \frac{190}{10} \\ &= 19\end{aligned}$$

Thus the variance is,

$$\begin{aligned}\sigma^2 &= \frac{1}{N} \sum f_i (\bar{x} - x_i)^2 = \frac{1}{10}(840) \\ &= 84\end{aligned}$$

37. The required probability is given by,

$$\begin{aligned}\frac{{}^{73}C_{28} + {}^{73}C_{43}}{{}^{75}C_{30}} &= \frac{\frac{73!}{28!45!} + \frac{73!}{43!30!}}{\frac{75!}{30!45!}} \\ &= \frac{\frac{1}{45(44)} + \frac{1}{30(29)}}{\frac{75(74)}{(30)(29)(45)(44)}} \\ &= \frac{2(29) + (44)(3)}{5(74)} \\ &= \frac{19}{37}\end{aligned}$$

38. From the given data,

$$\begin{aligned}\frac{(n-1)(1) + (n+1)\frac{1}{2}}{2n} &= \frac{41}{56} \\ \frac{2n-2+n+1}{4n} &= \frac{41}{56} \\ 42n-14 &= 41n \\ n &= 14\end{aligned}$$

So, number of unfair coins in the bag is given by,

$$\begin{aligned}n-1 &= 14-1 \\ &= 13\end{aligned}$$

39. The required probability is given by,

$$\begin{aligned}\frac{\frac{1}{4}\left(\frac{{}^6C_2 + {}^8C_2}{{}^{14}C_2}\right)}{\frac{1}{4}\left(\frac{{}^6C_2 + {}^8C_2}{{}^{14}C_2}\right) + \frac{3}{4}\left(\frac{{}^9C_2 + {}^5C_2}{{}^{14}C_2}\right)} &= \frac{(6)(5) + (8)(7)}{((6)(5) + (8)(7)) + 3[(9)(8) + (5)(4)]} \\ &= \frac{86}{86 + 276} \\ &= \frac{43}{181}\end{aligned}$$

40. Consider given data,

$X = x_i$	1	2	3	4	5	6
$P(X = x_i)$	0.2	0.3	0.12	0.1	0.2	0.08

Then,

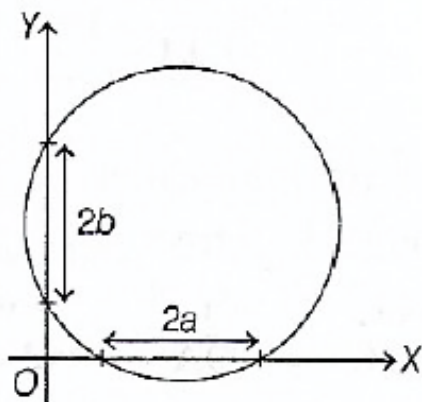
Now, $A = \{2, 3, 5\}$ and $B = \{1, 2, 3\}$.

$$\begin{aligned}P(A \cup B) &= P(X=1) + P(X=2) + P(X=3) + P(X=5) \\&= 0.2 + 0.3 + 0.12 + 0.2 \\&= 0.82\end{aligned}$$

41. For the unit mean,

$$\begin{aligned}\sum_{x=0}^{\infty} |x - \bar{x}| P(X = x) &= \sum_{x=0}^{\infty} |x - \bar{x}| \frac{e^{-1}}{x!} \\&= \frac{1}{e} \left[1 + \sum_{x=1}^{\infty} \frac{1}{(x-1)!} - \sum_{x=1}^{\infty} \frac{1}{x!} \right] \\&= \frac{1}{e} [1 + e - (e - 1)] \\&= \frac{2}{e}\end{aligned}$$

42. Consider the diagram,



Let the equation of circle be,

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

This implies,

$$2\sqrt{g^2 - c} = 2a$$

And,

$$2\sqrt{f^2 - c} = 2b$$

Then,

$$g^2 - a^2 = 0 \text{ and } f^2 - b^2 = 0$$

So,

$$g^2 - a^2 = f^2 - b^2$$

$$g^2 - f^2 = a^2 - b^2$$

When center is locus $(-g, -f)$ then,,

$$x^2 - y^2 = a^2 - b^2$$

43. After the rotation of the coordinate axes with given angle the relation between new (X, Y) and old coordinates (x, y) is given by,

$$(x, y) = [X \cos \theta - Y \sin \theta, (Y \cos \theta + X \sin \theta)],$$

The angle is $\theta = \frac{\pi}{4}$.

Then,

$$(x, y) = \left(\frac{X}{\sqrt{2}} - \frac{Y}{\sqrt{2}}, \left(\frac{Y}{\sqrt{2}} + \frac{X}{\sqrt{2}} \right) \right)$$

So,

$$25 \left(\frac{X - Y}{\sqrt{2}} \right)^2 + 9 \left(\frac{X + Y}{\sqrt{2}} \right)^2 = 225$$

$$34X^2 + 34Y^2 - 32XY = 450$$

$$17X^2 + 17Y^2 - 16XY = 225$$

Comparison of the coefficients gives,

$$\alpha = \gamma = 17, \beta = -16, \delta = 225$$

This implies,

$$\begin{aligned} (\alpha + \beta + \gamma - \sqrt{\delta})^2 &= (34 - 16 - 15)^2 \\ &= 3^2 \\ &= 9 \end{aligned}$$

44. The equation of line through the intersection of given lines is

$$(3x - 4y + 1) + \lambda(5x + y - 1) = 0$$

$$(3 + 5\lambda)x + (\lambda - 4)y = (\lambda - 1)$$

$$\frac{x}{\frac{\lambda - 1}{3 + 5\lambda}} + \frac{y}{\frac{\lambda - 1}{\lambda - 4}} = 1$$

From the given information,

$$\frac{\lambda - 1}{3 + 5\lambda} = \frac{\lambda - 1}{\lambda - 4} \text{ and } \lambda \neq 1.$$

Then,

$$\lambda - 4 = 5\lambda + 3$$

$$4\lambda = -7$$

$$\lambda = -\frac{7}{4}$$

So, equation of the required line is,

$$\left(3 - \frac{35}{4}\right)x + \left(-\frac{7}{4} - 4\right)y = \left(-\frac{7}{4} - 1\right)$$

$$-\frac{23}{4}x - \frac{23}{4}y = -\frac{11}{4}$$

$$23x + 23y = 11$$

45. The equation of the line that passes through $P(a, 2)$ and $a \neq 0$ making an angle 45° with positive X-axis is,

$$\frac{x-a}{\frac{1}{\sqrt{2}}} = \frac{y-2}{\frac{1}{\sqrt{2}}} = r$$

$$x = a + \frac{r}{\sqrt{2}} \text{ and } y = 2 + \frac{r}{\sqrt{2}}$$

So,

$$r = -a\sqrt{2}$$

For point C,

$$\begin{aligned} PB &= \sqrt{4+4} \\ &= 2\sqrt{2} \end{aligned}$$

And,

$$\begin{aligned} PC &= \sqrt{a^2 + a^2} \\ &= a\sqrt{2} \end{aligned}$$

From the points A and D,

$$\frac{\left(a + \frac{r}{\sqrt{2}}\right)^2}{9} + \frac{\left(2 + \frac{r}{\sqrt{2}}\right)^2}{4} = 1$$

$$4\left(a + \frac{r}{\sqrt{2}}\right)^2 + 9\left(2 + \frac{r}{\sqrt{2}}\right)^2 = 36$$

This implies,

$$13\left(\frac{r^2}{2}\right) + \left(\frac{8a}{\sqrt{2}} + \frac{36}{\sqrt{2}}\right)r + 4a^2 = 0$$

The roots are given by,

$$\begin{aligned} r_1 r_2 &= \frac{4a^2}{\frac{13}{2}} \\ &= \frac{8a^2}{13} \\ &= (PA)(PD) \end{aligned}$$

This implies,

PA, PB, PC, PD are in G.P so,

$$(PA)(PD) = (PB)(PC)$$

$$\frac{8a^2}{13} = 4a$$

$$2a = 13$$

46. The point $B(x_1, y_1)$ is the reflection of point A with respect to the line $x - y + 5 = 0$ so,

$$\begin{aligned}\frac{x_1 - 1}{1} &= \frac{y_1 + 2}{-1} \\ &= -2 \left(\frac{1 + 2 + 5}{2} \right) \\ &= -8 \\ x_1 &= -7\end{aligned}$$

And,

$$y_1 = 6$$

So, point is $B(-7, 6)$.

Similarly, for point $C(x_2, y_2)$ is reflection of A w.r.t line $x + 2y = 0$ so,

$$\begin{aligned}\frac{x_2 - 1}{1} &= \frac{y_2 + 2}{2} \\ &= -2 \left(\frac{1 - 4}{5} \right) \\ &= \frac{6}{5} \\ x_2 &= \frac{11}{5}\end{aligned}$$

And,

$$y_2 = \frac{2}{5}$$

So, point C is $\left(\frac{11}{5}, \frac{2}{5} \right)$

So equation of line BC is,

$$y - 6 = \frac{6 - \frac{2}{5}}{-7 - \frac{11}{5}}(x + 7)$$

$$\begin{aligned}y - 6 &= \frac{28}{-46}(x + 7) \\ &= \frac{-14}{23}(x + 7)\end{aligned}$$

Further simplify the above,

$$14x + 23y = 138 - 98$$

$$14x + 23y - 40 = 0$$

47. From the given data,

$$\frac{|4 - 3m|}{\sqrt{1 + m^2}} = 4$$

$$16 + 9m^2 - 24m = 16 + 16m^2$$

$$7m^2 + 24m = 0$$

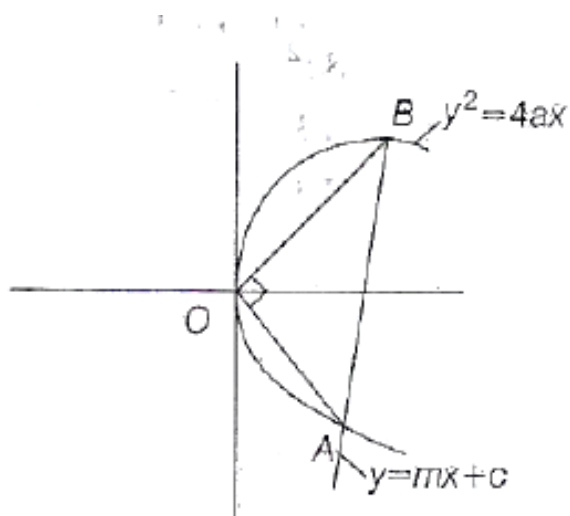
$$m = 0, -\frac{24}{7}$$

So, combined equation of the required line is,

$$y\left(y + \frac{24}{7}x\right) = 0$$

$$7y^2 + 24xy = 0$$

48. Consider the diagram,



Then,

$$y^2 - 4ax \left(\frac{y - mx}{c} \right) = 0$$

$$c + 4am = 0 \quad (I)$$

On substituting value of c it gives,

$$y = m(x - 4a)$$

It represents the family of lines passing through $(4a, 0)$.

49. Since it is given that abscissae of two points P and Q are roots of the equation $2x^2 + 4x - 7 = 0$ and their coordinates are the roots of the equation $3x^2 - 12x - 1 = 0$.

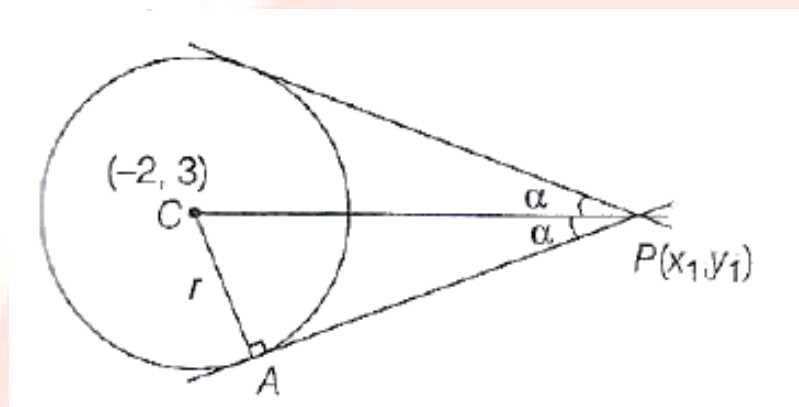
Let $P(x_1, y_1)$ and $Q(x_2, y_2)$ then,

$$x_1 + x_2 = -2 \text{ and } y_1 + y_2 = 4.$$

So, the center of circle with PQ as diameter is,

$$\left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right) = (-1, 2)$$

50. Consider the diagram,



Then,

$$\begin{aligned} \tan \alpha &= \frac{AC}{PA} \\ &= \frac{\sqrt{4 + 9 - 9\sin^2 \alpha + 13\cos^2 \alpha}}{\sqrt{x_1^2 + y_1^2 + 4x_1 - 6y_1 + 9\sin^2 \alpha + 13\cos^2 \alpha}} \\ &= \frac{\sqrt{13\sin^2 \alpha - 9\sin^2 \alpha}}{\sqrt{x_1^2 + y_1^2 + 4x_1 - 6y_1 + 9 + 4\cos^2 \alpha}} \\ &= \frac{\sqrt{4\sin^2 \alpha}}{\sqrt{x_1^2 + y_1^2 + 4x_1 - 6y_1 + 9 + 4\cos^2 \alpha}} \end{aligned}$$

This implies,

$$\frac{\sin^2 \alpha}{\cos^2 \alpha} = \frac{4\sin^2 \alpha}{x_1^2 + y_1^2 + 4x_1 - 6y_1 + 9 + 4\cos^2 \alpha}$$

$$x_1^2 + y_1^2 + 4x_1 - 6y_1 + 9 + 4\cos^2 \alpha = 4\cos^2 \alpha$$

$$x_1^2 + y_1^2 + 4x_1 - 6y_1 + 9 = 0$$

When taking locus of point $P(x_1, y_1)$, we get

$$x^2 + y^2 + 4x - 6y + 9 = 0$$

51. Consider the equation,

$$x^2 + y^2 - 4x - 6y - 12 = 0$$

The center of above is $C_1(2,3)$ and radius is,

$$\begin{aligned} r_1 &= \sqrt{4 + 9 + 12} \\ &= 5 \end{aligned}$$

Let the center of required circle be $C_2(h,k)$ and radius is given as 3 touches given circle as $A(-1,-1)$.

The point $A(-1,-1)$ divides the line joining the centers $C_1(2,3)$ and $C_2(h,k)$ externally in 5:3 so,

$$\begin{aligned}
 (-1, -1) &= \left(\frac{5h - 3(2)}{5 - 3}, \frac{5k - 3(3)}{5 - 3} \right) \\
 &= \left(\frac{5h - 6}{2}, \frac{5k - 9}{2} \right)
 \end{aligned}$$

From above,

$$5h - 6 = -2 \text{ and } 5k - 9 = -2.$$

This implies,

$$h = \frac{4}{5} \text{ and } k = \frac{7}{5}$$

So, equation of required circle is,

$$\left(x - \frac{4}{5}\right)^2 + \left(y - \frac{7}{5}\right)^2 = 3^2$$

$$x^2 + y^2 - \frac{8x}{5} - \frac{14y}{5} + \frac{65}{25} = 9$$

$$x^2 + y^2 - \frac{8x}{5} - \frac{14y}{5} + \frac{13}{5} = 9$$

$$5x^2 + 5y^2 - 8x - 14y - 32 = 0$$

52. Consider the equation of circle,

$$x^2 + y^2 + 2gx + 2fy + c = 0 \quad (I)$$

When center is $(-g, -f)$ then it must lie on the line,

$$2x + 3y - 7 = 0$$

$$2g + 3f + 7 = 0 \quad (II)$$

Since equation (I) cuts the given circle so,

$$x^2 + y^2 - 4x - 6y + 11 = 0$$

$$x^2 + y^2 - 10x - 4y + 21 = 0$$

Orthogonally,

$$2g(-2) + 2f(-3) = c + 11$$

$$4g + 6f + c + 11 = 0 \quad (III)$$

And,

$$2g(-5) + 2f(2) = c + 21$$

$$10g + 4f + c + 21 = 0 \quad (IV)$$

From equation (III) and (IV),

$$6g - 2f + 10 = 0 \quad (V)$$

From equation (II) and (V),

$$11f + 11 = 0$$

$$f = -1$$

So,

$$g = -2, c = 3$$

Then,

$$\begin{aligned}
 5g - 10f + 3c &= 5(-2) - 10(-1) + 33 \\
 &= -10 + 10 + 9 \\
 &= 9
 \end{aligned}$$

53. Consider the equation of circle,

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

And,

$$2x^2 + 2y^2 + 3x + 8y + 2c = 0$$

The radical axis of the above circles is,

$$(4g - 3)x + (4f - 8)y = 0$$

The radical axis touches the circle,

$$x^2 + y^2 + 2x + 2y + 1 = 0$$

So,

$$\frac{|-(4g - 3) - (4f - 8)|}{\sqrt{(4g - 3)^2 + (4f - 8)^2}} = \sqrt{1 + 1 - 1}$$

$$(4g - 3)^2 + (4f - 8)^2 + 2(4g - 3)(4f - 8) = (4g - 3)^2 + (4f - 8)^2$$

$$8(4g - 3)(f - 2) = 0$$

$$(4g - 3)(f - 2) = 0$$

54. The length of the chord intercepted from line $y = mx + c$ by parabola $x^2 = 4ay$ is,

$$4\sqrt{a(1+m^2)(c+am^2)}$$

So,

$$\sqrt{40} = 4\sqrt{a(1+4)(1+a(4))}$$

$$40 = 16(5a(1+4a))$$

$$1 = 2a(1+4a)$$

$$8a^2 + 2a - 1 = 0$$

Further simplify the above,

$$8a^2 + 4a - 2a - 1 = 0$$

$$4a(2a+1) - 1(2a+1) = 0$$

$$a = \frac{1}{4} \text{ or } -\frac{1}{2}$$

$$4a = 1 \text{ or } -2$$

55. The equation of the normal that have slope m to the parabola $y^2 = 4ax$ is,

$$y = mx - 2am - am^3$$

It passes through (h, k) then,

$$k = mh - 2am - am^3$$

Let the roots be m_1, m_2, m_3 so,

$$m_1 m_2 m_3 = -\frac{k}{a}$$

And,

$$m_1 m_2 = -1$$

So,

$$m_3 = \frac{k}{a}$$

Then,

$$k = \frac{k}{a}h - 2a\frac{k}{a} - a\left(\frac{k}{a}\right)^3$$

$$1 = \frac{h}{a} - 2 - \frac{k^2}{a^2}$$

$$\frac{k^2}{a^2} = \frac{h}{a} - 3$$

$$k^2 = a(h - 3a)$$

By taking locus of (h, k) ,

$$y^2 - ax + 3a^2 = 0$$

56. Consider the ellipse,

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$$

Let $P(a \cos \theta, b \sin \theta)$ be the point of above ellipse.

The area of ΔPF_1F_2 is given by,

$$\begin{aligned}\Delta PF_1F_2 &= \frac{1}{2}(2ae)b|\sin \theta| \\ &= aeb|\sin \theta| \\ &= A\end{aligned}$$

For the maximum value of A , $\theta = \frac{\pi}{2}$ or $\frac{3\pi}{2}$ so,

$$A_{\max} = aeb$$

57. The equation of chord that joins $A(\alpha)$ and $B(\beta)$ on the ellipse,

$$\frac{x^2}{25} + \frac{y^2}{9} = 1$$

Is,

$$\frac{x}{5} \cos \frac{\alpha + \beta}{2} + \frac{y}{3} \sin \frac{\alpha + \beta}{2} = \cos \frac{\alpha - \beta}{2} \quad (I)$$

Above equation passes through $(4,0)$ so,

$$\frac{4}{5} \cos \frac{\alpha + \beta}{2} = \cos \frac{\alpha - \beta}{2}$$

$$4 \left(\cos \frac{\alpha}{2} \cos \frac{\beta}{2} - \sin \frac{\alpha}{2} \sin \frac{\beta}{2} \right) = 5 \left(\cos \frac{\alpha}{2} \cos \frac{\beta}{2} + \sin \frac{\alpha}{2} \sin \frac{\beta}{2} \right)$$

$$4 \left(\cot \frac{\alpha}{2} \cot \frac{\beta}{2} - 1 \right) = 5 \left(\cot \frac{\alpha}{2} \cot \frac{\beta}{2} + 1 \right)$$

$$\cot \frac{\alpha}{2} \cot \frac{\beta}{2} = -9$$

58. Consider the equation.

$$16x^2 - 25y^2 - 96x + 100y - 356 = 0$$

$$\frac{(x-3)^2}{25} - \frac{(y-2)^2}{16} = 1$$

The equation of the tangent to above with slope 1 is,

$$y - 2 = 1(x - 3) + \sqrt{25(1) - 16}$$

$$y - 2 = x - 3 + 3$$

$$x - y + 2 = 0$$

59. Consider the given points,

$$PQ = \sqrt{1+1+16}$$

$$= 3\sqrt{2}$$

And,

$$\begin{aligned}QR &= \sqrt{9+9+0} \\ &= 3\sqrt{2}\end{aligned}$$

And

$$\begin{aligned}PR &= \sqrt{16+4+16} \\ &= \sqrt{36} \\ &= 6\end{aligned}$$

This implies,

$$\begin{aligned}PQ^2 + QR^2 &= 18 + 18 \\ &= 36 \\ &= PR^2\end{aligned}$$

Also, $PQ = QR$

This implies, PQR is a right angled isosceles triangle.

60. Consider given points,

$$A(2,3,5), B(\alpha,3,3), C(7,5,\beta)$$

The mid-point of BC is,

$$D\left(\frac{\alpha+7}{2}, 4, \frac{3+\beta}{2}\right)$$

The direction ratios of the lines joining the points A and D is,

$$\left(\frac{\alpha+3}{2}, 1, \frac{\beta-7}{2}\right)$$

The line axed AD is equally inclined with coordinate axis is,

$$\frac{\alpha + 3}{2} = 1 = \frac{\beta - 7}{2}$$

This implies,

$$\alpha = -1 \text{ and } \beta = 9.$$

So,

$$\cos^{-1}\left(\frac{\alpha}{\beta}\right) = \cos^{-1}\left(-\frac{1}{9}\right)$$

61. The equation of the plane that passes through origin and given line is,

$$(\bar{r} - 0) \cdot \left[(\bar{i} + 2\bar{j} - 3\bar{k}) \times (2\bar{i} - 3\bar{j} + \bar{k}) \right] = 0$$

$$\bar{r} \cdot [\bar{i}(2 - 9) - \bar{j}(1 + 6) + \bar{k}(-3 - 4)] = 0$$

$$\bar{r} \cdot [-7\bar{i} - 7\bar{j} - 7\bar{k}] = 0$$

$$\bar{r} \cdot [\bar{i} + \bar{j} + \bar{k}] = 0 \text{ or } x + y + z = 0$$

62. Consider the given expression,

$$\lim_{x \rightarrow \infty} \left\{ \frac{x^3 + 1}{x^2 + 1} - (\alpha x + \beta) \right\} = 2$$

Then,

$$\lim_{x \rightarrow \infty} \left\{ \frac{x^3 + 1 - \alpha x^3 + \beta x^2 - \alpha x - \beta}{x^2 + 1} \right\} = 2$$

For the above limit to exist, $x^3 = 0$ so,

$$\alpha = 1$$

Then,

$$\lim_{x \rightarrow \infty} \frac{-\beta x^2 - x - \beta + 1}{x^2 + 1} = 2$$

$$\lim_{x \rightarrow \infty} \frac{-\beta - \frac{1}{x} - \frac{\beta}{x^2} + \frac{1}{x^2}}{1 + \frac{1}{x^2}} = 2$$

$$-\beta = 2$$

$$\beta = -2$$

This implies,

$$(\alpha, \beta) = (1, -2)$$

63. Consider the expression for $k > 0$,

$$\sum_{x=0}^{\infty} \frac{k^x}{x!} \lim_{x \rightarrow \infty} \frac{n!}{(n-x)!} \left(1 - \frac{k}{n}\right)^{n-x} \left(\frac{1}{n}\right)^x$$

Simplify the above,

$$\begin{aligned}
 \lim_{n \rightarrow \infty} \sum_{x=0}^{\infty} \frac{n!}{(n-x)!} \left(1 - \frac{k}{n}\right)^{n-x} \left(\frac{k}{n}\right)^x &= \lim_{n \rightarrow \infty} \sum_{x=0}^{\infty} {}^nC_x \left(1 - \frac{k}{n}\right)^{n-x} \left(\frac{k}{n}\right)^x \\
 &= \lim_{n \rightarrow \infty} \left(1 - \frac{k}{n} + \frac{k}{n}\right)^n \\
 &= \lim_{n \rightarrow \infty} 1^n \\
 &= 1
 \end{aligned}$$

64. Consider the function,

$$f(x) = \begin{cases} 5, & \text{if } x \leq 1 \\ a + bx, & \text{if } 1 < x < 3 \\ b + 5x, & \text{if } 3 \leq x < 5 \\ 30, & \text{if } x \geq 5 \end{cases}$$

If $f(x)$ is continuous at $x = 1$ then,

$$a + b = 5 \quad \text{(I)}$$

If continuous at $x = 3$ then,

$$a + 3b = b + 15$$

$$a + 2b = 15 \quad \text{(II)}$$

If continuous at $x = 5$ then,

$$b + 25 = 30$$

$$b = 5$$

Substitute 5 for b in equation (II).

$$a + 2(5) = 15$$

$$a = 5$$

But equation (I) is not satisfied by $a = 5$ and $b = 5$ then,

$f : R \rightarrow R$ is not continuous for any value of a and b .

65. Consider the function,

$$y = [x] + |1 - x|$$

The points of discontinuity are $x = 0, 1, 2, 3$ in $-1 \leq x \leq 3$.

So, function is not differentiable at 4 points.

66. Consider the equation,

$$\sqrt{1 - x^6} + \sqrt{1 - y^6} = a(x^3 - y^3)$$

$$\frac{\sqrt{1 - x^6} + \sqrt{1 - y^6}}{x^3 - y^3} = a$$

On differentiation both sides w.r.t x ,

$$\frac{\left[\left(x^3 - y^3 \right) \left(\frac{-6x^5}{2\sqrt{1-x^6}} - \frac{-6y^5}{2\sqrt{1-y^6}} \frac{dy}{dx} \right) \right] - \left(\sqrt{1-x^6} \right) + \left(\sqrt{1-y^6} \right) \left(3x^2 - 3y^2 \frac{dy}{dx} \right)}{(x^3 - y^3)^2} = 0$$

$$y^2 \frac{dy}{dx} \left[\frac{\left(\sqrt{1-x^6} + \sqrt{1-y^6} \right) + (1-y^6) - y^3 x^3 + y^6}{\sqrt{1-y^6}} \right] = x^2 \left[\frac{\left(\sqrt{1-x^6} + \sqrt{1-y^6} \sqrt{1-x^6} - y^3 x^3 + x^6 \right)}{\sqrt{1-x^6}} \right]$$

$$y^2 \frac{dy}{dx} = x^2 \sqrt{\frac{1-y^6}{1-x^6}}$$

67. Consider the expression,

$$\begin{aligned}\frac{d^2x}{dy^2} &= \frac{d}{dy} \left(\frac{dx}{dy} \right) \\ &= \frac{dx}{dy} \frac{d}{dx} \left(\frac{1}{\frac{dy}{dx}} \right) \\ &= \left(\frac{dx}{dy} \right) \left(\frac{-\frac{d^2y}{dx^2}}{\left(\frac{dy}{dx} \right)^2} \right) \\ &= - \frac{\left(\frac{d^2y}{dx^2} \right)}{\left(\frac{dy}{dx} \right)^3}\end{aligned}$$

This implies,

$$\begin{aligned}\left(\frac{d^2x}{dy^2} \right)_P &= - \frac{(-3)}{4^3} \\ &= \frac{3}{64}\end{aligned}$$

68. Consider the expression for time period of simple pendulum,

$$T = 2\pi \sqrt{\frac{L}{g}}$$

This implies,

$$\frac{\Delta T}{T} \times 100 = \frac{1}{2} \frac{\Delta L}{L}$$

$$\begin{aligned}\frac{\Delta L}{L} \% &= -\frac{2 \times 2}{24 \times 60} \times 100 \\ &= -\frac{10}{36} \\ &= -\frac{5}{18}\end{aligned}$$

69. Consider the function,

$$f(x) = \sum_{k=1}^n (x - a_k)^2$$

Then,

$$\begin{aligned}f(x) &= \sum_{k=1}^n (x^2 - 2xa_k + a_k^2) \\ &= nx^2 - 2x(a_1 + a_2 + a_3 + \dots + a_n) + (a_1^2 + a_2^2 + \dots + a_n^2)\end{aligned}$$

The minimum value at $x = -\frac{b}{2a}$ so,

$$\begin{aligned}
 x &= -\frac{-2(a_1 + a_2 + a_3 + \dots + a_n)}{2n} \\
 &= \frac{a_1 + a_2 + a_3 + \dots + a_n}{n} \\
 &= A
 \end{aligned}$$

70. Consider the function,

$$f(x) = x^{2m-1}(a-x)^{2n}$$

The Rolle's theorem is applicable on the above function in the interval $[0, a]$.

$$f'(x) = (2m-1)x^{2m-2}(a-x) - 2n(a-x)^{2n-1}x^{2m-1}$$

When $x = c$ then $m > 1, n > 1$ and $f'(c) = 0$ so,

$$(2m-1)c^{2m-2} = 2nc^{2m-1}(a-c)^{2n-1}$$

$$\frac{(2m-1)}{c} = \frac{2n}{a-c}$$

$$c = \frac{a(2m-1)}{2n+2m-1}$$

71. Consider the function,

$$f(x) = \begin{cases} 2^x + 1, & \text{for } x \in [-1, 0) \\ 1, & \text{for } x = 0 \\ 2^x - 1, & \text{for } x \in (0, 1] \end{cases}$$

The function is strictly increasing in $x \in (-1, 0)$ and $f(0^-) \rightarrow 2$ but $f(0) = 1$.

And in interval $x \in (0, 1)$ again it is strictly increasing but $f(0^+) = 0$.

So, the function is neither maximum nor minimum.

72. Consider the function,

$$\int \frac{x-1}{(x+1)\sqrt{x^3+x^2+x}} dx$$

Simplify the expression

$$\begin{aligned} \int \frac{x^2-1}{(x+1)^2 x \sqrt{1+\frac{1}{x}+x}} dx &= \int \frac{x^2-1}{\left(1+1+\frac{1}{x}+x\right) x^2 \sqrt{1+\frac{1}{x}+x}} dx \\ &= \int \frac{1-\frac{1}{x^2}}{1+\left(1+\frac{1}{x}+x\right) \sqrt{1+\frac{1}{x}+x}} dx \end{aligned}$$

$$\text{Let } 1 + \frac{1}{x} + x = t^2$$

Then,

$$\left(1 - \frac{1}{x^2}\right) dx = 2t dt$$

This implies,

$$\begin{aligned} \int -\frac{2t dt}{(1+t^2)t} &= 2 \int \frac{dt}{1+t^2} \\ &= 2 \tan^{-1}(t) + c \\ &= 2 \tan^{-1}\left(\sqrt{\frac{1+x+x^2}{x}}\right) + c \end{aligned}$$

73. Consider the integral,

$$I(x) = \int x^2 (\log x)^2 dx$$

Simplify the above using “by parts”.

$$\begin{aligned} I(x) &= \frac{x^3}{3}(\log x)^2 - \int \frac{x^3}{3} \frac{2 \log x}{x} dx \\ &= \frac{x^3}{3}(\log x)^2 - \frac{2}{3} \left[\frac{x^3}{3} \right] (\log x) - \int \frac{x^3}{3} \frac{1}{x} dx \\ &= \frac{x^3}{3}(\log x)^2 - \frac{2}{3} \left[\frac{x^3}{3} (\log x) - \frac{1}{3} \frac{x^3}{3} \right] + C \\ &= \frac{x^3}{27} \left[9(\log x)^2 - 6(\log x) + 2 \right] + C \end{aligned}$$

For $I(1) = 0$,

$$\frac{2}{27} + C = 0$$

$$C = -\frac{2}{27}$$

This implies,

$$I(x) = \frac{x^3}{27} \left[9(\log x)^2 - 6(\log x) + 2 \right] - \frac{2}{27}$$

74. Consider the given integral,

$$\begin{aligned}\int \frac{x^5 dx}{(x^2 + x + 1)(x^6 + 1)(x^4 - x^3 + x - 1)} \\&= \int \frac{x^5 dx}{(x^2 + x + 1)(x^6 + 1)(x - 1)(x^3 + 1)} \\&= \int \frac{x^5 dx}{(x^6 + 1)(x^3 - 1)(x^3 + 1)} \\&= \int \frac{x^5 dx}{(x^6 + 1)(x^6 - 1)}\end{aligned}$$

Let $x^6 = t$ then, $6x^5 dx = dt$

This implies,

$$\begin{aligned}\int \frac{x^5 dx}{(x^6 + 1)(x^6 - 1)} &= \frac{1}{6} \int \frac{dt}{(t + 1)(t - 1)} \\&= \frac{1}{12} \log_e \left| \frac{t - 1}{t + 1} \right| + c \\&= \frac{1}{12} \log_e \left| \frac{x^6 - 1}{x^6 + 1} \right| + c\end{aligned}$$

75. Consider the integral,

$$I = \int \frac{dx}{x + \sqrt{x - 1}}$$

Take $x - 1 = t^2$ then, $dx = 2tdt$

Substitute these values in the integral and solve.

$$\begin{aligned}
 I &= \int \frac{2t}{(t^2 + 1) + t} dt \\
 &= \int \frac{(2t + 1) - 1}{t^2 + t + 1} dt \\
 &= \int \frac{2t + 1}{t^2 + t + 1} dt - \int \left(\frac{dt}{t^2 + t + 1} \right) \\
 &= \log_e |t^2 + t + 1| - \int \frac{dt}{\left(t + \frac{1}{2}\right)^2 + \frac{3}{4}}
 \end{aligned}$$

Further simplify the above,

$$\begin{aligned}
 I &= \log_e |t^2 + t + 1| - \frac{2}{\sqrt{3}} \tan^{-1} \left(\frac{t + \frac{1}{2}}{\frac{\sqrt{3}}{2}} \right) + c \\
 &= \log_e |t^2 + t + 1| - \frac{2}{\sqrt{3}} \tan^{-1} \left(\frac{2t + 1}{\sqrt{3}} \right) + c
 \end{aligned}$$

Substitute the value of $t = \sqrt{x - 1}$.

$$I = \log_e |x + \sqrt{x - 1}| - \frac{2}{\sqrt{3}} \tan^{-1} \left(\frac{2\sqrt{x - 1} + 1}{\sqrt{3}} \right) + c$$

76. Consider the given integral,

$$\int_{\log_e 2}^x \frac{dt}{\sqrt{e^t - 1}} = \frac{\pi}{6}$$

$$\int_{\log_e 2}^x \frac{e^{-\frac{t}{2}}}{\sqrt{1 - \left(e^{-\frac{t}{2}}\right)^2}} dt = \frac{\pi}{6}$$

Take $e^{-\frac{t}{2}} = u$ then, $e^{-\frac{t}{2}} dt = -2du$

$$\text{At } t = \log_e 2, u = \frac{1}{\sqrt{2}}$$

$$\text{At } t = x, u = e^{-\frac{x}{2}}$$

Substitute these values in the above expression.

$$\int_{\frac{1}{\sqrt{2}}}^{e^{\frac{x}{2}}} \frac{-2du}{\sqrt{1-(u)^2}} = \frac{\pi}{6}$$

$$\sin^{-1}\left(e^{\frac{x}{2}}\right) - \frac{\pi}{4} = -\frac{\pi}{12}$$

$$\sin^{-1}\left(e^{\frac{x}{2}}\right) = \frac{\pi}{6}$$

$$e^{\frac{x}{2}} = \frac{1}{2}$$

$$-\frac{x}{2} = \log_e\left(\frac{1}{2}\right)$$

$$= -\log_e(2)$$

$$x = 2\log_e(2)$$

77. Consider the given integral.

$$I = \int_0^1 \frac{\log_e(1+x)}{1+x^2} dx$$

Take $x = \tan \theta$, then $dx = \sec^2 \theta d\theta$

Substitute in the above integral and change the limits accordingly.

$$\begin{aligned} I &= \int_0^{\frac{\pi}{4}} \frac{\log_e (1 + \tan \theta)}{\sec^2 \theta} \sec^2 \theta d\theta \\ &= \int_0^{\frac{\pi}{4}} \log_e (1 + \tan \theta) d\theta \end{aligned}$$

Now, using the property $\int_0^a f(x) dx = \int_0^a f(a-x) dx$,

$$\begin{aligned} I &= \int_0^{\frac{\pi}{4}} \log_e \left(1 + \tan \left(\frac{\pi}{4} - \theta \right) \right) d\theta \\ &= \int_0^{\frac{\pi}{4}} \log_e \left(1 + \frac{1 - \tan \theta}{1 + \tan \theta} \right) d\theta \\ &= \int_0^{\frac{\pi}{4}} \log_e \left(\frac{2}{1 + \tan \theta} \right) d\theta \\ &= \int_0^{\frac{\pi}{4}} \log_e (2) dx - \int_0^{\frac{\pi}{4}} \log_e (1 + \tan \theta) d\theta \end{aligned}$$

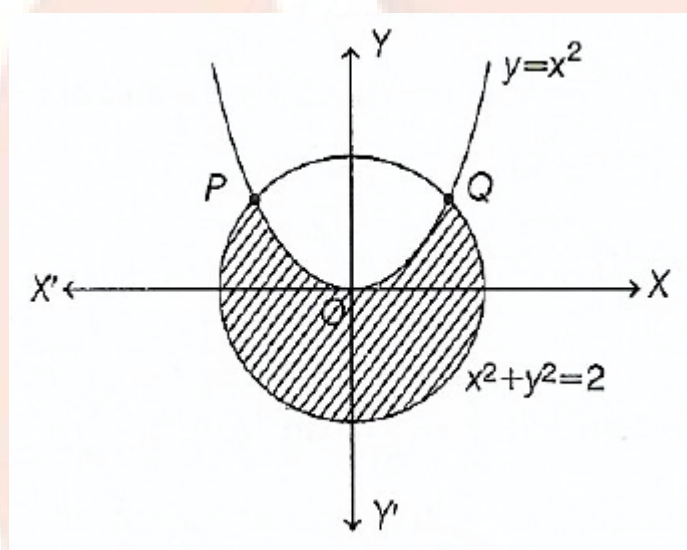
Further simplifying, we get

$$I = \frac{\pi}{4} \log_e(2) - I$$

$$2I = \frac{\pi}{4} \log_e(2)$$

$$I = \frac{\pi}{8} \log_e(2)$$

78. Consider the diagram,



Solve the given equations for the point of intersections.

$$y^2 + y - 2 = 0$$

$$y^2 + 2y - y - 2 = 0$$

$$y(y + 2) - 1(y + 2) = 0$$

$$y = 1 \quad [\because y > 0]$$

Thus, the area of the larger part will be,

$$\begin{aligned}\text{Area} &= \pi + 2 \int_0^1 \left(\sqrt{2-y^2} - \sqrt{y} \right) dy \\ &= \pi + 2 \left[\frac{y}{2} \sqrt{2-y^2} + \frac{2}{2} \sin^{-1} \frac{y}{\sqrt{2}} - \frac{2}{3} y^{\frac{3}{2}} \right]_0^1 \\ &= \pi + 2 \left[\frac{1}{2} + \frac{\pi}{4} - \frac{2}{3} \right] \\ &= \frac{3\pi}{2} - \frac{1}{3} \text{ sq. units}\end{aligned}$$

79. Consider the equation of family of curves.

$$\begin{aligned}x^2 &= c(y+c)^2 \\ x &= \sqrt{c}(y+c) \quad \dots\dots (1)\end{aligned}$$

After differentiating w.r.t to x , we get

$$\begin{aligned}1 &= \sqrt{c} \frac{dy}{dx} \\ \sqrt{c} &= \frac{dx}{dy}\end{aligned}$$

Now, from equation (1), we get

$$x = \left(\frac{dx}{dy} \right) \left[y + \left(\frac{dx}{dy} \right)^2 \right]$$

$$x \left(\frac{dy}{dx} \right)^3 = y \left(\frac{dy}{dx} \right)^2 + 1$$

$$x \left(\frac{dy}{dx} \right)^3 - y \left(\frac{dy}{dx} \right)^2 - 1 = 0$$

80. Simplify the given matrix.

$$\begin{vmatrix} f(x) & f'(x) \\ f'(x) & f''(x) \end{vmatrix} = 0$$

$$f(x)f''(x) - (f'(x))^2 = 0$$

$$f(x)f''(x) = (f'(x))^2$$

$$\frac{f''(x)}{f'(x)} = \frac{f'(x)}{f(x)}$$

Integrate both the sides.

$$\int \frac{f''(x)}{f'(x)} dx = \int \frac{f'(x)}{f(x)} dx$$

$$\log_e(f'(x)) = \log_e(f(x)) + c \quad \dots\dots (1)$$

Since $f(0) = 1$ and $f'(0) = 2$ then,

$$c = \log_e 2$$

Substituting the value of c in equation (1), we get

$$\log_e (f'(x)) = \log_e (f(x)) + \log_e 2$$

$$f'(x) = 2f(x)$$

$$\frac{f'(x)}{f(x)} = 2$$

Integrating both sides w. r. t. x ,

$$\int \frac{f'(x)}{f(x)} dx = \int 2 dx$$

$$\log_e f(x) = 2x + c$$

For $f(0) = 1$, $c = 0$.

This gives,

$$\log_e f(x) = 2x$$

$$f(x) = e^{2x}$$

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Solutions – Physic

81. The permeability of vacuum can be expressed as,

$$\begin{aligned}\mu_0 &\propto e^a m^b c^c h^d \\ &= k e^a m^b c^c h^d\end{aligned}$$

Here, k is the proportionality constant.

Substituting the dimensions of different physical quantities in the above expression, we get

$$\begin{aligned}[MLT^{-2}A^{-2}] &= [AT]^a [M]^b [LT^{-1}]^c [ML^2T^{-1}]^d \\ [MLT^{-2}A^{-2}] &= [M^{b+d}L^{c+2d}T^{a-c-d}A^a]\end{aligned}$$

The values of a , b , c and d is calculated by comparing the power on both sides.

$$a = -2$$

$$b + d = 1$$

$$c + 2d = 1$$

$$a - c - d = -2$$

On solving the above equations we get,

$$a = -2$$

$$b = 0$$

$$c = -1$$

$$d = 1$$

So, the permeability of free space can be expressed as,

$$\mu_0 = \frac{h}{ce^2}$$

82. The velocity of an object is given as,

$$v = 6t - 3t^2$$

As velocity is the rate of change of displacement therefore,

$$v = \frac{dx}{dt}$$

$$dx = vdt$$

Integrating both sides by taking the limit from $t = 0$ to $t = 2$, we get

$$\begin{aligned}x &= \int_0^2 v dt = \int_0^2 (6t - 3t^2) dt \\&= \left[3(2)^2 - (2)^3 \right] \\&= 12 - 8 = 4 \text{ m}\end{aligned}$$

The average velocity is given as,

$$\begin{aligned}v_{\text{avg}} &= \frac{\text{Total displacement}}{\text{Total time}} \\&= \frac{4}{2} = 2 \text{ m/s}\end{aligned}$$

83. The time taken by bullet to hit the target can be calculated as,

$$\begin{aligned}x &= vt \\t &= \frac{x}{v} \\&= \frac{600 \text{ m}}{500 \text{ m/s}} = 1.2 \text{ s}\end{aligned}$$

The value of h can be calculated using the second equation of motion.

$$\begin{aligned}h &= ut + \frac{1}{2}gt^2 \\&= 0 \times 1.2 + \frac{1}{2}(10 \text{ m/s}^2)(1.2 \text{ s})^2 \\&= 7.2 \text{ m}\end{aligned}$$

84. Let u is the velocity when angle is 60° and v is the velocity when angle is 45° with the horizontal.

In projectile motion, the horizontal component of velocity remains constant throughout the motion. Therefore,

$$\begin{aligned}v \cos 45^\circ &= u \cos 60^\circ \\v \times \frac{1}{\sqrt{2}} &= 140 \times \frac{1}{2} \\v &= 70\sqrt{2} \text{ m/s}\end{aligned}$$

If t is the time after which projectile makes an angle of 45° , then

$$v \sin 45^\circ = u \sin 60^\circ - gt$$

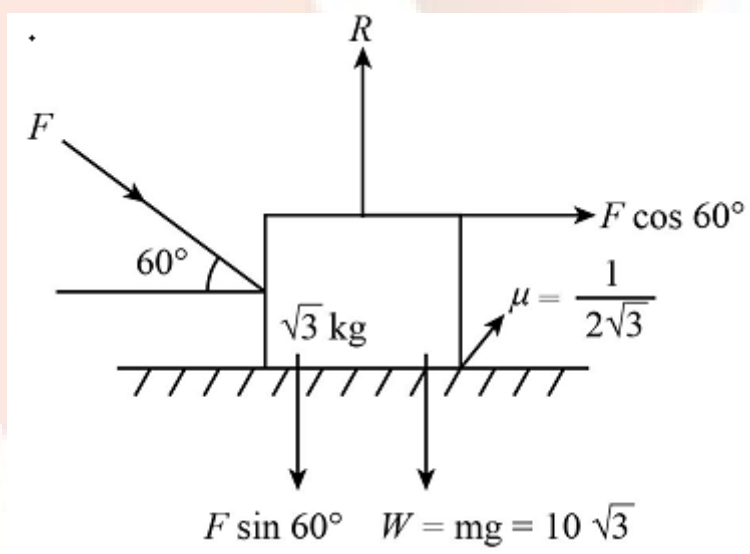
$$(70\sqrt{2}) \frac{1}{\sqrt{2}} = \frac{\sqrt{3}}{2}(140) - 10t$$

$$10t = 70\sqrt{3} - 70$$

$$t = 7(\sqrt{3} - 1)$$

$$= 5.124 \text{ s}$$

85. The figure below shows the free body diagram.



Applying the equilibrium along vertical direction,

$$R = (W + F \sin 60^\circ)$$

Applying the equilibrium along horizontal direction,

$$F \cos 60^\circ = \mu R$$

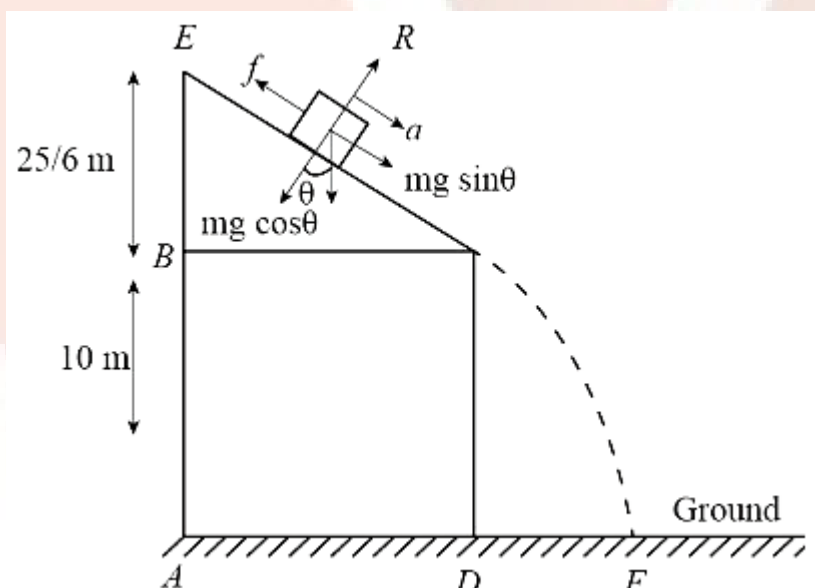
$$F \cos 60^\circ = \mu(W + F \sin 60^\circ)$$

Substituting the values in the above expression, we get

$$\frac{F}{2} = \frac{1}{2\sqrt{3}} \left[10\sqrt{3} + F \left(\frac{\sqrt{3}}{2} \right) \right]$$

$$F = 20 \text{ N}$$

86. The figure below shows the free body diagram of the block.



The friction force acting on the block is given as,

$$f = \mu_k (mg \cos \theta)$$

The force equation for the block along the plane of inclination,

$$mg \sin \theta - f = ma$$

$$mg \sin \theta - \mu_k (mg \cos \theta) = ma$$

$$a = 10 \times 0.6 - \frac{1}{8} \times 10 \times 0.8$$

$$a = 5 \text{ m/s}^2$$

The distance s is given from $\triangle EBC$,

$$EC = s = \frac{BE}{\sin \theta} = \frac{25 / 6}{0.6} = \frac{125}{18} \text{ m}$$

The velocity at final position is give as,

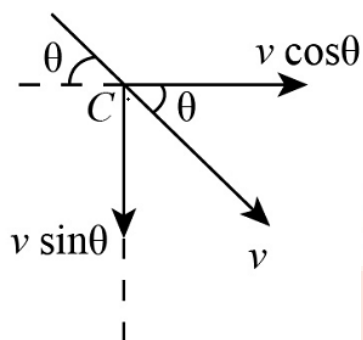
$$v^2 = 2as$$

Substituting the values in above expression, we get

$$v^2 = 2 \times 5 \text{ m/s}^2 \times \frac{125}{18} \text{ m}$$

$$v = \frac{25}{3} \text{ m/s}$$

The component of velocity at point C is given as,



Using second equation of motion, we get

$$h = ut + \frac{1}{2}gt^2$$

$$10 = ut \sin \theta + \frac{1}{2}gt^2$$

$$5t^2 + 5t - 10 = 0$$

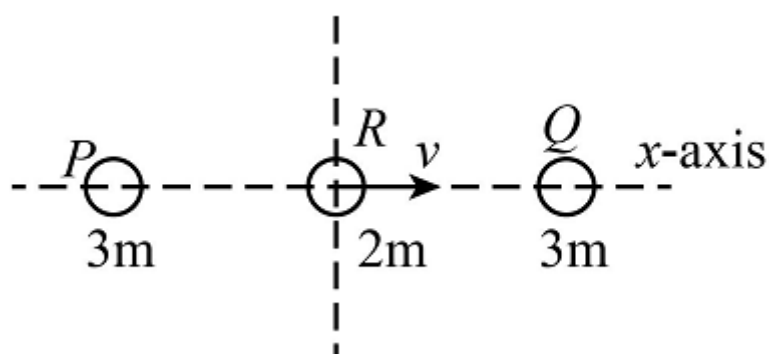
$$t^2 + t - 2 = 0$$

On solving the time comes out to be $t = 1$ s

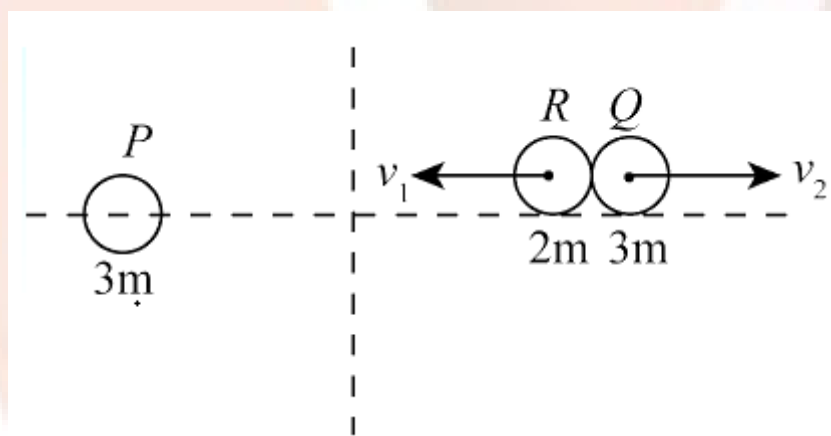
The distance DF is,

$$DF = v \cos \theta = \frac{25}{3} \times 0.8 = \frac{20}{3} \text{ m}$$

87. Consider the diagram given below,



As the three particles collide elastically and head on therefore, the collision of particle R and Q will be as shown below.



Thus, the particle R and Q will never collide after 1st collision.

Therefore, the total number of collision is 2.

88. The velocity of water jet can be calculated as,

$$\begin{aligned}v &= \frac{q}{A \times 2d} \\&= \frac{1800 \times 10^{-3}}{(30 \times 10^{-4}) \times 2 \times 30} \\&= 10 \text{ m/s}\end{aligned}$$

The work done by engine is,

$$\begin{aligned}W &= mgd + \frac{1}{2}mv^2 \\W &= 1800 \times 10 \times 30 + \frac{1}{2} \times 1800 \times (10)^2 \\&= 630000 \text{ J}\end{aligned}$$

The power of the engine is,

$$P = \frac{W}{t} = \frac{630000 \text{ J}}{60} = 10.5 \text{ kW}$$

89. The moment of inertia of solid sphere is calculated as,

$$I_s = \frac{2}{5}mr^2 = \frac{2}{5} \times 100 \times (10)^2 = 4000 \text{ kg} \cdot \text{m}^2$$

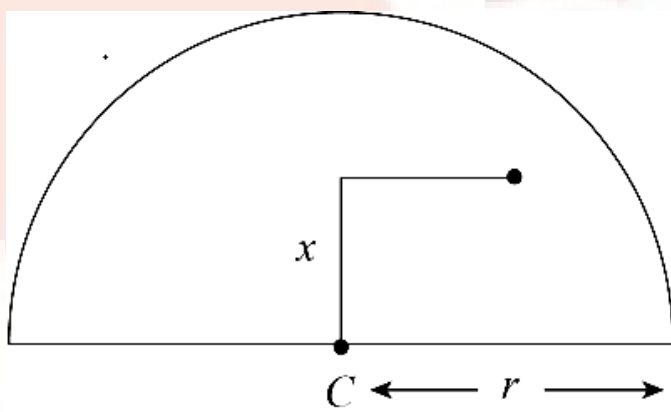
The moment of inertia of disc is calculated as,

$$I_d = \frac{1}{2}mr^2 = \frac{1}{2} \times 100 \times (20)^2 = 20000 \text{ kg} \cdot \text{m}^2$$

The rate of change of moment of inertia is,

$$\begin{aligned} \Delta I &= \frac{I_d - I_s}{t} = \frac{20000 - 4000}{60 \times 60} = \frac{16000}{60 \times 60} \\ &= \frac{40}{9} \text{ kg} \cdot \text{m}^2 \text{s}^{-1} \end{aligned}$$

90. Consider the diagram shown below,



The moment of inertia about the point C is,

$$I_C = \frac{mr^2}{2}$$

According to parallel axis theorem,

$$I_C = I_{cm} + mx^2$$

$$\begin{aligned} I_{cm} &= I_C - mx^2 \\ &= \frac{mr^2}{2} - mx^2 \end{aligned}$$

Here, the moment of inertia at point C is I_C

The expression for moment of inertia is given as,

$$I_C = \frac{mr^2}{2}$$

91. The expression for gravitational potential energy is given as,

$$PE = \frac{-Gm_1m_2}{r^2}$$

The negative sign shows the decrease in potential energy, which gives rise to increase in kinetic energy.

By applying energy conservation law, we get

$$\begin{aligned} \frac{1}{2}m_1v_1^2 &= \frac{Gm_1m_2}{r} \\ v_1 &= \sqrt{\frac{2Gm_2}{r}} \end{aligned}$$

And,

$$\frac{1}{2}m_2v_2^2 = \frac{Gm_1m_2}{r}$$

$$v_1 = \sqrt{\frac{2Gm_1}{r}}$$

The relative velocity of approach is,

$$v_1 + v_2 = \sqrt{\frac{2Gm_2}{r}} + \sqrt{\frac{2Gm_1}{r}}$$

$$= \sqrt{\frac{2G}{r}}(m_1 + m_2)$$

$$= \left[\frac{2G(m_1 + m_2)}{r} \right]^{1/2}$$

92. Applying the law of conservation of momentum, we get

$$mu_A r_A \sin \theta_A = mu_B r_B \sin \theta_B$$

$$\frac{u_A}{u_B} = \frac{r_B \sin \theta_B}{r_A \sin \theta_A}$$

$$\frac{u_A}{u_B} = \frac{60 \times 10^6 \sin 60^\circ}{90 \times 10^6 \sin 30^\circ}$$

$$= \frac{2}{\sqrt{3}}$$

Thus, the ratio of velocities comes out to be $2\sqrt{3}$.

93. From the expression of bulk modulus,

$$\beta = \frac{p}{\left(\frac{\Delta V}{V}\right)} = \frac{pV}{\Delta V}$$

$$\Delta V = \frac{pV}{\beta}$$

Substituting the values in above expression, we get

$$\Delta V = \frac{8000 \times 10^3 \times (7)^3}{140 \times 10^9} = 19.6 \times 10^{-3} \text{ cm}^3$$

94. The water can not enter the pinhole only when the hydrostatic pressure is equal to the excess pressure.

$$h\rho g = \frac{2T}{r}$$

Here, h is the depth upto which the cylinder is immersed, T is the surface tension of the water and r is the radius of the pinhole.

Substituting the given value in above expression, we get

$$\begin{aligned}h &= \frac{2T}{\rho gr} \\&= \frac{2 \times 0.07}{0.1 \times 10^{-3} \times 1000 \times 10} \\&= 0.14 \text{ m} \\&= 14 \text{ cm}\end{aligned}$$

95. From the expression of linear expansion,

$$\Delta R = R\alpha\Delta T$$

$$\frac{\Delta R}{R} = \alpha\Delta T$$

If f is the initial focal length and f' is the final focal length, then

$$\frac{\Delta f}{f} = \alpha\Delta T$$

$$\Delta f = f\alpha\Delta T$$

$$f' - f = f\alpha\Delta T$$

$$f' = f(1 + \alpha\Delta T)$$

Substituting the given value in above expression, we get

$$f' = 150(1 + 12 \times 10^{-6} \times 200)$$

$$f' = 150.36 \text{ cm}$$

96. The rate of heat flow can be calculated as,

$$\begin{aligned}\Delta Q &= \frac{ms \times 4.184 \text{ J}}{t} \\ &= \frac{20 \times 80 \times 4.184 \text{ J}}{300 \text{ s}} \\ &= 22.314 \text{ J/s}\end{aligned}$$

The thermal conductivity of rod can be calculated as,

$$\begin{aligned}\Delta Q &= \frac{kA\Delta T}{l} \\ 22.314 &= \frac{k(2.8 \times 10^{-4}) \times 80}{0.1} \\ k &= 99.61 \approx 100 \text{ J/s} \cdot \text{m} \cdot \text{K}\end{aligned}$$

97. Differentiation the given equation, we get

$$V = kT^{2/3}$$

$$dV = \frac{2}{3}kT^{-1/3}dT$$

The expression for work done is given as,

$$dW = PdV = \frac{RT}{V}dV$$

$$dW = \frac{RT}{kT^{2/3}}dV$$

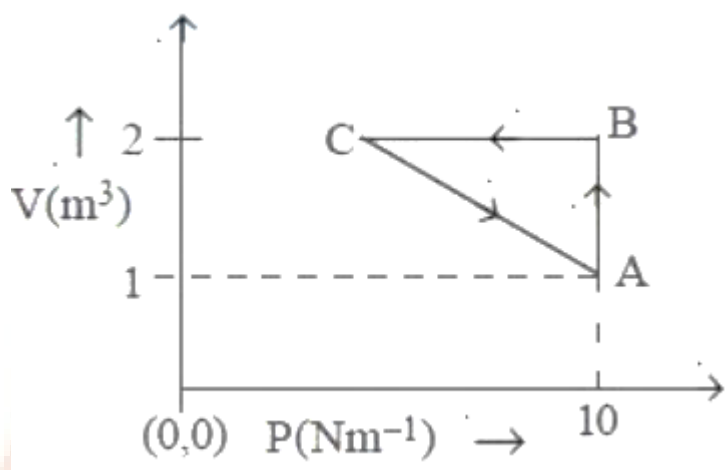
$$dW = \frac{R}{kT^{-1/3}}\left(\frac{2}{3}kT^{-1/3}dT\right)$$

$$dW = \frac{2}{3}RdT$$

Integrating the above expression, we get

$$\begin{aligned} W &= \frac{2}{3} \int_{T_1}^{T_2} RdT \\ &= \frac{2}{3}R(T_2 - T_1) \\ &= \frac{2}{3}R(60 - 0) \\ &= 40R \end{aligned}$$

98. Consider the given diagram..



The work done from A to B can be calculated as,

$$\begin{aligned}W_{AB} &= P \times dV \\&= 10 \times (2 - 1) \\&= 10 \text{ J}\end{aligned}$$

As the change in volume is zero during the process BC therefore, work done is zero.

From first law of thermodynamics,

$$\Delta Q = \Delta U + W$$

The internal energy change in whole process comes out to be 0.
Therefore,

$$Q = W$$

$$Q = W_{AB} + W_{BC} + W_{CA}$$

Substituting the values in the above equation, we get

$$\begin{aligned} W_{CA} &= 5 - W_{AB} - W_{BC} \\ &= 5 - 10 - 0 = -5\text{J} \end{aligned}$$

Thus, the magnitude of work in the process A to C is 5 J.

99. The expression for average translational kinetic energy is,

$$E = \frac{3}{2}kT$$

Substituting the given value in the above expression, we get

$$0.69 \text{ eV} = \frac{3}{2} \times 1.38 \times 10^{-23} T$$

$$0.69 \times 1.6 \times 10^{-19} \text{ V} = \frac{3}{2} \times 1.38 \times 10^{-23} T$$

$$T = 5333 \text{ K} = 5060^\circ\text{C}$$

100. Let v_p is the velocity of P wave, v_s is the velocity of S wave and d is the distance of epicenter from the point of observation. Then,

$$d = v_p t_p = v_s t_s$$

$$t_p = \frac{v_s}{v_p} t_s$$

$$t_p = \frac{4.5}{8} t_s$$

It is given that the first P wave arrives 3.5 min earlier than the first S -wave.

$$t_s - t_p = 3.5 \times 60$$

$$t_s - \frac{4.5}{8} t_s = 210$$

$$\begin{aligned} t_s &= \frac{210 \times 8}{3.5} \\ &= 480 \text{ s} \end{aligned}$$

The distance d comes out to be,

$$\begin{aligned} d &= v_s t_s \\ &= 4.5 \times 480 = 2160 \text{ km} \end{aligned}$$

101. The apparent frequency heard is given as,

$$f' = \left(\frac{v + v_0}{v - v_s} \right) f \quad \dots\dots (1)$$

Here, the source is stationary, it means $v_s = 0$

It is given that the speed of source is equal to $\frac{1}{5}$ times the speed of sound that is, $v_0 = \frac{v}{5}$

Therefore, from equation (1), we get

$$f' = \left(\frac{v + \frac{v}{5}}{v} \right) f = 1.2f$$

The wavelength remains unchanged irrespective of the motion of the observer.

102. The power of the mirror is calculated as,

$$\begin{aligned}P_m &= -\frac{1}{f_m} = -\frac{2}{(-R)} \\&= -\frac{2}{(-22)} = \frac{1}{11} \text{ D}\end{aligned}$$

The power of the lens is,

$$P_l = \frac{1}{20} \text{ D}$$

Now, the net power comes out to be,

$$\begin{aligned}P_{net} &= P_m + 2P_l \\&= \frac{1}{11} + 2\left(\frac{1}{20}\right) \\&= \frac{21}{110}\end{aligned}$$

So, the focal length of equivalent mirror is,

$$f = \frac{110}{21} \text{ cm}$$

From lens formula,

$$\frac{1}{f} = \frac{1}{v} + \frac{1}{u}$$
$$-\frac{21}{110} = \frac{1}{v} - \frac{1}{10}$$
$$v = -11 \text{ cm}$$

103. The condition for interference maxima is given as,

$$d \sin \theta = n\lambda \quad \dots (I)$$

Substituting the given values in above expression, we get

$$2\lambda \sin \theta = n\lambda$$

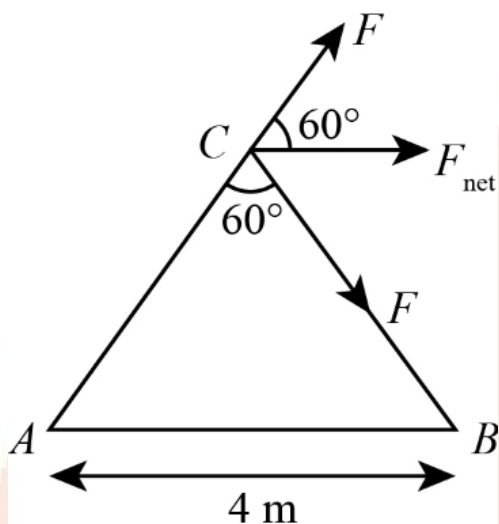
$$2 \sin \theta = n$$

The maximum value of $\sin \theta$ comes out to be 1.

The value of n comes out to be: $n = 2 \times 1 = 2$

Thus, total 5 integer is possible for equation (I) which are $-2, -1, 0, 1, 2$.

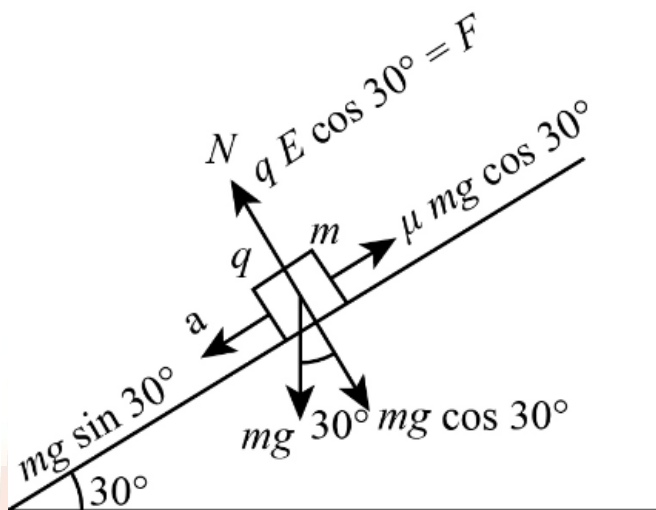
104. Consider the figure shown below,



The magnitude of total force acting on charge at C is,

$$\begin{aligned} F_{net} &= \frac{kQ_1Q_2}{r^2} \\ &= \frac{9 \times 10^9 \times (100 \times 10^{-6})^2}{(4)^2} \\ &= 5.625 \text{ N} \end{aligned}$$

105. The figure below shows the free body diagram.



Write the force equation along the plane of inclination.

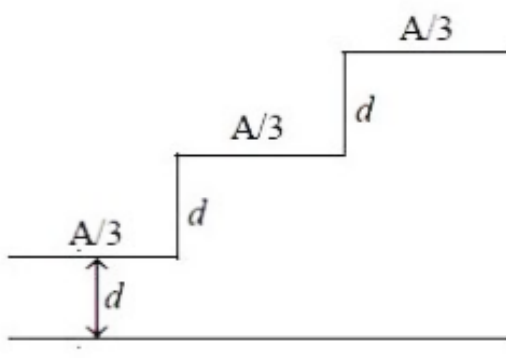
$$mg \sin 30^\circ - \mu mg \cos 30^\circ - qE \cos 30^\circ = ma$$

$$1 \times 10 \times \frac{1}{2} - 0.2 \times 1 \times \frac{\sqrt{3}}{2} - 0.01 \times 100 \times \frac{\sqrt{3}}{2} = 1 \times a$$

$$a = (5 - \sqrt{3}) \text{ m/s}^2$$

$$= 2.3 \text{ m/s}^2$$

106. Consider the figure shown below,



The capacitance for 1st capacitor is,

$$C_1 = \frac{\epsilon_0 A}{3d}$$

The capacitance for 2nd capacitor is,

$$C_2 = \frac{\epsilon_0 A}{3(2d)} = \frac{\epsilon_0 A}{6d}$$

The capacitance for 3rd capacitor is,

$$C_3 = \frac{\epsilon_0 A}{3(3d)} = \frac{\epsilon_0 A}{9d}$$

The equivalent capacitance of capacitor is,

$$\begin{aligned}
 C_{eq} &= C_1 + C_2 + C_3 \\
 &= \frac{\epsilon_0 A}{3d} + \frac{\epsilon_0 A}{6d} + \frac{\epsilon_0 A}{9d} \\
 &= \frac{11\epsilon_0 A}{18d}
 \end{aligned}$$

107. The expression for electric potential is given as,

$$V = \frac{1}{4\pi\epsilon_0} \frac{\sigma \cdot A}{r} = \frac{1}{4\pi\epsilon_0} \left(\frac{\sigma \cdot 4\pi r^2}{r} \right) = \frac{\sigma r}{\epsilon_0}$$

$$\sigma = \frac{\epsilon_0 V}{r}$$

The pressure due to surface tension inside the bubble is given as,

$$p_i = \frac{4S}{r} \quad \dots (i)$$

The electrostatic pressure outside the soap bubble is,

$$p_0 = \frac{\sigma^2}{2\epsilon_0} \quad \dots (ii)$$

From equation (i) and (ii) we get,

$$\frac{4S}{r} = \frac{\sigma^2}{2\epsilon_0} = \frac{\left(\frac{\epsilon_0 V}{r}\right)^2}{2\epsilon_0}$$

$$V = \sqrt{\frac{8Sr}{\epsilon_0}}$$

108. It is given that the ratio of heat generated through the shunt and galvanometer is 7:5.

The expression for heat generated through shunt is,

$$p_s = \frac{V^2}{R_s} \quad \dots\dots (1)$$

The expression for heat generated through galvanometer is,

$$p_g = \frac{V^2}{R_g} \quad \dots\dots (2)$$

Dividing equation (2) by equation (1), we get

$$\begin{aligned} R_s &= \frac{5}{7} \times R_g = \frac{5}{7} \times 112 \\ &= 80 \, \Omega \end{aligned}$$

109. The inductive reactance is calculated as,

$$X_L = \omega L = 2\pi f \times L$$

$$X_L = 2\pi \times 50 \times \frac{6}{\pi} = 600 \Omega$$

The capacitance reactance is calculated as,

$$X_C = \frac{1}{\omega C} = \frac{1}{2\pi f \times C}$$

$$X_C = \frac{1}{\omega C} = \frac{\pi \times 10^6}{2\pi \times 50 \times 50} = 200 \Omega$$

The resistance is calculated as,

$$Z^2 = R^2 + (X_L - X_C)^2$$

$$(500)^2 = R^2 + (600 - 200)^2$$

$$R^2 = 250000 - 160000$$

$$R = 300 \Omega$$

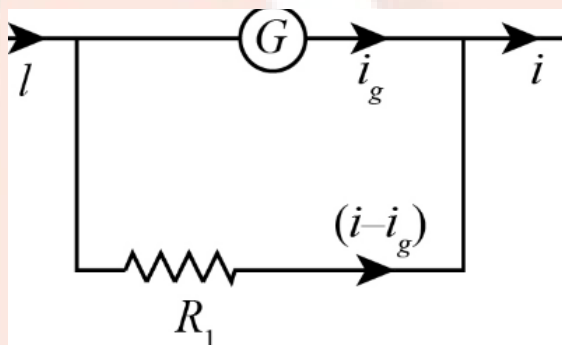
The impedance is calculated as,

$$Z = \frac{V_{rms}}{I_{rms}} = \frac{220}{440} \times 10^3$$

$$Z = 500 \Omega$$

110. As the charge on the neutron is zero therefore, it will not move in the circular path. Thus Assertion is not correct but the reason is correct.

111. For case I, consider the figure shown below.

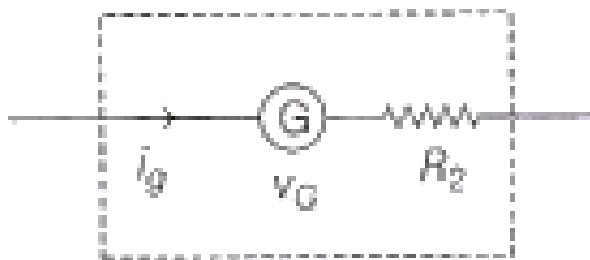


In this case, the shunt resistance is,

$$R_1 = \frac{G}{i - i_g}$$

$$R_1 = \frac{G}{51 - 1} = \frac{G}{50}$$

For case II, Consider the figure shown below.



The resistance in this case is given as,

$$\begin{aligned} R_2 &= G(V_G - 1) \\ &= G(11 - 1) \\ &= 10G \end{aligned}$$

The ratio of resistances comes out to be,

$$\frac{R_2}{R_1} = 500$$

112. If B_1 is the magnetic field due to straight part and B_2 is the magnetic field due to circular part, then the magnetic field at point C is given as,

$$B_C = B_1 + B_2$$

$$\begin{aligned} B_C &= \frac{\mu_0 i}{4\pi r} + \frac{\mu_0 i}{4r} \\ &= \frac{\mu_0 i}{4\pi r} (1 + \pi) \end{aligned}$$

113. The expression for magnetic flux is given as,

$$\phi = BNA$$

The expression for net charge is given as,

$$\begin{aligned} Q_{net} &= \frac{\phi}{R} \\ Q_{net} &= \frac{BNA}{R} \end{aligned}$$

Substituting the given values in above expression, we get

$$7.5 \times 10^{-3} = \frac{B \times 30 \times (2.5 \times 10^{-4})}{0.3}$$

$$B = 0.3 \text{ T}$$

114. The expression for energy stored in the capacitor is given as,

$$U = \frac{1}{2} \frac{Q^2}{C}$$

As, energy is equally distributed in electric field and magnetic field so, the energy stored in the capacitor is half of maximum.

$$\frac{1}{2} \times \frac{1}{2} \frac{Q^2}{C} = \frac{1}{2} \times \frac{Q'^2}{C}$$

$$\frac{1}{4} \frac{Q^2}{C} = \frac{1}{2} \times \frac{Q'^2}{C}$$

$$Q' = \frac{Q}{\sqrt{2}}$$

115. The energy density is calculated as,

$$u = \frac{1}{2} \epsilon_0 E_0^2$$

$$u = \frac{1}{2} \times 8.8 \times 10^{-12} \times (4)^2$$

$$u = 70.4 \times 10^{-12} \text{ J/m}^3$$

116. The expression for energy of photon when switch S_1 is closed is,

$$E = h\nu_0 + (5 + 1) \text{ eV} \dots\dots (I)$$

The expression for energy of photon when switch S_2 is closed and S_1 is open is,

$$2E = h\nu_0 + (20 - 5) \text{ eV} \dots\dots (II)$$

On solving equation (I) and (II), we get

$$h\nu_0 = 3 \text{ eV}$$

$$\nu_0 = \frac{3 \text{ eV}}{h}$$

$$\nu_0 = \frac{3 \times 1.6 \times 10^{-16} \text{ V}}{6.626 \times 10^{-34} \text{ J} \cdot \text{s}}$$

$$\nu_0 = 7.25 \times 10^{14} \text{ Hz}$$

The wavelength is calculated as,

$$\lambda = \frac{c}{\nu_0}$$

$$\lambda = \frac{3 \times 10^8 \text{ m/s}}{7.25 \times 10^{14} \text{ Hz}}$$

$$\lambda = 4133.3$$

117. According to Bohr's model,

$$\begin{aligned} mvr &= \frac{nh}{2\pi} \\ m\omega r^2 &= \frac{nh}{2\pi} \end{aligned} \quad \text{..... (1)}$$

The electrostatic force acting on particle A is balanced by the centripetal force. Therefore,

$$\begin{aligned} mr\omega^2 &= \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{r^2} \\ &= \frac{1}{4\pi\epsilon_0} \frac{(2q)(q)}{r^2} \\ &= \frac{1}{4\pi\epsilon_0} \frac{4q^2}{r^2} \end{aligned} \quad \text{..... (2)}$$

Solving equation (1) and (2), we get

$$\omega = \frac{2\pi m q^4}{\epsilon_0^2 h^3}$$

118. The disintegration constant can be calculated as,

$$\begin{aligned}\lambda &= \frac{1}{\tau} \\ &= \frac{1}{50 \times 60} \text{ s}^{-1} \\ &= \frac{1}{3000} \text{ s}^{-1}\end{aligned}$$

If N is the total number of nuclide, then the activity of the radioactive substance is given as,

$$\begin{aligned}\frac{dN}{dt} &= \lambda N \\ 2000 &= \frac{1}{3000} N \\ N &= 60 \times 10^5\end{aligned}$$

Thus, the number of nuclides is 60×10^5 .

119. The wavelength of the light photon can be calculated using the Planck's quantum theory.

$$E = \frac{hc}{\lambda}$$

Substituting the values in the above expression, we get

$$50 \times 10^{-3} \times 1.6 \times 10^{-19} = \frac{6.626 \times 10^{-34} \times 3 \times 10^8}{\lambda}$$
$$\lambda = 24.75 \mu\text{m}$$

120. If f_c is the carrier frequency and f_s is the signal frequency, then the side band frequencies are given as,

$$f = f_c \pm f_s$$

$$f = (1 \times 10^6) \pm (10 \times 10^3)$$

$$f = (1000 \pm 10) 10^3 \text{ Hz}$$

$$f = 1010 \text{ kHz or } 990 \text{ kHz}$$

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121. The de-Broglie equation is,

$$\lambda = \frac{h}{mv}$$

$$v = \frac{h}{m \times \lambda}$$

$$v = \frac{6.625 \times 10^{-34} \text{ Js}}{(9.1 \times 10^{-31} \text{ kg})(182 \times 10^{-9} \text{ m})}$$
$$= 4 \times 10^3 \text{ m/s}$$

The kinetic energy of the particle is given by,

$$\text{KE} = \frac{1}{2}mv^2$$
$$= \frac{1}{2} \times (9.1 \times 10^{-31} \text{ kg})(4 \times 10^3 \text{ m/s})^2$$
$$= 7.28 \times 10^{-24} \text{ J}$$

122. The relation of the radius of orbit is given as

$$r_n = r_o \times \frac{n^2}{Z}$$

$$52.9 \text{ pm} = 52.9 \text{ pm} \times \frac{n^2}{Z}$$

$$n^2 = Z$$

The energy is given by,

$$E_n = E_o \times \frac{Z^2}{n^2}$$

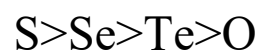
$$\begin{aligned} E_n &= (-2.18 \times 10^{-18}) \times \frac{n^4}{n^2} \\ &= (-2.18 \times 10^{-18}) \times n^2 \end{aligned}$$

For $n = 2$, the energy is given by,

$$\begin{aligned} E_n &= (-2.18 \times 10^{-18}) (2^2) \\ &= -8.72 \times 10^{-18} \text{ J} \end{aligned}$$

123. When we move down in the periodic table the enthalpy of electron gain decreases.

Therefore, the correct order is,



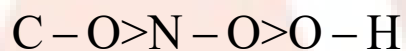
124. The value of dipole moment varies with the electronegativity of the elements in a compound.

The correct order of the dipole moment is,



The electronegativity difference increases then bond length decreases.

The correct order is,



Therefore, the statement II and III are correct.

125. He_2^+ has bond order of 0.5. It has one unpaired electron. So it is paramagnetic.

Li_2^+ has bond order of 0.5. It has also paramagnetic nature.

B_2 has bond order one. It has paramagnetic nature.

C_2 has bond order 2. It is diamagnetic in nature.

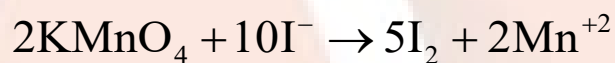
126. The kinetic energy is given by

$$v_{rms} = \sqrt{\frac{2KE}{M}}$$

$$KE = \frac{3RT}{2} \text{ and } = \sqrt{\frac{2(4 \times 10^3)}{32 \times 10^{-3}}}$$

$$= 500 \text{ m/s}$$

127. The chemical reaction is as follows.



n -factor of $\text{KMnO}_4 = 5$

The conversion I^- to I_2 then, n -factor = 2

Number of equivalent of KMnO_4 = Number of equivalent of I^- .

$$0.02 \times V \times 5 = 60 \times 0.01 \times 2$$

$$v = 6 \text{ mL}$$

128. The energy required by 12 g of graphite is 248 kJ/mol.

Therefore, the energy required by 6 g is 124 kJ/mol.

The heat required is given by

$$\begin{aligned}124 \text{ kJ/mol} &= c_v (T_2 - T_1) \\&= c_v (31 - 25) \\c_v &= 20.6 \text{ kJ/K}\end{aligned}$$

129. Because of common ion effect, the solubility of AgCl in 0.1M of KCl will be less.

130. Only H_2O , HSO_4^- are the only compound which acts as both bronsted acid and base as well.

131. The bond length depends upon the electrostatic force of attraction. It does not vary for the isotope. Therefore, the given element will have same bond length.

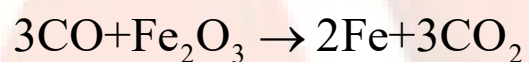
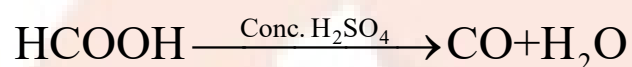
132. The bigger atom can stabilise the bigger superoxide ion with less negative field.

Since, the size of fluorine ion is smaller, it has high lattice energy and hence less solubility.

Therefore, statement II and III are correct.

133. Out of the given element only Ga can exist in liquid phase at room temperature, because of the presence of strong van der Waals' force.

134. The chemical reaction is as follows.



Thus, the element D is Fe.

135. The non-biodegradable waster from the fertiliser industries is Gypsum.

136. The hydrogen atom does not show hybridisation the compound $(\text{CH}_3)_4\text{C}$ has 20 hybridised orbital. Therefore, option (2) is correct.

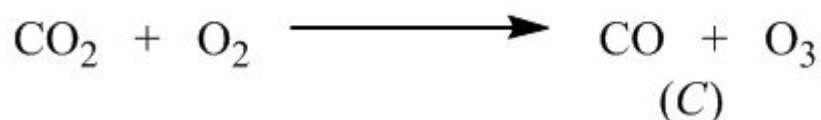
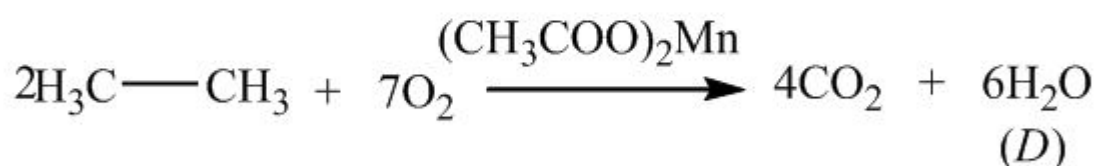
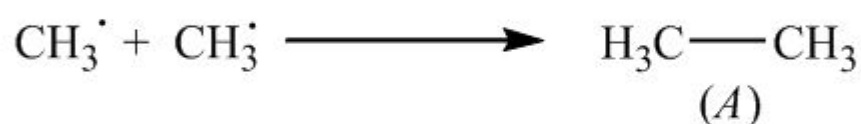
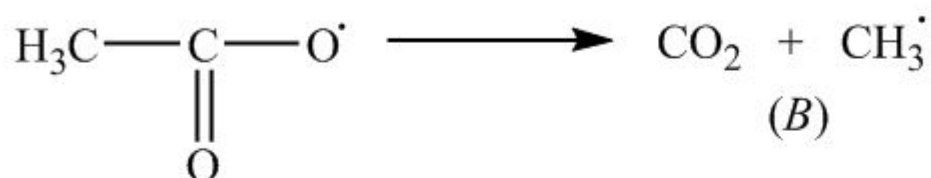
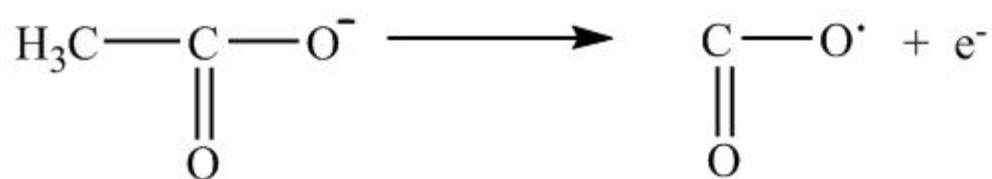
137. By the gas equation,

$$\frac{p_1 V_1}{T_1} = \frac{p_2 V_2}{T_2}$$
$$\frac{725 \times 50}{300} = \frac{760 \times V_2}{273}$$
$$V_2 = 434 \text{ mL}$$

$$\text{mass of } 43.4 \text{ L of } \text{N}_2 \text{ is } \frac{28 \times 434}{22400} = 5.4 \times 10^{-2} \text{ g of } \text{N}_2.$$

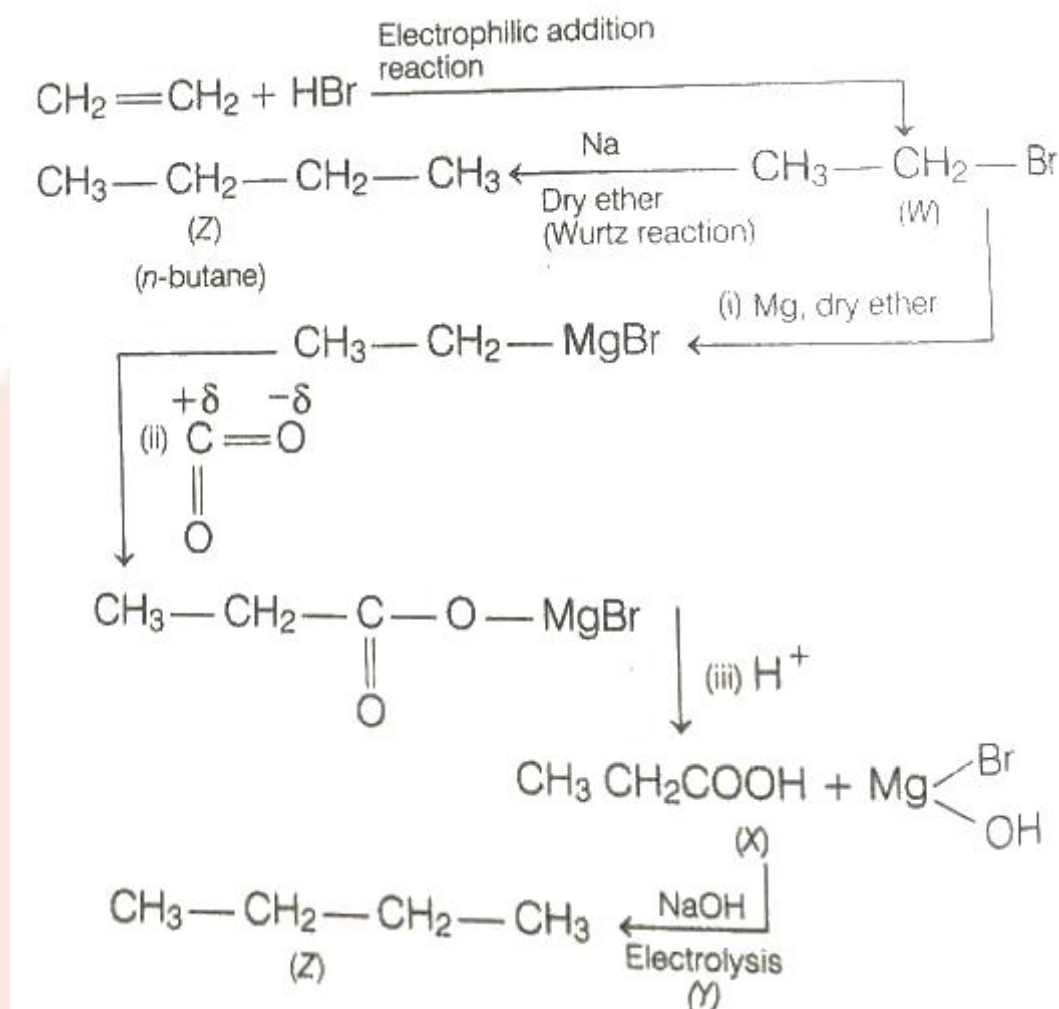
$$\% \text{ mass of } \text{N}_2 = \frac{5.4 \times 10^{-2}}{1} \times 100 = 5.4\%$$

138. The Kolbe, electrolysis of $\text{CH}_3\text{COO}^-\text{Na}^+$ is given by the reaction,



Therefore, option (3) is correct

139.



Therefore, option (d) is correct.

140. The density is given by

$$\rho = \frac{Z \times \text{atomic weight}}{a^3 \times N_A}$$

Substitute the values in the above equation.

$$d = \frac{4 \times 63.5}{x^3 \times 6.023 \times 10^{23} \times (10^{-8})^3}$$

$$= \frac{423}{x^3}$$

141. The number of moles of NaOH = $\frac{0.2 \times 500}{1000} = 0.1 \text{ mol}$

The number of moles of $\text{H}_2\text{SO}_4 = \frac{0.1 \times 200}{1000 \times 2} = 0.01 \text{ M}$

The number of moles of urea = $\frac{6}{60} = 0.1 \text{ M}$

Therefore, option (a) is correct.

142. Let, the assume the amount of naphthalene is $x \text{ g}$.

The expression of the molality is given by,

$$\frac{\Delta T}{K_f} = \frac{\frac{x}{w_{\text{C}_{10}\text{H}_8}} + \frac{6-x}{w_{\text{C}_{14}\text{H}_{10}}}}{300} \times 1000$$

$$\frac{0.7}{5.1} = \frac{\frac{x}{128} + \frac{6-x}{178}}{300} \times 1000$$

$$x = 3.4 \text{ g}$$

The mass of anthracene is $6 - 3.4 = 2.6 \text{ g}$

$$143. E_{cell}^{\circ} = 0.34 - (-0.76) = 1.1 \text{ V}$$

The Nernst equation is given by,

$$E = E^{\circ} - \frac{2.303RT}{nF} \log Q$$

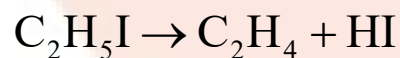
$$E = E^{\circ} - \frac{2.303RT}{nF} \log \left(\frac{\text{Zn}^{2+}}{\text{Cu}^{2+}} \right)$$

$$E = 1.1 - \frac{0.059}{n} \log \left(\frac{C_2}{C_1} \right)$$

The emf will be maximum when $\frac{C_2}{C_1}$ should be minimum. The minimum possible value will be 0.1.

Therefore, option (1) is correct.

144. The governing reaction is,



The above reaction is of first order,

$$\begin{aligned} k &= \frac{2.303}{t} \log \frac{p}{p_o - x} \\ &= \frac{2.303}{1000} \log \frac{2}{0.1} \\ &= 3 \times 10^{-3} \text{ min}^{-1} \end{aligned}$$

145. I. Sulphur is multi-molecular colloid.

II. The diameter of the particle must not be less than the wavelength.

III. Peptisation is the process of forming stable colloid by using an electrolyte to break up a precipitate and dispersed into the colloids.

IV. Eosin and glectin are negatively charged solution.

146. The Siderite and Calamine are the carbonate ores.

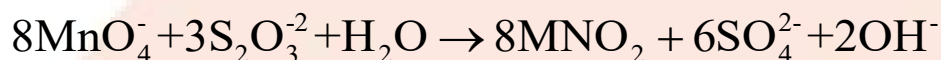
147. The S_8 molecule has both monoclinic as well as rhombic.

148. Gold and platinum are soluble in aqua-regia. It is a mixture of HCl and HNO_3 in the ratio of 1:3.

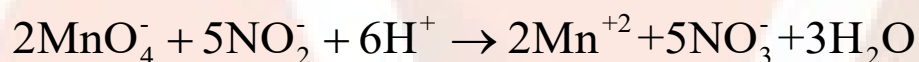
149. Mn_2O_7 , Cr_2O_3 , V_2O_5 , since this group has higher state of central metal.

150. The wavelength absorbed by the complexes is inversely proportional to the strength of ligands. Therefore, the correct order is $\lambda_1 > \lambda_3 > \lambda_2$.

151. The chemical reaction in neutral solution is given as follows



In acidic medium the reaction is,



Therefore, option (4) is correct

152. The correct match of the given lists is

A	B	C	D
II	V	I	III

153. The Option (2) is correct.

Hydrolysis of the X Y and Z gives the given products.

154. Chloramphenicol, Ofloxacin and Ampicillin are the broad spectrum antibiotics.

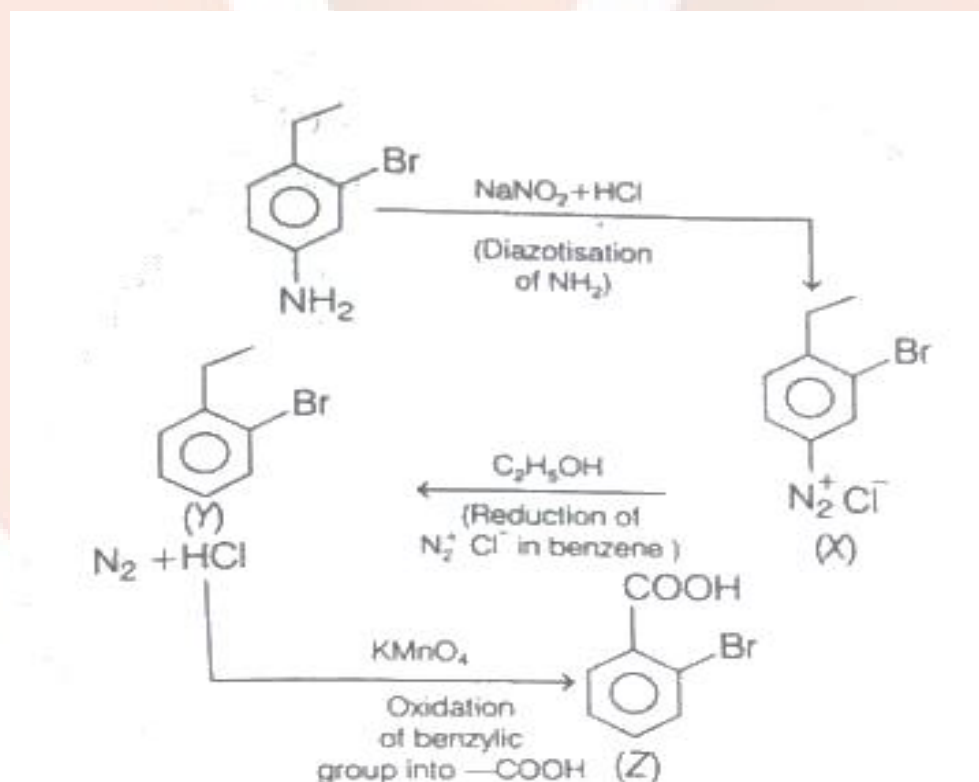
155. The rate of S_N2 reaction is inversely proportional to the sterically hindrance.

Therefore, the correct order of reactivity is

$$Q > R > P.$$

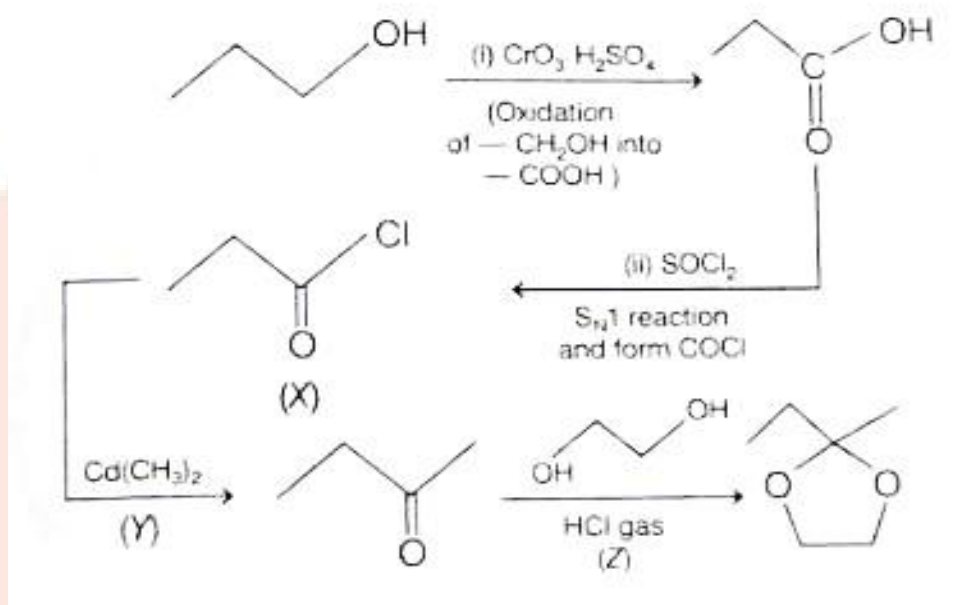
156. In alcohol, it is sp^3 hybridisation and the value of the bond angle is 109° .

157. The chemical reaction is as follows.

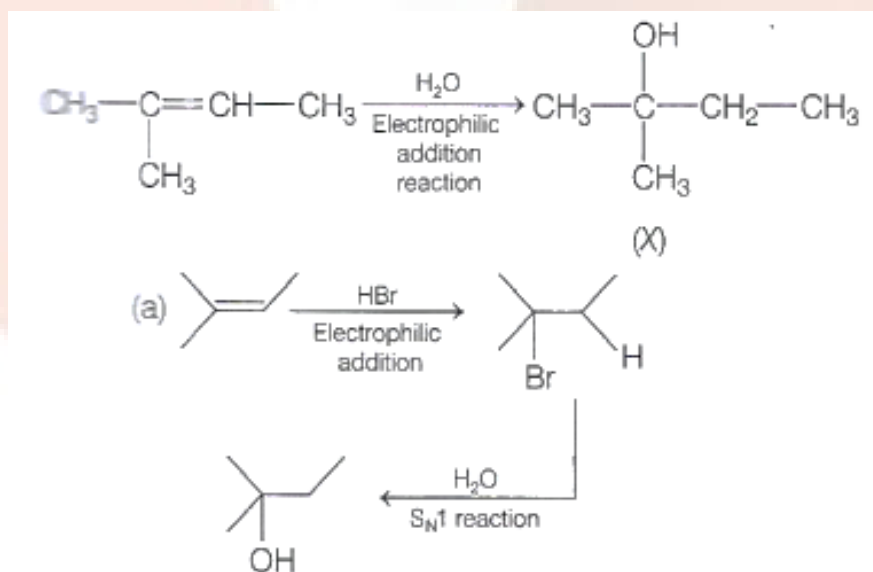


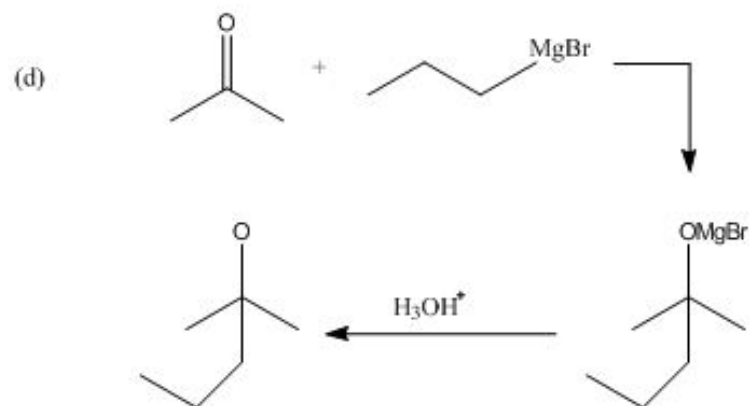
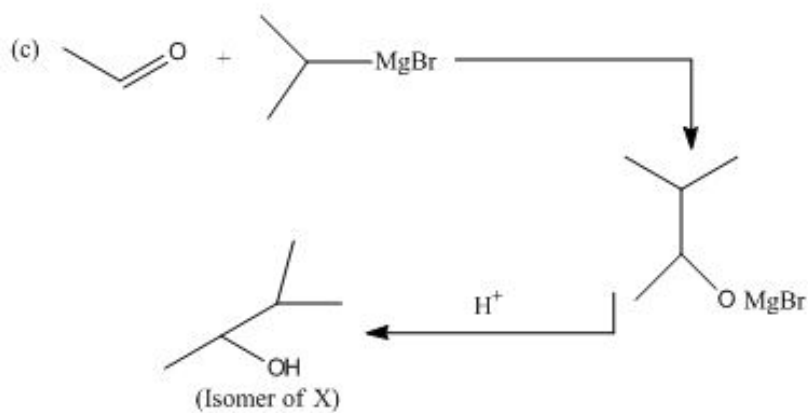
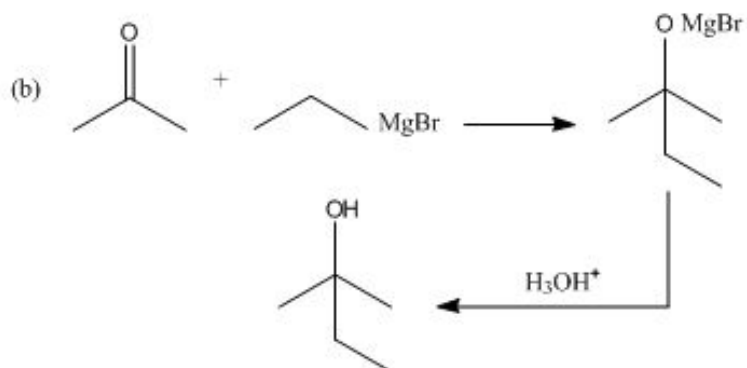
Therefore, option (4) is correct.

158. The given reaction is,

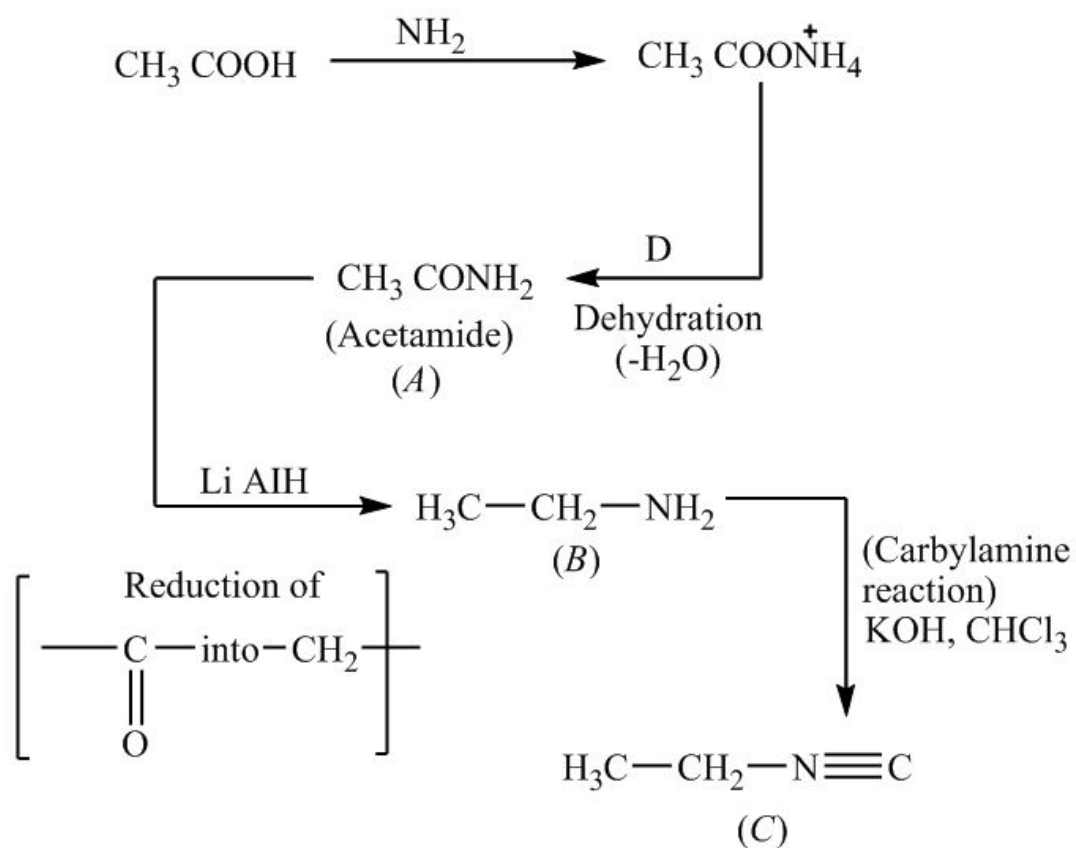


159. The complete reaction is,





160. The complete reaction is,



Therefore, option (2) is correct.